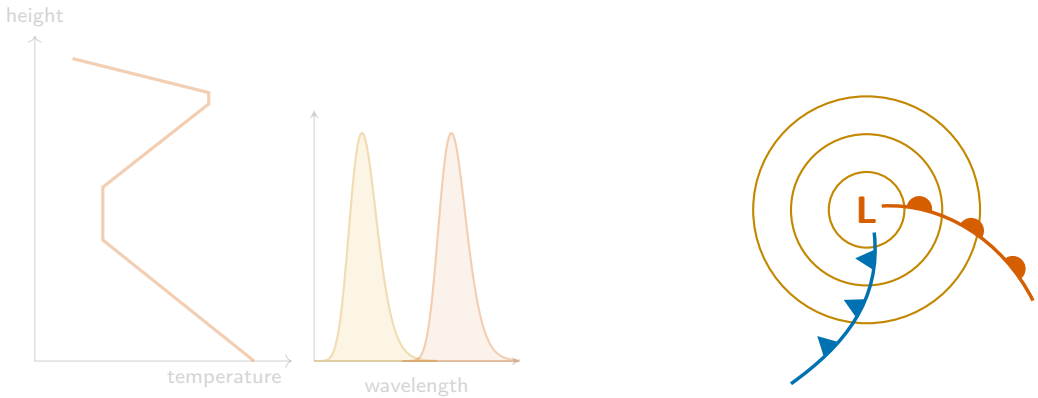
The Atmosphere: Part III

Atmospheric moisture, atmospheric motion and weather map analysis

A worked, step-by-step guide with figures, examples and answers



Part I Part II Part III

Part III of The Atmosphere

One part of The Atmosphere, with the same numbering as the full book. Every question, data table, figure and worked answer is on the page.

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Chapter 1

How to use this book

1.1 What is in it

This book teaches the atmosphere through questions you work yourself. It has three parts, each following on from the one before: Part I describes the structure of the atmosphere, Part II explains how the Sun heats it, and Part III how that uneven heating fills it with moisture and sets it in motion. Everything you need is on the page: every question, data table, map and figure, with its worked answer.

Table 1.1. The three parts of the book.

Part	What it covers	Chapters	Appendix
I	**Vertical structure of the atmosphere.** Pressure and the halving rule, oxygen at altitude, the standard atmosphere, soundings, inversions and the tropopause, lapse rates, stability, climates.	2–10	A
II	**Solar radiation.** The laws of radiation, black bodies and planetary temperatures, the Earth's energy balance and the greenhouse effect, sun angle and the seasons.	11–19	B
III	**Moisture, motion and weather maps.** Humidity and the dew point, the forces that move the air, the geostrophic wind and friction, station models, streamlines, fronts and a full weather-map analysis.	20–30	C

Every part opens with a chapter that maps each question to the chapter that answers it, and closes with a chapter that gathers all the answers in one table, a list of mistakes to avoid and a self-test. Its appendix holds the formulas, derivations and symbols. Appendix D lists the shared constants and units, and the references are at the end.

1.2 How each chapter is laid out

1. **The ideas first,** in plain language. Every important equation is numbered, for example (3.4), and followed by a *where* list of its symbols, units and values.
2. **The question**, in a box, with its data beside it.
3. **The method**, one step per line, with every number and unit shown.
4. **The answer**, in a green box, followed by a **check** that it is sensible.
5. **Worked examples**, **key ideas**, points to **take care** over, and **questions that come up**.
6. **Try it yourself**: a practice set, with answers in the green box that follows.
7. **Python**: the equations as functions, which you can run and change. They are also collected in one tested package.

How to get the most out of the book

Cover the green answer box, work the question on paper, then uncover it. If your answer differs, find the step where you diverged before reading on. That step is where the learning is.

1.3 Conventions used throughout

- **Numbering.** "Eq. (4.2)" is equation 2 of Chapter 4; figures and tables are numbered the same way. "§ 4.3" is section 3 of Chapter 4. Appendices are lettered A to D, one per part and one for shared constants.
- **Units** are SI, written as powers: $\mathrm{W\,m^{-2}}$ means watts per square metre, $\mathrm{m\,s^{-1}}$ metres per second. Pressure is in hectopascals, which are the same as millibars ($1\ \mathrm{hPa} = 1\ \mathrm{mb} = 100\ \mathrm{Pa}$). Heights are in metres or kilometres.
- **Temperature.** Temperatures in the data are in degrees Celsius. Any formula in which temperature is multiplied, divided or raised to a power (the gas law, the radiation laws, the potential temperature) needs kelvin: $T_\mathrm{K} = T_{\circ\mathrm{C}} + 273.15$. A *difference* of temperature is the same in both scales, so lapse rates can be written in $^\circ\mathrm{C\,km^{-1}}$ or $\mathrm{K\,km^{-1}}$.
- **Constants.** Each part uses rounded values, stated where they first appear; Appendix D gives precise values and sources.
- **Rounding.** Intermediate results keep at least four significant figures; answers are rounded at the end, usually to three or four.
- **Cross-references between parts** are written in the same way as within a part, with the part named: "Part II, § 17.7".
- **References** to the literature are given as author and year, for example Stull (2017), and listed in full at the end.

1.4 What to have beside you

A calculator with powers, logarithms and trigonometric functions, set to **degrees**; graph paper, a ruler, a soft pencil and three or four coloured pencils, so that each line on a graph has its own colour. Python is optional.

1.5 Acknowledgement

The questions and data in this book grew out of the practicals of EGM305 The Atmosphere at Ulster University, and are reused with the permission of the module staff. The explanations, figures, code and any errors are the author's.

1.6 Reading

Good textbooks for going further are Lutgens et al. (2019) and Ahrens and Henson (2022). A free textbook with worked examples is Stull (2017). For more depth, Wallace and Hobbs (2006) treats the physics at the next level up.

Part III

Moisture, motion and weather maps

Chapter 20

Part III at a glance

20.1 What this part covers

This part has three sections, each building a skill that the next one needs, so it pays to work through them in order. It brings together the two parts before it: the vertical structure of Part I gives the pressure surfaces whose slopes drive the wind, and the uneven solar heating of Part II is what makes them slope.

Moisture. How do we say how much water vapour the air holds? You meet four different measures (vapour pressure, mixing ratio, relative humidity and dew point) and learn to move between them with a table and a graph.

Motion. Why does the air move at all, and why does it move the way it does? You study the three forces that matter for large-scale winds (the pressure gradient force, the Coriolis force and friction) and see how their balance gives the geostrophic wind high up and a wind that crosses the isobars near the ground.

Weather maps. You decode the station model that every weather map uses, draw streamlines from wind barbs, and analyse a whole map: isobars, the low, the fronts, the cloud and the rain.

The moisture questions are numbered 1 to 6. The numbering starts again at 1 for the motion questions, which run to 11, and the weather map questions carry on from 12 to 14. A question of this part is always named with its section, for example *Moisture Q3*, *Motion Q7* or *Map Q12*.

Table 20.1. Where each question is answered, and the equations it needs.

Question	What it asks	Chapter	Equations
Moisture Q1	relative humidity of five samples	21	(21.3)
Moisture Q2	heating cold air indoors in winter	21	(21.3)
Moisture Q3	ranking five samples by actual vapour	21	(21.4)
Moisture Q4	the weakness of relative humidity	21	
Moisture Q5	dew point of 20 °C, 10 g kg^{-1}	22	(21.2)
Moisture Q6	dew point of 34 °C at 62 %	22	(21.4), (22.1)
Motion Q1–Q6	pressure surfaces, Saskatoon and Monterrey	24	(24.1), (23.1)
Motion Q7	geostrophic wind, north and south	25	(25.1)
Motion Q8–Q9	friction and the forces on a box	26	(26.1)
Motion Q10–Q11	four circulation systems	26	(26.1)
Map Q12	decode four station models	27	(27.1)
Map Q13	streamlines	28	
Map Q14	fronts and the full analysis	29	(25.1)

Chapter 23 builds the physics of the motion section before its questions; Chapter 30 gathers every answer in one table. The formulas, the 34 stations of the weather map, derivations and weather symbols are in Appendix C.

20.2 What to have beside you

- A calculator that has `ln` and `exp` keys, though most questions need only multiplication and division.
- Four coloured pencils for the force diagrams: red for the pressure gradient force, blue for the Coriolis force, green for friction, and black for the wind.
- A ruler and a soft pencil for the weather map.

20.3 Symbols and units

Table 20.2. The symbols used in Part III.

Symbol	Meaning	Unit
T, T_d	air temperature, dew-point temperature	°C (°F on the station models)
e, e_s	vapour pressure, saturation vapour pressure	hPa (= mb)
w, w_s	mixing ratio, saturation mixing ratio	$\mathrm{g\,kg^{-1}}$
RH	relative humidity	%
p	air pressure	hPa (= mb)
z	height above the ground	m
ρ	air density (about 1.2 $\mathrm{kg\,m^{-3}}$ near the ground)	$\mathrm{kg\,m^{-3}}$
f	Coriolis parameter, $f = 2\Omega \sin\varphi$	$\mathrm{s^{-1}}$
V, V_g	wind speed, geostrophic wind speed	$\mathrm{m\,s^{-1}}$ (or knots)
Δn	distance across the isobars (or height lines)	km

Millibar and hectopascal. Older charts say *mb*; modern texts say *hPa*. They are the same: 1 mb = 1 hPa = 100 Pa. This part uses both.

Knots. Wind barbs use knots: 1 kt = 0.514 $\mathrm{m\,s^{-1}} \approx 1.85$ $\mathrm{km\,h^{-1}}$.

20.4 About the weather map

The weather map shows 34 stations in the central United States on one day. Wind speeds were obtained by counting barbs and are approximate. The fronts were placed by hand from the temperature, dew-point and wind changes, as an analyst would. § C.2 lists every station.

Chapter 21

Moisture in the air: mixing ratio and relative humidity

This chapter answers Moisture Q1 to Q4.

21.1 Why water vapour matters

Water vapour is the invisible, gaseous form of water. It makes up between a fraction of a percent and about four percent of the air by volume, yet it is the ingredient of every cloud, every shower and every fog. It also sets how comfortable a day feels, because sweat evaporates slowly into air that is already moist. To talk about weather precisely we need a way to say *how much* vapour the air contains. There are four common measures. This chapter covers the first three; the fourth, the dew point, has its own chapter.

21.2 Vapour pressure and saturation

Air is a mixture of gases, and each gas contributes a share of the total pressure, in proportion to how many of its molecules are present: the partial pressure met for oxygen in Part I (§ 4.4). The share due to water vapour is the **vapour pressure**, written e. In the lowest kilometres it is about 1 to 30 hPa, out of a total pressure of roughly 1000 hPa.

Imagine a closed dish of water with air above it. Molecules leave the water surface (evaporation) and return to it (condensation). At first more leave than return, so the vapour pressure above the water rises. It stops rising when the two flows are equal. The air is then **saturated**, and the vapour pressure has reached its upper limit for that temperature: the **saturation vapour pressure**, e_s.

The warmer the water, the more energetic its molecules and the more easily they escape, so e_s rises with temperature, and it rises steeply. Table 21.1 gives it as the "mb" column: 6.1 mb at 0 °C, 23.4 mb at 20 °C and 73.8 mb at 40 °C. Plotted, it is the curve of Figure 21.1 (the Clausius–Clapeyron curve).

A question that comes up

Why does warm air "hold" more water than cold air?
Air does not really hold water the way a sponge does. The saturation vapour pressure is a property of water itself, how readily it evaporates at a given temperature, and the other gases hardly affect it. "Holding capacity" is a convenient picture, and this part uses it, but it is the temperature that sets the limit.

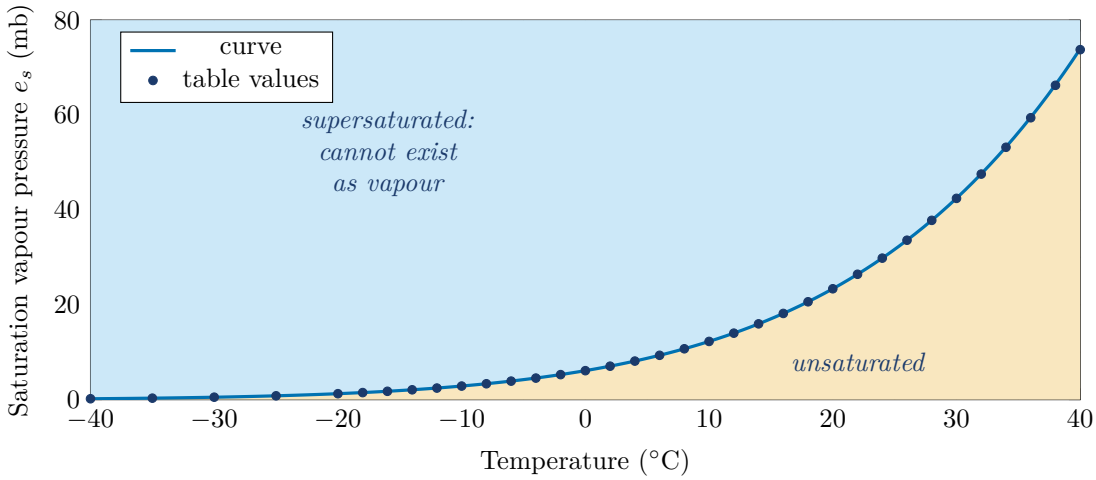


Figure 21.1. Saturation vapour pressure against temperature. The dots are the values of Table 21.1. Air whose vapour pressure and temperature fall below the curve is unsaturated; points above it are not possible for long, because the surplus vapour condenses.

21.3 Mixing ratio

Vapour pressure is not the only way to measure moisture. The **mixing ratio**, w, compares the *mass* of vapour with the mass of the *dry air* that goes with it:

$$w = \frac{\text{mass of water vapour}}{\text{mass of dry air}}. \tag{21.1}$$

where

w	mixing ratio ($\mathrm{g\,kg^{-1}}$: grams of vapour per kilogram of dry air)	calculated

Typical values run from $0.1\,\mathrm{g\,kg^{-1}}$ in the polar winter to $20\,\mathrm{g\,kg^{-1}}$ in the humid tropics. When the air is saturated the ratio is the **saturation mixing ratio**, w_s, the second column of numbers in Table 21.1 (the "g/kg" column).

The two measures are linked. For a total pressure p and a vapour pressure e,

$$w \approx 622\,\frac{e}{p-e}. \tag{21.2}$$

where

w	mixing ratio ($\mathrm{g\,kg^{-1}}$)	calculated
e	vapour pressure (hPa)	
p	total air pressure (hPa)	995.5 for Table 21.1
622	ratio of the molar masses of water and dry air, 18/29, times 1000 (to give $\mathrm{g\,kg^{-1}}$)	

The factor 0.622 is the ratio of the molar masses of water (18) and dry air (29): a molecule of vapour is lighter than the average molecule of dry air.

A question that comes up

Why use the mass of *dry* air, and why grams per kilogram?
The mass of dry air in a rising or sinking parcel stays fixed, while the total mass of moist air does not, as soon as vapour condenses or evaporates. Dividing by the dry mass gives a number that stays constant for a parcel until water is added or removed. Grams per kilogram

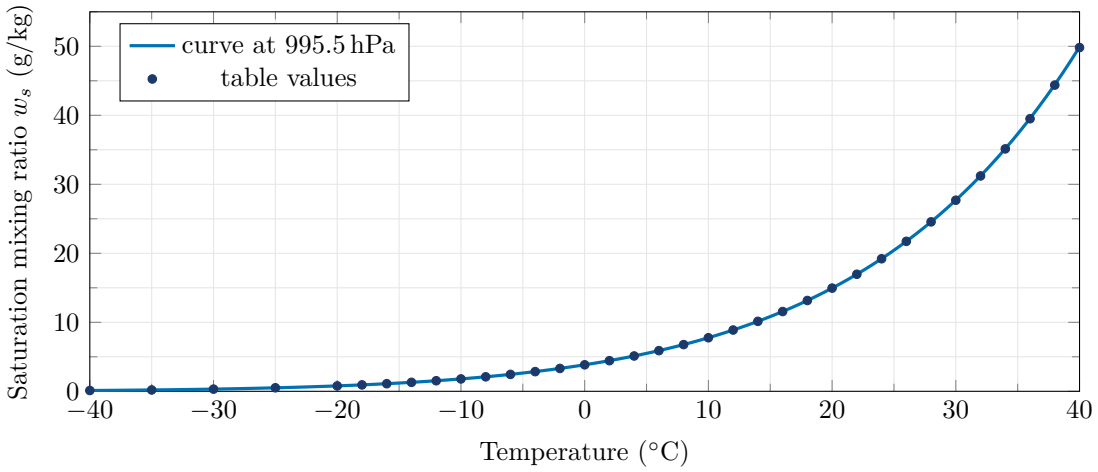


Figure 21.2. Saturation mixing ratio against temperature. The curve is computed from the saturation vapour pressure at a pressure of 995.5 hPa, the pressure of Table 21.1.

simply keeps the numbers handy: a few units to a few tens.

A question that comes up

Table 21.1 is for sea level. Which pressure is that?
Working backwards from the table, $p = e_s + 622\, e_s/w_s$ comes out at about 995.5 hPa for every row from −20 to 40 °C. That is a typical low-ish sea-level pressure, not the standard 1013.25 hPa. At 1013.25 hPa the mixing ratios would be about 1.8 % smaller. The difference never matters for these questions; the calculations here use 995.5 hPa so that every answer agrees with the table.

21.4 Relative humidity

Relative humidity (RH) says how close the air is to saturation. It is defined in two equivalent ways:

$$\mathrm{RH} = \frac{e}{e_s} \times 100\,\% = \frac{w}{w_s} \times 100\,\%. \tag{21.3}$$

where

RH	relative humidity (%)	calculated
e, e_s	vapour pressure and saturation vapour pressure (hPa)	Table 21.1
w, w_s	mixing ratio and saturation mixing ratio ($\mathrm{g\,kg^{-1}}$)	Table 21.1

An RH of 50 % means the air holds half of what it could hold at its present temperature; 100 % means saturated. Because e_s and w_s both depend strongly on temperature, *the same air has a different relative humidity at a different temperature.* That fact is behind most of the puzzles in this chapter.

Relative humidity in one line

Relative humidity = (how much vapour is there) ÷ (how much there could be at this temperature) ×100.

21.5 A picture to hold on to: the glass

Picture the air as a glass. The **size of the glass** is the saturation mixing ratio at the air's temperature. The **water in the glass** is the actual mixing ratio. Relative humidity is how full the glass is. Warm air has a big glass, cold air a small one.

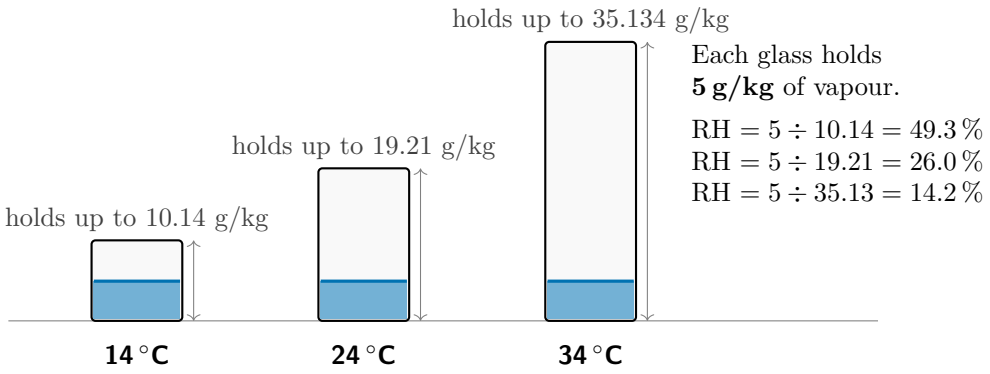


Figure 21.3. The glass picture. The same 5 g of vapour per kilogram of air fills the glass of 14 °C air to 49 %, the glass of 24 °C air to 26 %, and the big glass of 34 °C air to only 14 %. The amount of water is identical; only the size of the glass changes.

21.6 Reading the table

Three skills with Table 21.1 carry you through all the moisture questions.

Table 21.1. Saturation vapour pressure e_s and saturation mixing ratio w_s against temperature, at a sea-level pressure of 995.5 hPa.

Temp. (°C)	e_s **(mb)**	w_s (g kg^{-1})
−40	0.189	0.118
−35	0.314	0.195
−30	0.519	0.318
−25	0.807	0.510
−20	1.254	0.784
−18	1.488	0.931
−16	1.760	1.102
−14	2.076	1.300
−12	2.441	1.529
−10	2.863	1.794
−8	3.348	2.099
−6	3.906	2.450
−4	4.545	2.852
−2	5.275	3.313
0	6.108	3.840
2	7.055	4.439
4	8.129	5.120
6	9.347	5.894

Temp. (°C)	e_s (mb)	w_s (g kg^{-1})
8	10.722	6.771
10	12.272	7.762
12	14.017	8.882
14	15.977	10.140
16	18.173	11.560
18	20.630	13.162
20	23.373	14.956
22	26.430	16.963
24	29.831	19.210
26	33.608	21.734
28	37.796	24.557
30	42.430	27.694
32	47.551	31.213
34	53.200	35.134
36	59.422	39.502
38	66.264	44.381
40	73.777	49.815

Using the table. (1) To look up, read across from the temperature. (2) To interpolate, take the proportion of the way between two rows. (3) To look up backwards (for a dew point), find the two rows whose w_s bracket the value and interpolate for the temperature. Linear interpolation over a 2-degree step is accurate to about 0.1 %.

Skill 1. Look up a value at a temperature that is listed, for example $w_s(24\,^\circ\text{C}) = 19.21\,\text{g kg}^{-1}$.

Skill 2. Interpolate between two rows for a temperature that is not listed.

Skill 3. Look up backwards: given a mixing ratio, find the temperature at which it is the saturation value. This is how you find a dew point (Chapter 22).

Worked example: interpolating in the table

Find w_s at 25 °C. The table lists 24 °C (19.21) and 26 °C (21.734). 25 °C is half way between, so take the half-way value:

$$w_s(25) \approx \frac{19.21 + 21.734}{2} = 20.47\ \text{g kg}^{-1}.$$

The exact value from the formula is 20.44 g kg^{-1}. Because the curve is convex (it bends upwards), straight-line interpolation overestimates slightly, by about 0.1 % over a 2-degree step. That is far smaller than the precision of the question.

21.7 Question 1: relative humidity of five samples

The question

Using Table 21.1, determine the saturation mixing ratio and calculate the relative humidity of five air samples at the temperatures and mixing ratios shown.

Method.

Step 1. For each sample read the saturation mixing ratio w_s from Table 21.1 at the sample's temperature.

Step 2. Divide the sample's mixing ratio w by w_s and multiply by 100.

Step 3. Check: more vapour at the same temperature must give a higher RH; the same vapour in warmer air must give a lower RH.

Answer to Moisture Q1

Temp.	Sat. mixing ratio ($\mathrm{g\,kg^{-1}}$)	Mixing ratio ($\mathrm{g\,kg^{-1}}$)	RH (%)
14 °C	10.14	5	$5/10.14 = \mathbf{49.3}$
14 °C	10.14	9	$9/10.14 = \mathbf{88.8}$
24 °C	19.21	5	$5/19.21 = \mathbf{26.0}$
24 °C	19.21	2	$2/19.21 = \mathbf{10.4}$
34 °C	35.134	7	$7/35.134 = \mathbf{19.9}$

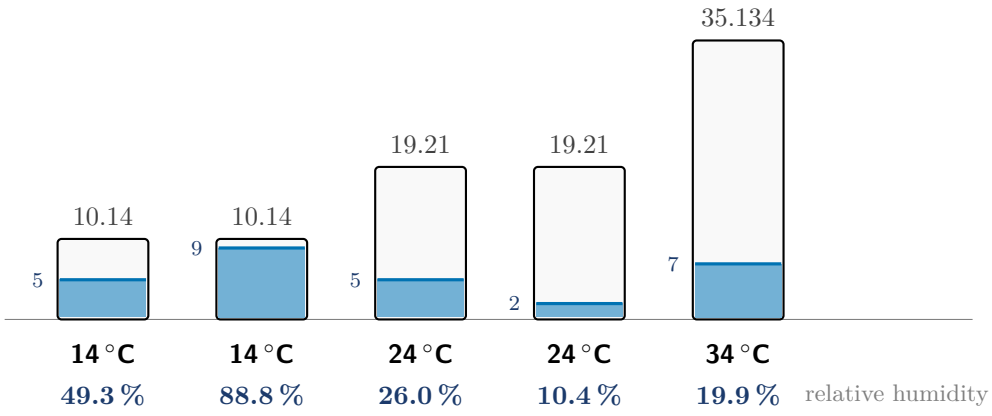


Figure 21.4. The five samples of Question 1 drawn as glasses on the same scale. The number at the top of each glass is its capacity (the saturation mixing ratio); the number just above the water is the sample's mixing ratio; the percentage below is the relative humidity.

Check. The two 14 °C samples have 5 and 9 $\mathrm{g\,kg^{-1}}$; the one with more vapour (49.3 % against 88.8 %) has the higher RH. The two 24 °C samples likewise. The 34 °C sample, with 7 $\mathrm{g\,kg^{-1}}$ in a huge glass, has the lowest RH of the five bar the 2 $\mathrm{g\,kg^{-1}}$ sample, as it should.

21.8 Question 2: cold air heated in a house

The question

In winter, cold air is brought into homes and heated. How does this change the humidity in the house?

Method. Follow one parcel of air. Heating changes its temperature, so it changes the size of the glass, but it adds no water, so the amount of vapour stays fixed.

Worked example: a cold day in Belfast

Outside, the air is at $0\,^\circ\mathrm{C}$ with a relative humidity of $80\,\%$. From Table 21.1 $w_s(0\,^\circ\mathrm{C}) = 3.84\,\mathrm{g\,kg^{-1}}$, so the air holds

$$w = 0.80 \times 3.84 = 3.07\ \mathrm{g\,kg^{-1}}.$$

Bring it indoors and heat it to $21\,^\circ\mathrm{C}$. The saturation mixing ratio at $21\,^\circ\mathrm{C}$ is about $15.9\,\mathrm{g\,kg^{-1}}$ (half way between the 16.0 of $20\,^\circ\mathrm{C}$ and 17.0 of $22\,^\circ\mathrm{C}$; the exact value is 15.93). Nothing has been added, so w is still $3.07\,\mathrm{g\,kg^{-1}}$ and

$$\mathrm{RH} = \frac{3.07}{15.9} \times 100 = 19\,\%.$$

To reach a comfortable $40\,\%$ the air would need $0.40 \times 15.9 = 6.4\,\mathrm{g\,kg^{-1}}$, which means adding $6.4 - 3.07 = 3.3\,\mathrm{g\,kg^{-1}}$ of water.

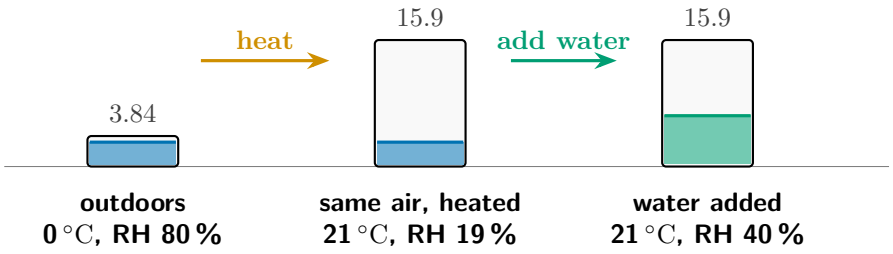


Figure 21.5. Question 2. Heating leaves the amount of vapour untouched but makes the glass four times bigger, so the relative humidity falls from $80\,\%$ to $19\,\%$.

Answer to Moisture Q2

The humidity in the house will **drop significantly**. Cold air holds little vapour, so the outside air has a low mixing ratio. Heating it without adding water leaves the mixing ratio unchanged but increases the saturation mixing ratio a great deal, so the relative humidity becomes very low.

21.9 Question 3: who holds the most water?

The question

Five samples are described by temperature and relative humidity. Complete the mixing ratios and rank them from 1 (most vapour) to 5 (least).

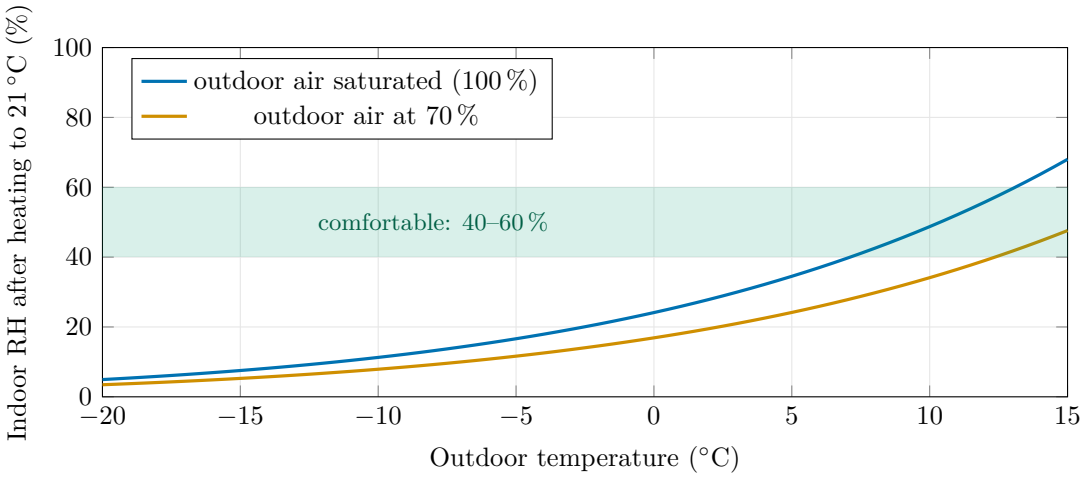


Figure 21.6. How dry the indoor air becomes. Outdoor air at −10 °C and 70 % humidity gives only about 6 % when heated to 21 °C with no water added; on a mild 10 °C day the same calculation gives 28 %.

Method. Turn each relative humidity back into an amount of vapour:

$$w = w_s \times \frac{\mathrm{RH}}{100}. \tag{21.4}$$

where

w	actual mixing ratio ($\mathrm{g\,kg^{-1}}$)	calculated
w_s	saturation mixing ratio at the air temperature ($\mathrm{g\,kg^{-1}}$)	Table 21.1
RH	relative humidity (%)	given

Step 1. Read w_s from Table 21.1 for each temperature.

Step 2. Multiply by the relative humidity written as a fraction (90 % is 0.90).

Step 3. Rank the results from the largest to the smallest.

Answer to Moisture Q3

Temp.	w_s ($\mathrm{g\,kg^{-1}}$)	RH	$w = w_s \times$RH ($\mathrm{g\,kg^{-1}}$)	Rank
14 °C	10.14	90 %	$10.14 \times 0.90 = 9.126$	3
20 °C	14.956	60 %	$14.956 \times 0.60 = 8.974$	4
24 °C	19.21	40 %	$19.21 \times 0.40 = 7.684$	5
30 °C	27.694	40 %	$27.694 \times 0.40 = 11.078$	1
34 °C	35.134	30 %	$35.134 \times 0.30 = 10.54$	2

Check. Make sure the ranking depends on the amount of vapour, not on the humidity. If you had ranked by RH you would have put 14 °C first. The glass picture shows why that is wrong: a small glass 90 % full holds less than a big glass 40 % full.

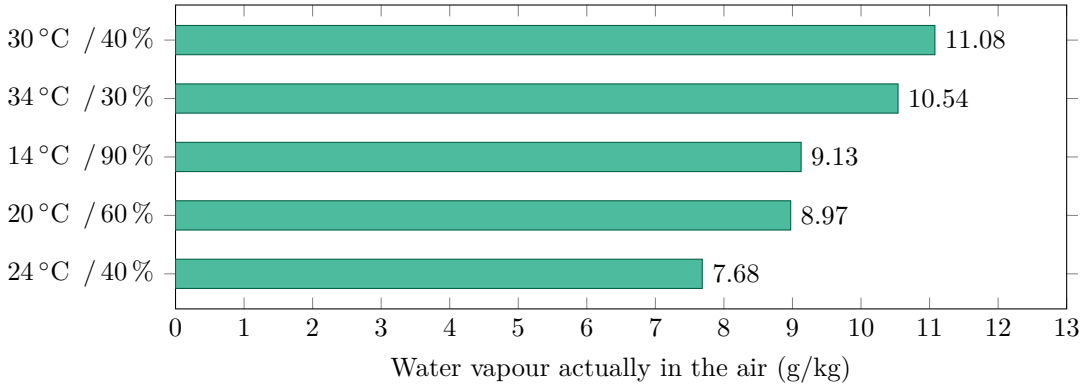


Figure 21.7. Question 3 ranked (rank 1 at the top). The 14 °C sample is the "dampest" at 90 % but is only third in actual water; the 30 °C sample, which feels "dry" at 40 %, holds the most.

21.10 Question 4: the disadvantage of relative humidity

Answer to Moisture Q4

Relative humidity does not tell you how much water vapour is actually in the air.

That is the short answer. The fuller explanation is worth knowing, because it recurs in weather forecasting.

1. **It depends on temperature as well as on the vapour.** The same air has an RH of 49 % at 14 °C and 26 % at 24 °C (Question 1). Nothing was added or removed.
2. **It changes through the day for no reason connected with moisture.** On a calm day the vapour content hardly changes, but the temperature rises by several degrees after dawn, so the RH falls.
3. **It cannot be used to compare two places.** The most humid-feeling sample in Question 3 (90 %) held less vapour than three drier-feeling ones.

Better measures. The mixing ratio and the dew point measure the vapour itself. Neither changes when air is heated or cooled, as long as nothing condenses. For that reason forecasters watch the dew point more closely than the relative humidity, and so do the station models you meet in Chapter 27.

21.11 Try it yourself: Practice Set 21

Practice Set 21: relative humidity and vapour amounts

21.1 Air at 28 °C has a mixing ratio of 15 g kg^{-1}. What is its relative humidity?

21.2 Air at 10 °C has a relative humidity of 70 %. How much water vapour does a kilogram of it contain?

21.3 Sample P: 32 °C, RH 25 %. Sample Q: 20 °C, RH 85 %. Which holds more water vapour, and how much more?

21.4 Air at 0 °C and RH 90 % is heated to 20 °C with no water added. What is its relative humidity now?

21.5 Air at 24 °C and RH 50 % is cooled to 14 °C with no water added (and no condensation). What is its RH now?

21.6 Use interpolation to find w_s at 31 °C, and hence the mixing ratio of air at 31 °C with RH 55 %.

Answers to Practice Set 21

21.1 $w_s(28) = 24.557$, so RH $= 15/24.557 = 61\,\%$.

21.2 $w_s(10) = 7.762$, so $w = 0.70 \times 7.762 = 5.43\,\mathrm{g\,kg^{-1}}$.

21.3 P: $31.213 \times 0.25 = 7.80\,\mathrm{g\,kg^{-1}}$. Q: $14.956 \times 0.85 = 12.71\,\mathrm{g\,kg^{-1}}$. Q holds more, by $4.91\,\mathrm{g\,kg^{-1}}$, although its temperature is 12 °C lower.

21.4 $w = 0.90 \times 3.84 = 3.456\,\mathrm{g\,kg^{-1}}$; $w_s(20) = 14.956$; RH $= 3.456/14.956 = 23\,\%$.

21.5 $w = 0.50 \times 19.21 = 9.605\,\mathrm{g\,kg^{-1}}$; $w_s(14) = 10.14$; RH $= 9.605/10.14 = 94.7\,\%$. The air is almost saturated: another degree or so of cooling brings the dew point.

21.6 $w_s(31) \approx (27.694 + 31.213)/2 = 29.45\,\mathrm{g\,kg^{-1}}$; $w = 0.55 \times 29.45 = 16.2\,\mathrm{g\,kg^{-1}}$.

21.12 Humidity in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def saturation_vapour_pressure(t_c):
    """Saturation vapour pressure (hPa) over water at t_c (deg C)."""
    return 6.112 * math.exp(17.67 * t_c / (t_c + 243.5))
```

```
def saturation_mixing_ratio(t_c, p_hpa=TABLE_PRESSURE_HPA):
    """Saturation mixing ratio (g/kg) at t_c and p_hpa."""
    e_s = saturation_vapour_pressure(t_c)
    return 622.0 * e_s / (p_hpa - e_s)
```

```
def relative_humidity(w, w_s):
    """Relative humidity (%) from the mixing ratio and the saturation mixing ratio."""
    return 100.0 * w / w_s
```

```
def mixing_ratio_from_rh(w_s, rh):
    """Mixing ratio (g/kg) from the saturation mixing ratio and the relative humidity (%)."""
    return w_s * rh / 100.0
```

`saturation_mixing_ratio(10)` returns about 7.76, the table's value at 10 °C.

Chapter 22

The dew point

This chapter answers Moisture Q5 and Q6.

22.1 Cooling the glass

Take a sample of air and cool it, taking nothing away and adding nothing. Two things happen. The vapour in the air stays the same, because no water has gone in or out. The glass (the saturation mixing ratio) shrinks, because colder air can hold less. So the glass gets fuller and the relative humidity rises. At one particular temperature the glass is exactly full. That temperature is the **dew point**, T_d.

Definition

The **dew point** is the temperature to which an air sample must be cooled, at constant pressure and without adding or removing water, to reach saturation. At that temperature the saturation mixing ratio equals the sample's actual mixing ratio, $w_s(T_d) = w$.

If you cool the air below its dew point the surplus vapour cannot stay as vapour. It condenses: into dew on cold grass, into fog near the ground, into cloud higher up. A cold can of drink on a summer day is covered with water for exactly this reason: the air against the can is cooled below its dew point.

22.2 Three ways to find a dew point

All three give the same answer. Question 5 asks for the first; the others are checks.

Method A. **Graph.** Plot the air as a point (temperature across, mixing ratio up). Cooling moves the point horizontally to the left, because only the temperature changes. Stop where the point meets the saturation curve. Read the temperature underneath.

Method B. **The table, backwards.** Find the two rows whose mixing ratios bracket the sample's, and interpolate for the temperature.

Method C. **Formula.** Compute the vapour pressure $e = wp/(622 + w)$ and invert the saturation formula (Appendix C.1).

22.3 Question 5: 20 °C and 10 g kg^{-1}

The question

Consider a sample of air with a temperature of 20 °C and a mixing ratio of 10 g kg^{-1}. Plot the point on a graph of saturation mixing ratio against temperature (Figure 21.2), and show on the graph to which temperature the sample must cool before it reaches saturation.

Method A: the graph.

Step 1. Mark the point: 20 °C on the horizontal axis, 10 g kg^{-1} on the vertical axis.

Step 2. Cooling moves the point to the left along the horizontal line at 10 g kg^{-1}. Draw an arrow in that direction.

Step 3. The arrow meets the curve where the air is saturated. (At that point the curve's height, w_s, equals 10 g kg^{-1}.)

Step 4. Draw a vertical line down from the meeting point to the temperature axis and read the value.

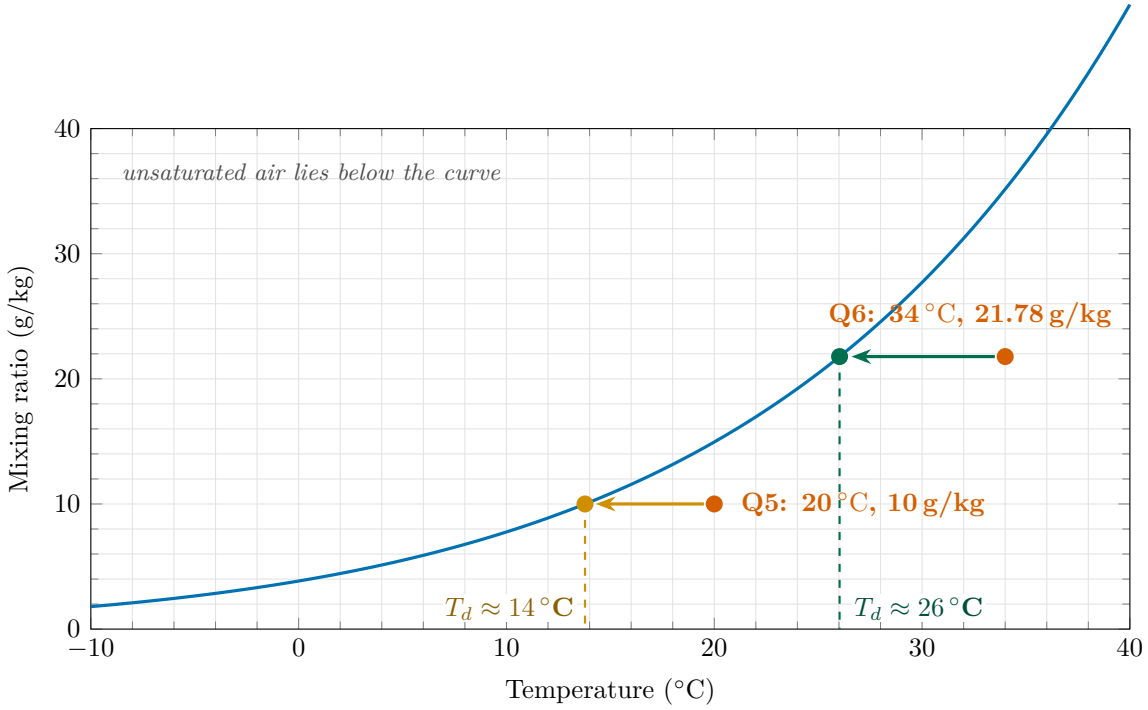


Figure 22.1. The graph solved for both questions. Each air sample is a red dot; cooling moves it horizontally to the curve; the dew point is read on the temperature axis. For Question 5 it is 13.8 °C (about 14 °C); for Question 6 it is 26.0 °C.

Method B: the table backwards. The sample holds 10 g kg^{-1}. Table 21.1 gives 8.882 g kg^{-1} at 12 °C and 10.14 g kg^{-1} at 14 °C, so the dew point is between 12 and 14 °C. The fraction of the way from 12 to 14 is

$$\frac{10 - 8.882}{10.14 - 8.882} = \frac{1.118}{1.258} = 0.89, \qquad \text{so} \qquad T_d \approx 12 + 0.89 \times 2 = 13.8 \text{ °C}.$$

Method C: formula. The vapour pressure is $e = \dfrac{10 \times 995.5}{622 + 10} = 15.75$ hPa, and inverting $e_s(T) = 6.112 \exp[17.67T/(T + 243.5)]$ (Bolton, 1980) gives $T_d = 13.78$ °C.

Answer to Moisture Q5

The sample cools along the horizontal line at 10 g kg^{-1} and meets the curve at about **14** °C (13.8 °C). That is its dew point.

Check. The dew point must be lower than the air temperature (20 °C), and 14 °C is. At 14 °C Table 21.1 says the air could hold 10.14 g kg^{-1}, only a little more than the 10 g kg^{-1} it has: so the air is just short of saturated at 14 °C, as it should be.

22.4 Question 6: 34 °C and 62 % humidity

The question

A sample of air has a temperature of 34 °C and a relative humidity of 62 %. What is its saturation mixing ratio, its mixing ratio and its dew point?

Method.

Step 1. Saturation mixing ratio at 34 °C, from Table 21.1: 35.134 g kg^{-1}.

Step 2. Mixing ratio: $w = w_s \times \mathrm{RH}/100 = 35.134 \times 0.62 = 21.78$ g kg^{-1}.

Step 3. Dew point: find the temperature at which $w_s = 21.78$ g kg^{-1}. Table 21.1 has 21.734 at 26 °C and 24.557 at 28 °C. The value 21.78 is just above the 26 °C entry:

$$\frac{21.78 - 21.734}{24.557 - 21.734} = \frac{0.046}{2.823} = 0.016, \qquad T_d \approx 26 + 0.016 \times 2 = 26.03\ ^\circ\mathrm{C}.$$

Answer to Moisture Q6

Saturation mixing ratio: **35.134** g kg^{-1}. Mixing ratio: **35.134 × 0.62 = 21.78** g kg^{-1}. Dew point: about **26** °C.

Why 26 °C, not 25 °C

A dew point of 25 °C is an easy slip here, but it cannot be right, and there are three independent ways to see it.

1. **The table.** At 25 °C air can hold at most about 20.4 g kg^{-1} (half way between 19.21 and 21.734, which is 20.47). The sample holds 21.78 g kg^{-1}, *more* than that. At 25 °C it would be oversaturated: a relative humidity of 21.78/20.44 = 107 %.
2. **The graph.** In Figure 22.1 the horizontal line at 21.78 g kg^{-1} meets the curve at 26 °C. At 25 °C the line lies above the curve, in the supersaturated region.
3. **The rule of thumb** (next section): RH ≈ 100 − 5 × ($T - T_d$). A dew point of 26 °C gives 100 − 5 × 8 = 60 %, close to the 62 % we started with. A dew point of 25 °C would give 55 %.

22.5 A quick check by rule of thumb

Each degree by which the dew point lies below the temperature lowers the relative humidity by about 5 %. Written as a formula,

$$\mathrm{RH} \approx 100 - 5\,(T - T_d), \tag{22.1}$$

where

RH	relative humidity (%)	estimated
T, T_d	air temperature and dew point (°C)	

good to within a few percent for RH above about 50 %. Turned round, $T - T_d \approx (100 - \mathrm{RH})/5$: air at 80 % is about 4 °C above its dew point; at 50 % about 10 °C.

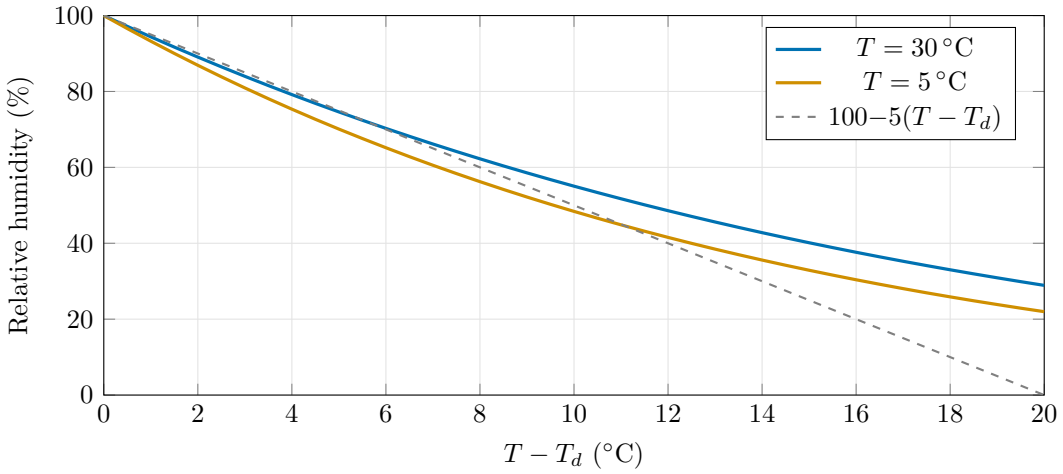


Figure 22.2. The rule of thumb against the exact relationship. For ordinary temperatures the exact curves stay close to the straight line.

22.6 What happens below the dew point

Continue cooling the Question 6 sample below 26 °C. The air can no longer hold all its vapour, so the surplus condenses. At each temperature the vapour left is $w_s(T)$, and the difference is liquid water:

$$\text{condensed water} = w - w_s(T). \tag{22.2}$$

where

w	mixing ratio of the air before cooling ($\mathrm{g\,kg^{-1}}$)	
$w_s(T)$	saturation mixing ratio at the new, lower temperature ($\mathrm{g\,kg^{-1}}$)	Table 21.1

Cooled to 20 °C, the sample holds only $14.956\,\mathrm{g\,kg^{-1}}$ as vapour, so $21.78 - 14.956 = 6.8\,\mathrm{g\,kg^{-1}}$ has turned to liquid. That is dew on the ground, or fog, or cloud droplets. Condensation also releases latent heat, which is why cloud keeps air warmer than it would otherwise be as it cools.

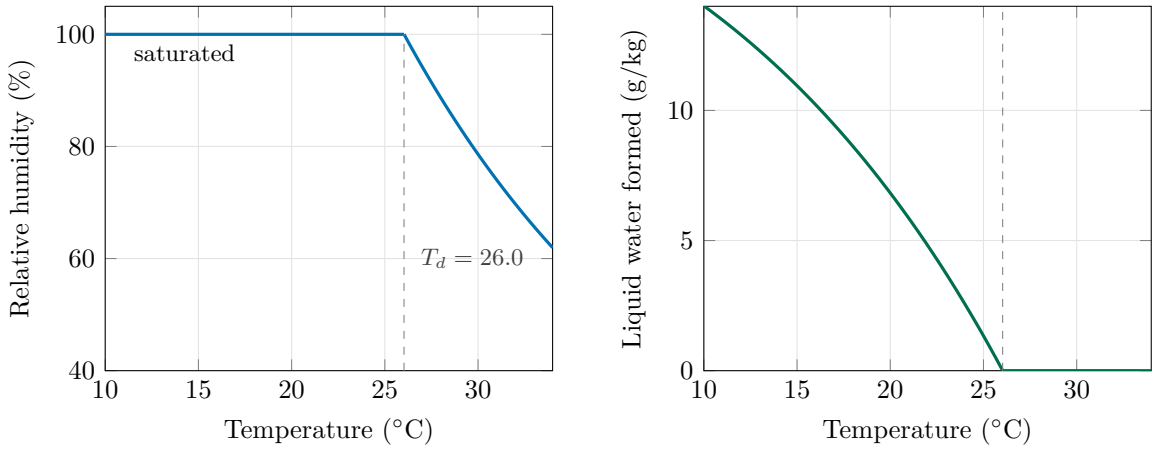


Figure 22.3. The Question 6 sample cooled from 34 °C. Above 26 °C nothing condenses and the relative humidity rises as the glass shrinks. Below 26 °C the air stays at 100 % and the surplus vapour condenses: $6.8\,\mathrm{g\,kg^{-1}}$ by the time it reaches 20 °C.

22.7 Why meteorologists prefer the dew point

- **It measures the vapour itself.** Heating or cooling the air leaves the dew point unchanged (until condensation), unlike the relative humidity.
- **It is a temperature,** so it is easy to compare with the air temperature: the gap $T - T_d$ tells you at a glance how close the air is to making dew, fog or cloud.
- **It describes comfort.** Most people find a dew point below about 10 °C dry and comfortable and one above about 20 °C oppressive.
- **It appears on every station model** at the bottom left of the circle, as you will see in Chapter 27.

22.8 The dew point and the height of cloud base

Air that rises expands and cools at about 9.8 K per kilometre (the dry adiabatic lapse rate, derived in Part I, Eq. (8.2)). Its dew point falls more slowly, by about 1.8 K per kilometre, because the pressure falls as it rises. The temperature and the dew point therefore approach each other by 8 K per kilometre, and when they meet the air is saturated and cloud forms. Starting with a gap of $T - T_d$ degrees, they meet after rising

$$z \approx \frac{T - T_d}{8} \text{ km} \approx 125\,(T - T_d) \text{ m}. \tag{22.3}$$

where

z	height of cloud base above the ground	calculated
T, T_d	surface temperature and dew point (°C)	
8	rate at which T and T_d approach in rising air (9.8 − 1.8, K km^{-1})	

Worked example: where will cloud form?

The air of Question 6 has $T - T_d = 34 - 26 = 8$ °C. If it is lifted (by a hill or by convection), cloud forms after $125 \times 8 = 1000$ m: the cloud base is about one kilometre above the surface. A day with $T = 30$ °C and $T_d = 18$ °C has a gap of 12 °C and a cloud base near 1500 m.

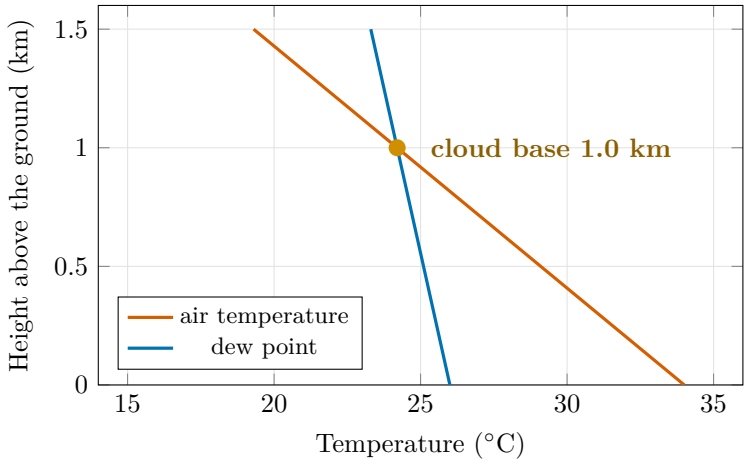


Figure 22.4. Rising air, starting at 34 °C with a dew point of 26 °C. The temperature falls 9.8 K per kilometre and the dew point 1.8 K per kilometre. They meet at a height of 1.0 km.

22.9 Questions that come up

A question that comes up

Does the dew point change when the day warms up?
Not because of the warming. If no water is added or removed, the mixing ratio is fixed, and so is the dew point. In the afternoon the *relative humidity* falls because the temperature rises. The dew point changes only when the air gains moisture (evaporation from soil and plants, an incoming sea breeze) or loses it (rain, a dry wind).

A question that comes up

Can the dew point be higher than the temperature?
Not in the open air for more than a moment: it would mean the air is oversaturated, and the surplus vapour would condense at once, lowering the dew point. At most the two are equal (RH $= 100\,\%$), as in fog.

A question that comes up

What if the dew point is below freezing?
Then the vapour deposits directly as ice (frost) when the air is cooled to it. Strictly the temperature of deposition is the *frost point*, which is a degree or two above the dew point. Table 21.1 is for saturation over liquid water; the questions here never need the ice case.

22.10 Try it yourself: Practice Set 22

Practice Set 22: the dew point

22.1 Air at $30\,^\circ\mathrm{C}$ has a mixing ratio of $12\,\mathrm{g\,kg^{-1}}$. Find its dew point (a) from Table 21.1 by interpolation and (b) with the rule of thumb after finding its RH.

22.2 Air at $20\,^\circ\mathrm{C}$ has a relative humidity of $50\,\%$. What is its dew point?

22.3 On an afternoon the air is at $28\,^\circ\mathrm{C}$ with a relative humidity of $40\,\%$. Overnight the temperature falls to $12\,^\circ\mathrm{C}$ with no change in the amount of vapour. Will dew form?

22.4 Find the cloud base for air with $T = 30\,^\circ\mathrm{C}$ and $T_d = 18\,^\circ\mathrm{C}$.

22.5 Air at $25\,^\circ\mathrm{C}$ has a dew point of $15\,^\circ\mathrm{C}$. What are its mixing ratio and relative humidity?

22.6 Cool the Question 6 sample ($34\,^\circ\mathrm{C}$, $21.78\,\mathrm{g\,kg^{-1}}$) to $20\,^\circ\mathrm{C}$: how much liquid water forms per kilogram of air?

Answers to Practice Set 22

22.1 (a) Table 21.1: $16\,^\circ\mathrm{C}$ holds 11.56 and $18\,^\circ\mathrm{C}$ holds 13.162, so $T_d = 16 + 2 \times (12 - 11.56)/(13.162 - 11.56) = 16.55 \approx 16.6\,^\circ\mathrm{C}$. (b) $w_s(30) = 27.694$, RH $= 12/27.694 = 43\,\%$; the rule gives $T - T_d = (100 - 43)/5 = 11.4$, so $T_d \approx 18.6\,^\circ\mathrm{C}$. The rule is only rough at low humidity; the table answer is better.

22.2 $w = 0.50 \times 14.956 = 7.478\,\mathrm{g\,kg^{-1}}$. Table 21.1: $8\,^\circ\mathrm{C}$ holds 6.771 and $10\,^\circ\mathrm{C}$ holds 7.762, so $T_d = 8 + 2 \times (7.478 - 6.771)/(7.762 - 6.771) = 9.4\,^\circ\mathrm{C}$. (The rule of thumb gives $10\,^\circ\mathrm{C}$.)

22.3 $w = 0.40 \times 24.557 = 9.82\,\mathrm{g\,kg^{-1}}$. At $12\,^\circ\mathrm{C}$ the air can hold only $8.882\,\mathrm{g\,kg^{-1}}$, which is less: so yes, dew (or fog) forms. The dew point is $13.5\,^\circ\mathrm{C}$, so dew begins when the air

cools through 13.5 °C.

22.4 $125 \times (30 - 18) = 1500$ m.

22.5 $w = w_s(15\,°\text{C})$ (the air is just saturated at its dew point) $= 10.83\,\text{g kg}^{-1}$. $w_s(25) = 20.44$, so RH $= 10.83/20.44 = 53\,\%$. (Rule: $100 - 5 \times 10 = 50\,\%$.)

22.6 $21.78 - 14.956 = 6.8\,\text{g kg}^{-1}$ of liquid water.

22.11 The dew point in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def vapour_pressure(w, p_hpa=TABLE_PRESSURE_HPA):
    """Vapour pressure (hPa) from the mixing ratio (g/kg)."""
    return w * p_hpa / (622.0 + w)
```

```
def dew_point(w, p_hpa=TABLE_PRESSURE_HPA):
    """Dew point (deg C) from the mixing ratio (g/kg)."""
    x = math.log(vapour_pressure(w, p_hpa) / 6.112)
    return 243.5 * x / (17.67 - x)
```

```
def rh_rule_of_thumb(t_c, t_d):
    """Approximate relative humidity (%) from the temperature and the dew point."""
    return 100.0 - 5.0 * (t_c - t_d)
```

```
def cloud_base(t_c, t_d):
    """Approximate height of the cloud base (m)."""
    return 125.0 * (t_c - t_d)
```

`dew_point(10)` returns about 13.8, the answer to Moisture Q5.

Chapter 23

Why air moves: the three forces

This chapter is the background for the motion questions.

23.1 Newton's law for a parcel of air

Wind is air that is moving, and air moves for the same reason anything moves: forces act on it. Newton's second law says that the acceleration of a parcel equals the total force on it divided by its mass. It is convenient to talk about the force *per unit mass* (that is, as an acceleration, in $\mathrm{m\,s^{-2}}$), so we never need the mass of the parcel.

For large-scale winds only three horizontal forces matter:

1. the **pressure gradient force**, which starts the air moving;
2. the **Coriolis force**, which steers it once it is moving;
3. **friction**, which slows it near the ground.

If the three add up to zero the wind is steady, and the whole of Part II is about what happens when they do.

23.2 Pressure, and why it falls with height

The pressure at any level is the weight of the air above it, spread over one square metre. At sea level that weight is about 10 tonnes per square metre, which is a pressure of about 1013 hPa. Climb, and there is less air above you, so the pressure drops. Half the atmosphere's mass lies below about 5.5 km, so the pressure there is about 500 hPa; at the cruising height of airliners it is about 250 hPa.

23.3 The pressure gradient force

Pressure also varies from place to place at the same height. Air is pushed from where the pressure is higher towards where it is lower, just as water runs downhill. The push is the **pressure gradient force** (PGF). Per unit mass it is

$$\mathrm{PGF} = -\frac{1}{\rho}\,\frac{\Delta p}{\Delta n}, \tag{23.1}$$

where

PGF	pressure gradient force per unit mass ($\mathrm{m\,s^{-2}}$)	calculated
ρ	air density ($\mathrm{kg\,m^{-3}}$)	about 1.2 near the ground
$\Delta p/\Delta n$	pressure change per unit distance measured *across* the isobars ($\mathrm{Pa\,m^{-1}}$)	from the map

The minus sign means that the force points down the gradient, from high to low pressure (Holton and Hakim, 2013, ch. 1).

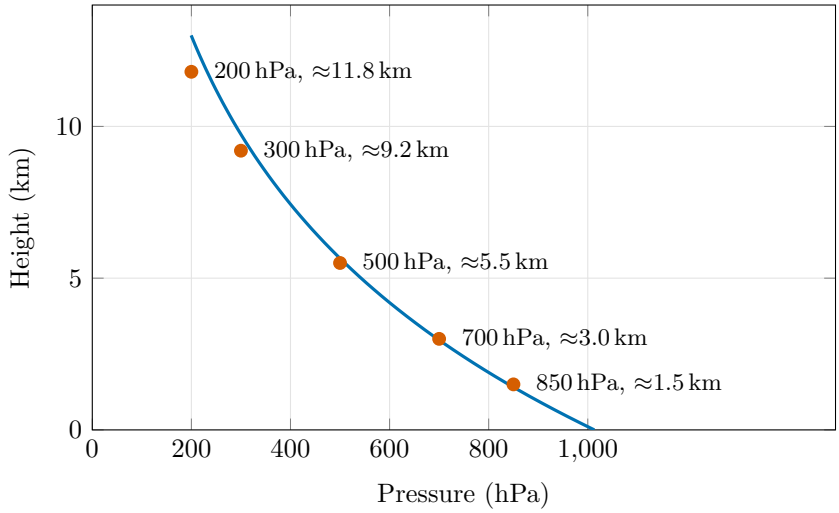


Figure 23.1. Pressure falls roughly exponentially with height, by a factor of e every 8 km or so: the halving rule of Part I (Chapter 3). The red dots mark the standard pressure levels used on upper-air charts; those used here (850, 500 and 300 hPa) are at about 1.5, 5.5 and 9 km.

- **Direction:** from higher to lower pressure, at right angles to the isobars.
- **Magnitude:** proportional to the pressure difference between two points, and inversely proportional to the distance between them. Closely spaced isobars mean a strong force and a strong wind.

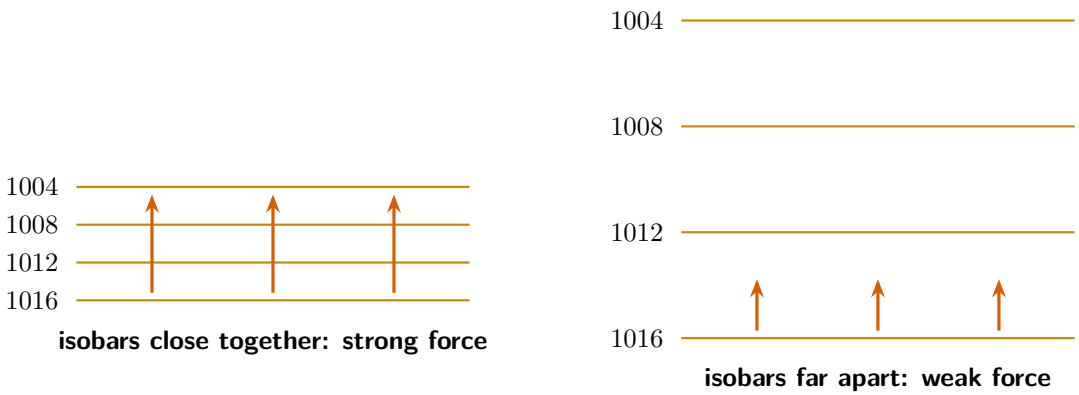


Figure 23.2. The same pressure difference (here 12 hPa) squeezed into a short distance (left) and spread over a long one (right). The force, shown by the red arrows, is proportional to the gradient.

Worked example: how big is the pressure gradient force?

The isobars of Figure 26.1 are 1012 and 1016 mb, that is 4 hPa = 400 Pa apart. Suppose they are 200 km apart (2×10^5 m). Then

$$\text{PGF} = \frac{1}{1.2} \times \frac{400}{2 \times 10^5} = 1.7 \times 10^{-3}\ \text{m s}^{-2}.$$

That looks tiny, but acting for an hour (3600 s) with nothing to oppose it, it would add 6 m s^{-1} to the wind speed. That is how quickly a pressure difference sets the air going.

23.4 The Coriolis force

The Earth turns once a day, and we live on its surface, so we measure motion from a spinning frame. An object that moves freely in a straight line (as seen from space) appears to us to curve. We describe the curving with a force, the **Coriolis force**, which has three properties you must remember.

1. **Direction:** at right angles to the motion. It acts to the **right** in the Northern Hemisphere and to the **left** in the Southern Hemisphere.
2. **Magnitude:** it increases with the wind speed V and with latitude: Coriolis $= f\,V$, where $f = 2\Omega \sin\varphi$ is the Coriolis parameter and $\Omega = 7.292 \times 10^{-5}\,\mathrm{s}^{-1}$ is the Earth's rotation rate.
3. **It does no work.** Being at right angles to the motion, it changes the direction of the wind but never its speed.



Figure 23.3. The Coriolis force always acts at right angles to the motion: to the right in the north and to the left in the south.

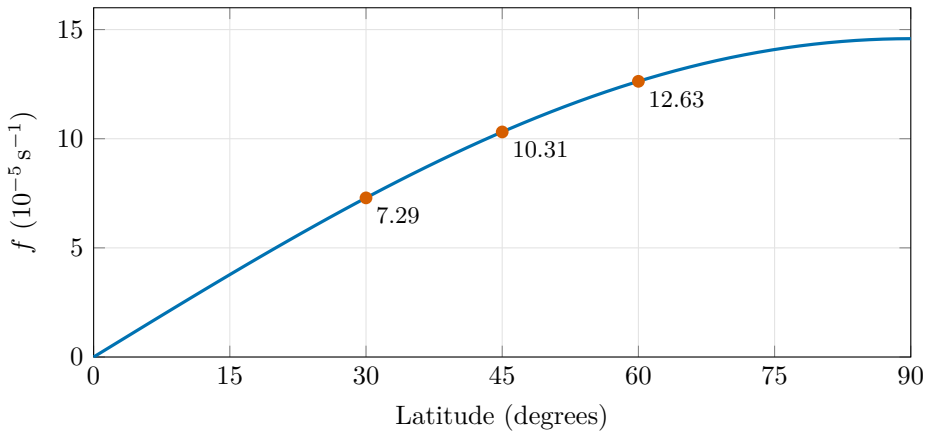


Figure 23.4. The Coriolis parameter against latitude. It is zero at the equator, where the force vanishes, and largest at the poles.

A question that comes up

Does the Coriolis force make water spin the opposite way in a bath in the two hemispheres?

No. For a bath or a sink the force is minute compared with the shape of the basin and the way the water was stirred. The Coriolis force matters for motions that last hours and cover hundreds of kilometres: winds, ocean currents, storms.

Table 23.1. The Coriolis parameter $f = 2\Omega \sin\varphi$ and the acceleration it gives to a wind of $10\,\mathrm{m\,s^{-1}}$.

Latitude	f ($\mathrm{s^{-1}}$)	Coriolis acceleration for $10\,\mathrm{m\,s^{-1}}$ ($\mathrm{m\,s^{-2}}$)
0°	0	0
10°	2.5×10^{-5}	2.5×10^{-4}
30°	7.3×10^{-5}	7.3×10^{-4}
45°	1.03×10^{-4}	1.03×10^{-3}
60°	1.26×10^{-4}	1.26×10^{-3}
90°	1.46×10^{-4}	1.46×10^{-3}

A question that comes up

Why is the Coriolis force zero at the equator?
It depends on the *vertical* component of the Earth's rotation, which is $\Omega \sin\varphi$. At the equator the rotation axis is horizontal, so that component is zero.

23.5 Friction

Near the ground the air rubs against the surface: grass, trees, buildings, hills, waves. The result is a **frictional force** that always acts **opposite to the wind**, slowing it. Its strength depends on how rough the surface is (a forest or a city is much rougher than a calm sea) and increases with the wind speed. Friction is important only in the lowest kilometre or two of the atmosphere, the *boundary layer*. Above that the air is nearly free of it, which is why the winds at 500 or 300 hPa are so much simpler than those at the surface.

23.6 The three forces side by side

Table 23.2. The three forces.

Force	Direction	Magnitude
Pressure gradient	From higher to lower pressure, at right angles to the isobars	Proportional to the pressure difference between two points (and inversely to their distance)
Coriolis	At right angles to the airflow: to the **right** in the Northern Hemisphere, to the **left** in the Southern Hemisphere	Increases with wind strength and with latitude (fV with $f = 2\Omega \sin\varphi$)
Friction	Opposite to the direction of the airflow	Depends on the surface characteristics that obstruct the airflow (and increases with speed)

23.7 Three situations to hold in your head

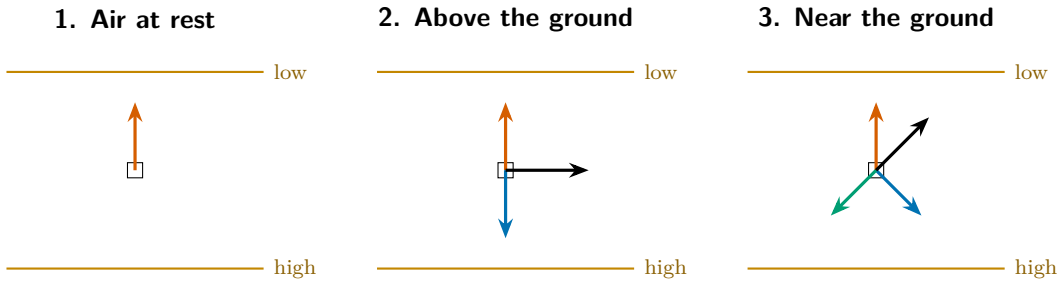


Figure 23.5. (1) At rest only the pressure gradient force acts. (2) High above the ground the Coriolis force balances it and the wind blows along the isobars: the geostrophic wind (Chapter 25). (3) Near the ground friction joins in, the three forces cancel, and the wind crosses the isobars towards low pressure (Chapter 26). Red: pressure gradient force; blue: Coriolis; green: friction; black: the wind.

23.8 Try it yourself: Practice Set 23

Practice Set 23: forces

23.1 Two isobars are 4 hPa apart and 100 km from each other. Find the pressure gradient force per unit mass, taking $\rho = 1.2\,\mathrm{kg\,m^{-3}}$. How does it compare with the 200 km case in the worked example?

23.2 Find the Coriolis parameter at 60° N and the Coriolis acceleration on a wind of $15\,\mathrm{m\,s^{-1}}$.

23.3 A wind is blowing northward in the Southern Hemisphere. Which way does the Coriolis force push it?

23.4 Why are strong winds found where the isobars are close together?

Answers to Practice Set 23

23.1 $\mathrm{PGF} = \dfrac{1}{1.2} \times \dfrac{400}{10^5} = 3.3 \times 10^{-3}\ \mathrm{m\,s^{-2}}$: exactly *twice* the 200 km value, because the distance is halved.

23.2 $f = 2 \times 7.292 \times 10^{-5} \times \sin 60^\circ = 1.26 \times 10^{-4}\ \mathrm{s^{-1}}$; $fV = 1.26 \times 10^{-4} \times 15 = 1.9 \times 10^{-3}\ \mathrm{m\,s^{-2}}$.

23.3 To the left of its motion, which is **west**.

23.4 Close isobars mean a steep pressure gradient, so the pressure gradient force is large; the Coriolis force (and, near the ground, friction) must grow to balance it, which needs a faster wind.

Chapter 24

Pressure surfaces and the pressure gradient

This chapter answers Motion Q1 to Q6.

24.1 What a pressure surface is

Upper-air charts do not show pressure at a fixed height; they show the *height at which the pressure has a chosen value*. The chart for 500 hPa, for example, is a map of the height of the surface on which the pressure is 500 hPa. Such a surface is called a **pressure surface** (or isobaric surface). Where the pressure surface is high, the pressure at a fixed level nearby is high; where it is low, the pressure is low. The height contours on the chart therefore play the part of isobars, and the wind blows along them (Chapter 25).

Pressure surfaces are not flat. They are high over warm air and low over cold air, and that single fact explains Questions 1 to 6.

24.2 Why cold air gives low pressure surfaces

Pressure at a point is the weight of the air above it. A column of cold air is denser, so a given amount of air (a given pressure drop) occupies less height. A column of warm air is less dense, so the same pressure drop is spread over a greater height: the warm column is thicker. Start both columns with the same pressure at the ground and the 500 hPa surface will be lower over the cold one than over the warm one, and the difference grows with height.

The exact relationship is the **hypsometric equation** (Wallace and Hobbs, 2006, ch. 3). It is the exponential law of Part I, $p = p_0 e^{-z/H}$ with $H = R_d T/g$ (Part I, Eq. (3.3)), written for a layer: for the layer between pressures p_1 (lower) and p_2 (upper),

$$\Delta z = z_2 - z_1 = \frac{R_d \overline{T}}{g} \ln \frac{p_1}{p_2}, \tag{24.1}$$

where

Δz	thickness of the layer (m)	calculated
$\overline{T}$	mean temperature of the layer (K)	
R_d	gas constant for dry air ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	287
g	acceleration due to gravity ($\mathrm{m\,s^{-2}}$)	9.81
$p_1,\ p_2$	pressures at the bottom and top of the layer (hPa)	

The thickness Δz is directly proportional to the mean temperature. Part I used the same law the other way round, to find the scale height of a layer from a sounding (Part I, Eq. (8.1)), and to explain why the troposphere is thickest over the warm tropics (Part I, § 7.4).

Worked example: thickness of the 1000 to 500 hPa layer

For a cold layer with $\overline{T} = -10\,°\text{C} = 263.15\,\text{K}$,

$$\Delta z = \frac{287 \times 263.15}{9.81} \ln \frac{1000}{500} = 7699 \times 0.693 = 5339\ \text{m}.$$

For a warm layer with $\overline{T} = +10\,°\text{C} = 283.15\,\text{K}$ the result is 5745 m. A 20-degree difference in mean temperature makes the layer 406 m thicker, about 20 m per degree.

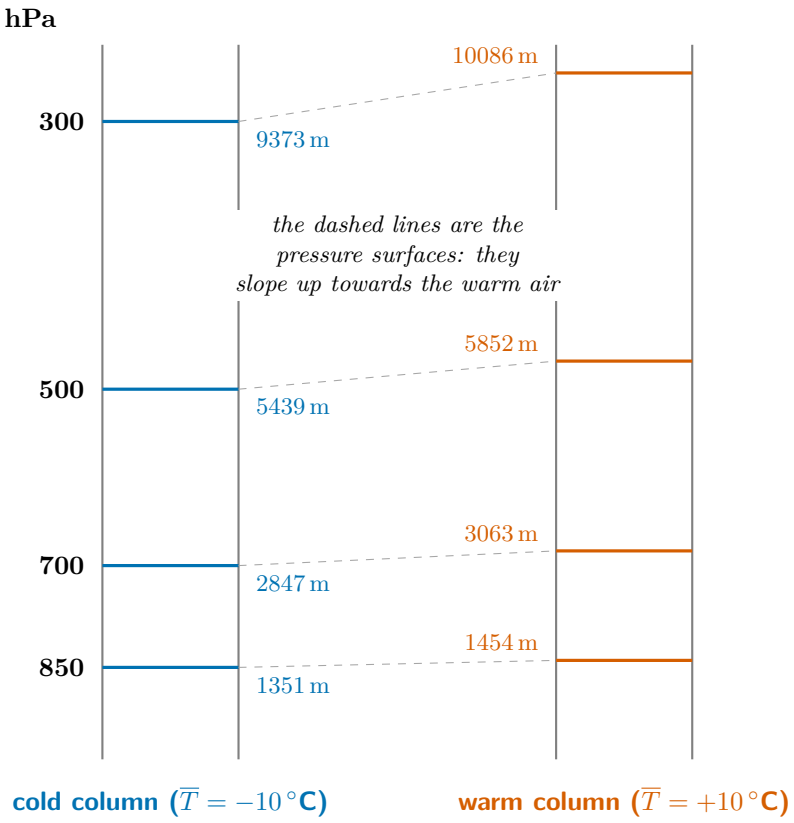


Figure 24.1. Two columns of air that start with the same pressure at the ground (1013 hPa). Each pressure level sits higher over the warm column, and the difference (the slope of the surface) grows with height: 103 m at 850 hPa, 216 m at 700 hPa, 413 m at 500 hPa and 712 m at 300 hPa. (The heights come from the hypsometric equation with a constant mean temperature.)

24.3 The data

Table 24.1 gives the pressure level, the height and the temperature at two places, Saskatoon in Canada and Monterrey in Mexico (about 2 990 km apart, Monterrey to the south):

Notice that Monterrey is warmer at every level, and every one of its pressure surfaces is higher.

Table 24.1. Pressure levels over Saskatoon and Monterrey.

	Saskatoon		Monterrey	
Pressure	**Height**	**Temp.**	**Height**	**Temp.**
300 mb	8700 m	−47 °C	9500 m	−38 °C
500 mb	5400 m	−25 °C	5700 m	−12 °C
850 mb	1360 m	−1 °C	1500 m	15 °C

24.4 Question 1: plot the pressure surfaces

The question

Following the example of 850 mb, plot the heights of the 500 and 300 mb levels on a diagram with Saskatoon on the left and Monterrey on the right, draw lines connecting the two points for each level, and label each line.

Method.

Step 1. The diagram has Saskatoon on the left and Monterrey on the right, with height up the sides (0 to 10 000 m). The 850 mb line, from 1360 m to 1500 m, is the example: from 1360 m to 1500 m.

Step 2. For 500 mb put a dot at 5400 m above Saskatoon and a dot at 5700 m above Monterrey. Join them with a straight line and label it "500 mb".

Step 3. For 300 mb put dots at 8700 m (Saskatoon) and 9500 m (Monterrey), join them, and label the line "300 mb".

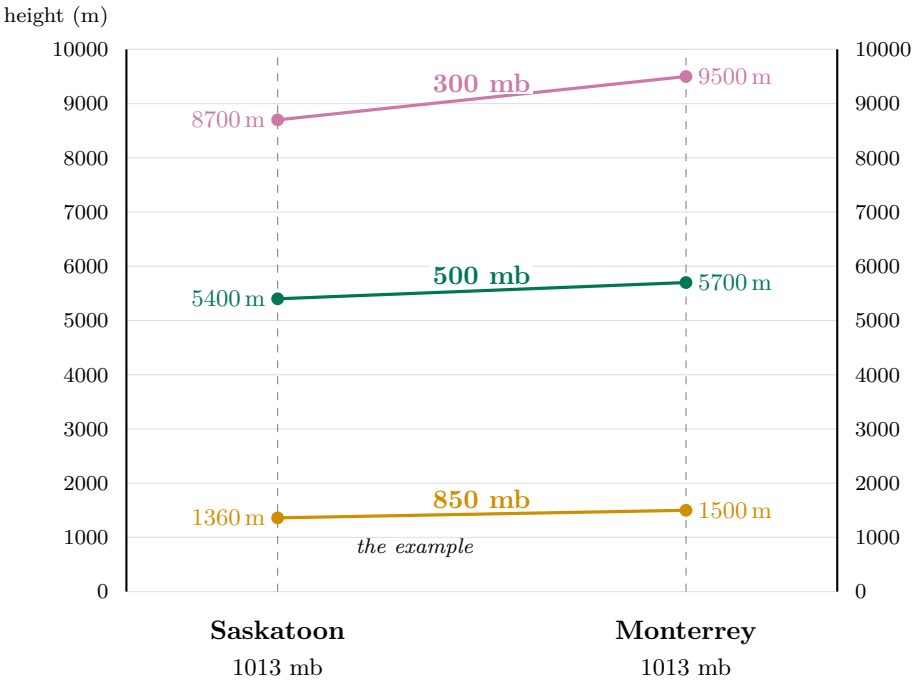


Figure 24.2. The diagram completed for Question 1. The 500 and 300 mb lines slope up towards Monterrey, and the slope is greater for the higher level.

Answer to Motion Q1

Two lines are added to the diagram: 500 mb from 5400 m (Saskatoon) to 5700 m (Monterrey), and 300 mb from 8700 m to 9500 m. Each rises to the right, more steeply than the one below it.

24.5 Questions 2 and 3: where does pressure fall fastest?

Answer to Motion Q2

Saskatoon. Its pressure surfaces are lower: the 500 mb surface is reached at 5400 m instead of 5700 m, the 300 mb surface at 8700 m instead of 9500 m. Less height is needed to use up the same pressure, so pressure falls faster with height.

Answer to Motion Q3

Saskatoon is colder at each height (-1 against $15\,^\circ$C at 850 mb, -25 against $-12\,^\circ$C at 500 mb, -47 against $-38\,^\circ$C at 300 mb). Cold air is denser, so a column of it is thinner: the pressure falls faster with height.

Check with the hypsometric equation. Use the mean of the two temperatures that bound each layer.

$$\text{Saskatoon: } \Delta z_{850\to300} = 29.27\,[\,260.15\ln\tfrac{850}{500} + 237.15\ln\tfrac{500}{300}\,] = 29.27\,[138.0 + 121.1] = 7587\text{ m},$$
$$\text{Monterrey: } \Delta z_{850\to300} = 29.27\,[\,274.65\ln\tfrac{850}{500} + 248.15\ln\tfrac{500}{300}\,] = 29.27\,[145.7 + 126.8] = 7976\text{ m}.$$

The rounded heights give $8700 - 1360 = 7340$ m and $9500 - 1500 = 8000$ m. Monterrey agrees to 0.3 %; Saskatoon is 3 % off, because the figure's numbers are rounded. The warmer column is thicker, as predicted.

24.6 Question 4: which point has the higher pressure?

The question

Both points x and z are at 9000 m. Which has the highest pressure? At 9000 m write H where the pressure is higher and L where it is lower, then draw an arrow between them showing the direction of the horizontal pressure gradient force.

Method.

Step 1. Pressure decreases with height. At a given height, being *below* a pressure surface means a *higher* pressure than the surface's value, and being *above* it means a lower pressure.

Step 2. Point x is above Saskatoon at 9000 m. The 300 mb surface there is at 8700 m, so x is above it: its pressure is *less than* 300 mb.

Step 3. Point z is above Monterrey at 9000 m. The 300 mb surface there is at 9500 m, so z is below it: its pressure is *more than* 300 mb.

Step 4. So z has the higher pressure: write H at z (the Monterrey side) and L at x (the Saskatoon side).

Step 5. The pressure gradient force points from high to low: draw the arrow from z towards x, that is towards Saskatoon.

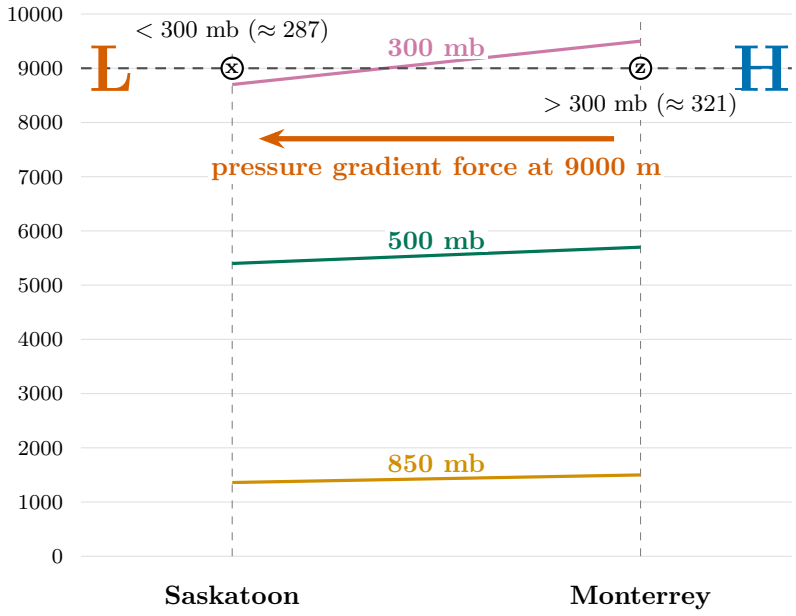


Figure 24.3. Question 4. At 9000 m x is above the 300 mb line (pressure below 300 mb, L) and z is below it (pressure above 300 mb, H). The pressure gradient force points from z to x: from Monterrey towards Saskatoon.

Answer to Motion Q4

z has the higher pressure, because it lies below the 300 mb line at 9000 m. Write **H at z** and **L at x**, and draw the pressure gradient force from z towards x (towards Saskatoon).

How big are the pressures? You can estimate them. Above the 300 mb level at Saskatoon, pressure falls about 1 % every 66 m, so 300 m above the 8700 m level it is about $300\,e^{-0.045} = 287$ mb. At Monterrey, 9000 m is 500 m below the 9500 m level of 300 mb, so the pressure is higher, about 321 mb. The difference between the two places is therefore about 34 mb, over 2 990 km.

24.7 Question 5: where is the pressure gradient greater?

Answer to Motion Q5

The pressure gradient is greater at **9000 m**. The pressure surfaces differ in height by $9500 - 8700 = 800$ m between the two places at the 300 mb level, but only by $1500 - 1360 = 140$ m at the 850 mb level (about 1500 m).

The slope of the surfaces is what drives the wind. The slope grows with height: 140 m at 850 mb, 300 m at 500 mb and 800 m at 300 mb. If you compare pressures at a fixed height instead, you find the same thing: about 15 mb between the two places at 1500 m, but about 34 mb at 9000 m.

A question that comes up

The pressure itself is smaller at 9000 m. How can the gradient be greater? Because what drives the wind is the *change* of pressure from place to place, and that depends on the temperature difference accumulated in the whole layer below. On a pressure surface the force per unit mass is $-g\,\Delta z/\Delta n$, and it does not depend on the air density at all. The slope $\Delta z/\Delta n$ grows with height as the colder column's thickness deficit adds up, so the force does too.

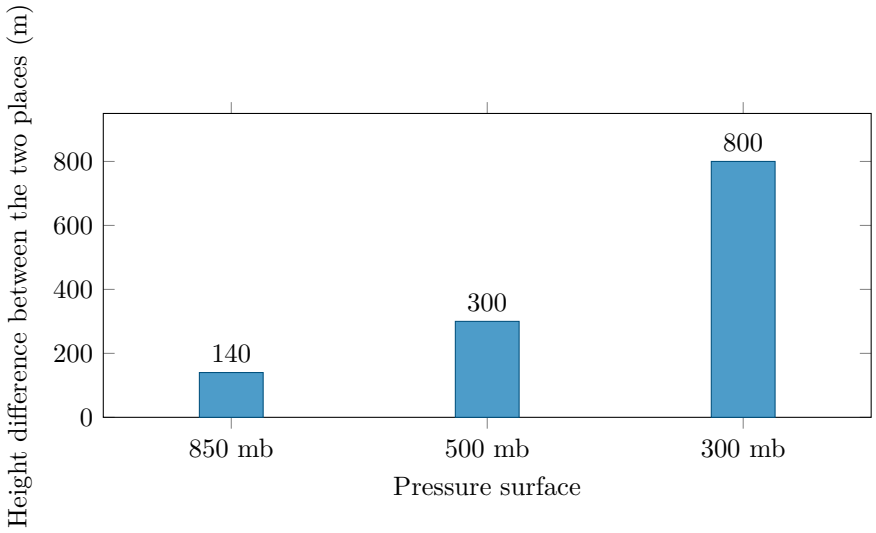


Figure 24.4. The slope of the pressure surfaces between Saskatoon and Monterrey increases with height.

24.8 Question 6: what this means for the wind

Answer to Motion Q6

Wind speeds at **9000 m should be greater**, because the pressure gradient force is greater there.

You can put numbers on this. The geostrophic wind (Chapter 25) is $V_g = (g/f)\,\Delta z/\Delta n$, Eq. (25.1), where Δz is the difference in height of a pressure surface over a distance Δn. The two places are $\Delta n = 2\,990$ km apart, at a mean latitude of 39°, where $f = 2 \times 7.292 \times 10^{-5} \times \sin 39° = 9.16 \times 10^{-5}\,\mathrm{s}^{-1}$.

Worked example: wind estimates from the slopes

$$850\,\mathrm{mb}: \quad V_g = \frac{9.81}{9.16 \times 10^{-5}} \times \frac{140}{2.99 \times 10^{6}} = 5.0\ \mathrm{m\,s^{-1}},$$

$$500\,\mathrm{mb}: \quad V_g = \frac{9.81}{9.16 \times 10^{-5}} \times \frac{300}{2.99 \times 10^{6}} = 10.7\ \mathrm{m\,s^{-1}},$$

$$300\,\mathrm{mb}: \quad V_g = \frac{9.81}{9.16 \times 10^{-5}} \times \frac{800}{2.99 \times 10^{6}} = 28.6\ \mathrm{m\,s^{-1}}.$$

The wind blows from west to east (Monterrey, with the higher surfaces, is to the south, on the right of the wind in the Northern Hemisphere).

These numbers are rough: the heights are rounded, and only two places are used. Real winds at 300 mb are often 50 to 80 $\mathrm{m\,s^{-1}}$. The lesson is nevertheless exact. **The faster the temperature falls towards the pole, the faster the westerly winds aloft, and they strengthen with height.** This is the **thermal wind**, and it is why the jet streams sit near 10 km above the zone where the temperature contrast is greatest.

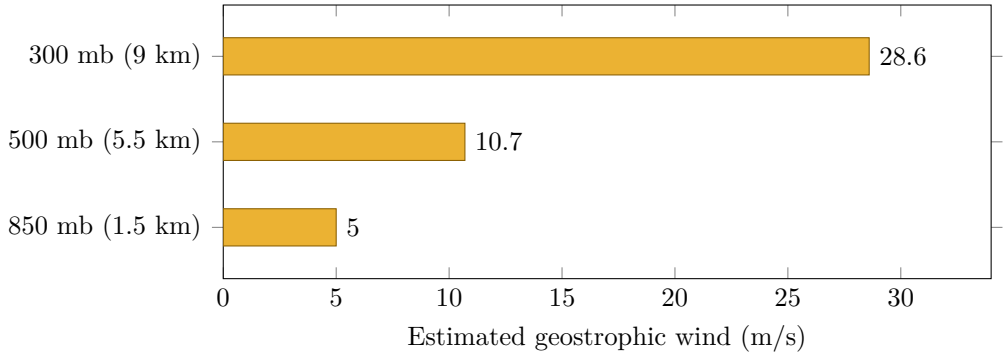


Figure 24.5. Geostrophic wind estimated from the slope of the pressure surfaces between the two places. The wind speeds up fivefold from 850 to 300 mb.

24.9 An experiment: change the temperatures

Replace Saskatoon and Monterrey by two columns that both start at 1013 hPa and have constant mean temperatures T_1 and T_2. The hypsometric equation then gives the height of each surface. If $T_1 = T_2$ every surface is flat: there is no horizontal pressure gradient and no wind, however cold the air. Increase the contrast and the surfaces tilt, increasingly steeply with height.

Table 24.2. Surface slopes and geostrophic winds for two columns with mean temperatures of −25 °C and 0 °C, 2 000 km apart at 45° latitude.

Level	Height, cold column	Height, warm column	Slope / wind
850 hPa	1274 m	1403 m	128 m / 6 m s^{-1}
700 hPa	2685 m	2955 m	270 m / 13 m s^{-1}
500 hPa	5129 m	5645 m	517 m / 25 m s^{-1}
300 hPa	8839 m	9729 m	890 m / 42 m s^{-1}

24.10 Try it yourself: Practice Set 24

Practice Set 24: pressure surfaces

24.1 Find the thickness of the 1000 to 500 hPa layer for mean temperatures of −10 °C and +10 °C, and the difference.

24.2 At 40° N the 500 hPa surface is at 5520 m over town A and 5640 m over town B, which lies 800 km due south of A. Estimate the geostrophic wind and say from which direction it blows.

24.3 The 500 mb heights over two stations are 5400 m (P) and 5600 m (Q). Which has the colder air beneath it?

24.4 The 500 hPa surface slopes up from north to south. Which way does the wind blow in the Northern Hemisphere?

Answers to Practice Set 24

24.1 5339 m and 5745 m (see the worked example); the warm layer is 406 m thicker.

24.2 $f = 2 \times 7.292 \times 10^{-5} \sin 40^\circ = 9.37 \times 10^{-5}\ \mathrm{s^{-1}}$; $V_g = \dfrac{9.81}{9.37 \times 10^{-5}} \times \dfrac{120}{8 \times 10^5} = 15.7\ \mathrm{m\,s^{-1}}$ (about 30 kt). B is to the south and has the higher surface, so the wind has high pressure on its right: it is a **westerly** (blowing from west to east).

24.3 P, because a column that is colder is thinner and has lower pressure surfaces.

24.4 With the low heights to the north (on the left of the wind), the wind blows from west to east, a westerly. Pressure surfaces that rise towards the equator give westerly winds, which is the pattern over mid-latitudes.

24.11 Thickness in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def thickness(mean_t_k, p1_hpa, p2_hpa):
    """Hypsometric equation: thickness (m) of the layer between two pressures."""
    return R_D * mean_t_k / G * math.log(p1_hpa / p2_hpa)
```

A warmer mean temperature gives a larger result: the warmer column is thicker.

Chapter 25

The geostrophic wind

This chapter answers Motion Q7.

25.1 Two forces in balance

High above the ground, friction can be ignored. Only the pressure gradient force and the Coriolis force act. Suppose they are equal in size and opposite in direction. Then the total force is zero, the air does not accelerate, and the wind is steady. That wind is the **geostrophic wind** ("earth-turning" wind).

The geostrophic wind

The geostrophic wind is the wind for which the pressure gradient force and the Coriolis force exactly balance. It blows *along* the isobars (or the height contours on a pressure-surface chart), with low pressure on its left in the Northern Hemisphere and on its right in the Southern Hemisphere.

Writing the two forces per unit mass, the balance is

$$\underbrace{\frac{1}{\rho}\frac{\Delta p}{\Delta n}}_{\text{pressure gradient force}} = \underbrace{f\,V_g}_{\text{Coriolis force}},$$

so that the geostrophic wind speed is (Holton and Hakim, 2013, ch. 3)

$$V_g = \frac{1}{\rho f}\frac{\Delta p}{\Delta n} = \frac{g}{f}\frac{\Delta z}{\Delta n}. \tag{25.1}$$

where

V_g	geostrophic wind speed ($\mathrm{m\,s^{-1}}$)	calculated
ρ	air density ($\mathrm{kg\,m^{-3}}$)	about 1.2 near the ground
f	Coriolis parameter, $2\Omega\sin\varphi$ ($\mathrm{s^{-1}}$)	$\Omega = 7.292\times10^{-5}\,\mathrm{s^{-1}}$
$\Delta p/\Delta n$	pressure gradient across the isobars ($\mathrm{Pa\,m^{-1}}$)	from a surface map
$\Delta z/\Delta n$	slope of a pressure surface ($\mathrm{m\,m^{-1}}$)	from an upper-air chart
g	acceleration due to gravity ($\mathrm{m\,s^{-2}}$)	9.81

The last form is for a pressure surface, where Δz is the difference in height of the surface over the distance Δn. It is the form used with the 500 hPa heights in this part.

25.2 How the balance comes about: the four stages

Figure 25.1 follows a parcel of air as the balance is established. Think of a parcel of air that is at rest between a region of higher pressure (south) and lower pressure (north) in the Northern Hemisphere.

Stage 1. At rest. The parcel is on the high-pressure side. No motion means no Coriolis force, so only the pressure gradient force acts, pointing north, towards low pressure. The parcel starts to move northward.

Stage 2. Moving northward. As soon as it moves, the Coriolis force appears, at right angles and to the right of the motion: towards the east. The parcel's path bends to the right.

Stage 3. Moving north-east. The speed has grown, so the Coriolis force is larger, and since it is at right angles to the motion it now points partly south, against the pressure force. The path keeps bending.

Stage 4. Moving east. The parcel is now moving along the isobars. The Coriolis force points exactly south, equal and opposite to the pressure gradient force pointing north. The forces balance, the speed stops growing, and the wind blows steadily to the east.

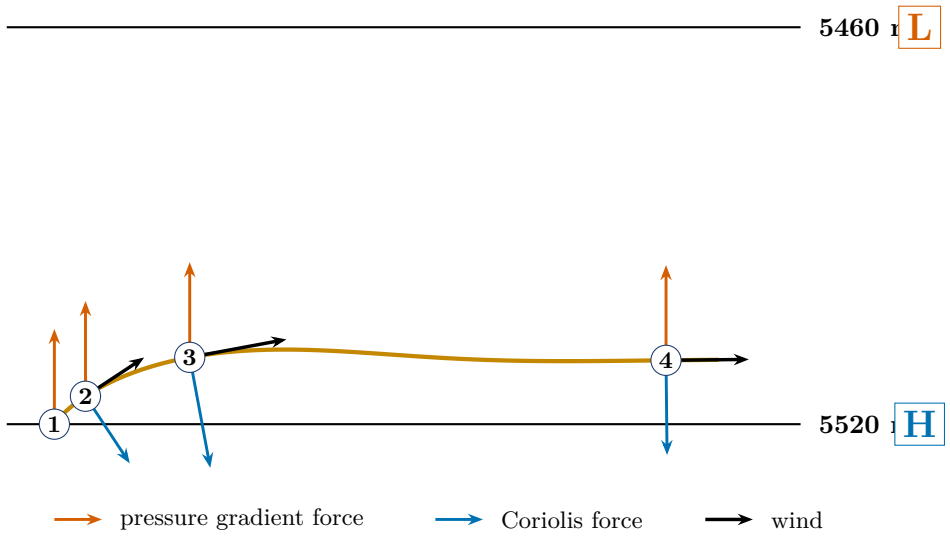


Figure 25.1. The four stages of the adjustment, computed for two 500 hPa height lines 60 m and 500 km apart at 45° N. The parcel starts at rest on the high side. At each numbered position the red arrow is the pressure gradient force, the blue arrow is the Coriolis force and the black arrow is the wind. The path bends to the right until the wind is along the isobars and the forces balance (stage 4).

A question that comes up

Why does the parcel not just keep accelerating towards low pressure?
Because as soon as it speeds up, the Coriolis force grows in proportion, and it turns the motion aside. Without any damping the parcel would overshoot and loop around the balance state (an *inertial oscillation* with a period of $2\pi/f$, about 17 hours at 45°). In the real atmosphere the oscillations are damped as the adjustment radiates energy away in waves, and the air settles to the balance. The figure above includes a little damping, which pulls the velocity towards the geostrophic wind, so that the settling is visible. Its final state is the geostrophic wind.

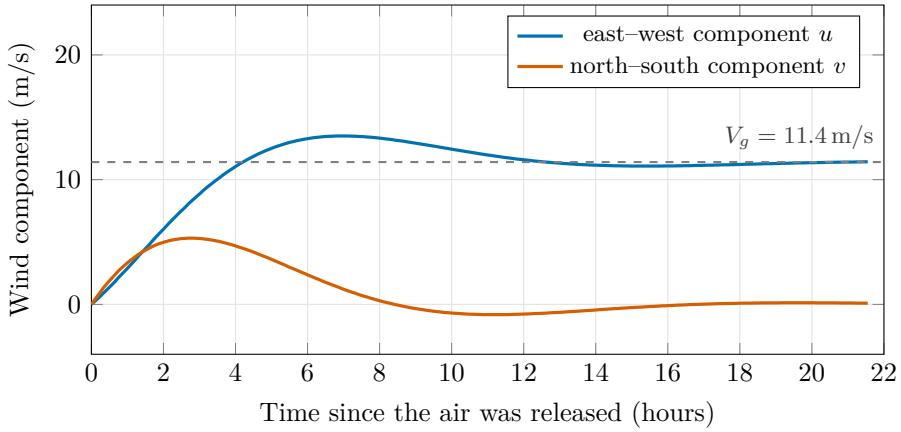


Figure 25.2. The two components of the parcel's velocity against time. The eastward component rises to the geostrophic value (dashed) and the northward component dies away to zero: the air ends up blowing along the isobars.

25.3 Question 7: the Northern Hemisphere

The question

Mark the side with the higher pressure with an H and the side with the lower pressure with an L.

Method.

Step 1. The lines are labelled with the *height* of the 500 hPa surface: 5460 m on the upper line and 5520 m on the lower line.

Step 2. A pressure surface is higher where the pressure is higher. So the line with the larger height, 5520 m, is on the high-pressure side. Write **H** at the 5520 m line (bottom).

Step 3. Write **L** at the 5460 m line (top).

Step 4. Check with the arrows: the pressure gradient force F_{pg} is drawn pointing up, from H to L, as it should.

Answer to Motion Q7, part 1

H at the 5520 m line (south, bottom); L at the 5460 m line (north, top). The pressure gradient force points from H to L (north); the Coriolis force points to the right of the wind, south; the geostrophic wind blows to the east, with low pressure on its left.

25.4 Question 7: the Southern Hemisphere

The question

Now consider the Southern Hemisphere. Mark H and L, and draw the pressure gradient force, the Coriolis force and the resulting geostrophic wind.

What changes. In the Southern Hemisphere the Coriolis force acts to the **left** of the motion. Everything else is the same: the pressure gradient force still points from high to low, and the balance must still be between two opposite forces. The result is a mirror image of Figure 25.1.

Step 1. The two lines are 5640 m (top) and 5580 m (bottom). The larger height is the high-pressure side: **H** at the 5640 m line (top), **L** at the 5580 m line (bottom).

Step 2. Draw the pressure gradient force F_{pg} from H to L: **downwards**.

Step 3. The Coriolis force must be equal and opposite, so it points **upwards**, towards H. (To check: an eastward wind has its left-hand side to the north, which is up the page. The Coriolis force acts to the left, so it points up. ✓)

Step 4. The wind blows between them, to the **east**, with the low pressure on its right.

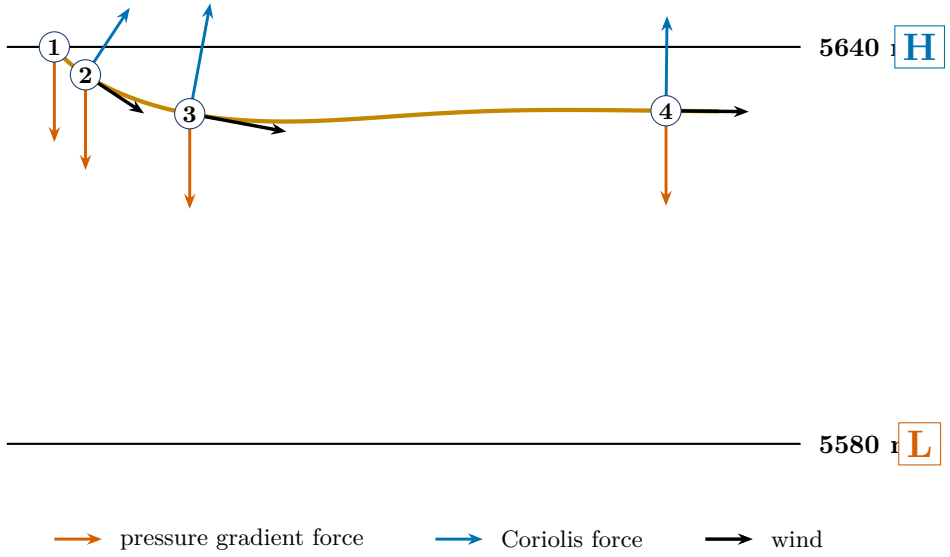


Figure 25.3. The Southern Hemisphere. The Coriolis force turns the parcel to the left. At the end (stage 4) the pressure gradient force points down towards L, the Coriolis force points up towards H, and the wind blows to the east with the low pressure on its right.

Answer to Motion Q7, part 2

H at the 5640 m line (top); L at the 5580 m line (bottom). Draw F_{pg} **down** (towards L) and the Coriolis force **up** (towards H), of equal length; the geostrophic wind **W** points **east** between them.

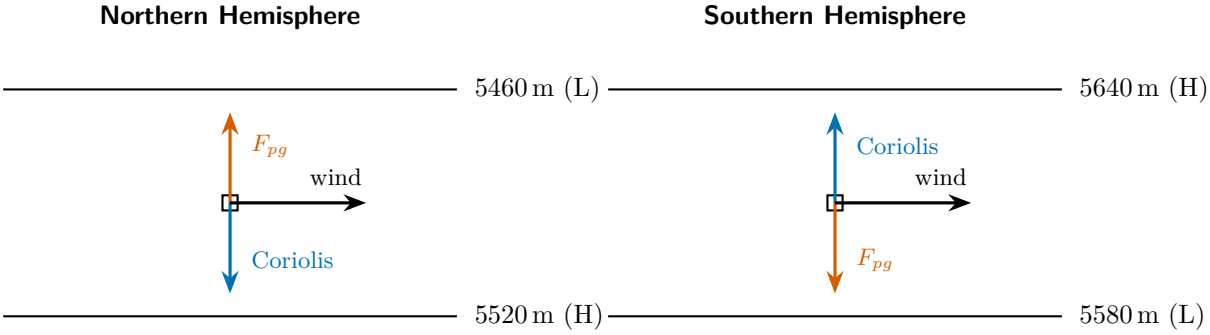


Figure 25.4. The two answers side by side. In both figures the wind blows east. In the north the low pressure (5460 m) is on its left; in the south the low pressure (5580 m) is on its right.

25.5 Buys-Ballot's law

You can find where the low pressure lies, without a map, from the wind alone:

- **Northern Hemisphere:** stand with your back to the wind; low pressure is on your **left**.
- **Southern Hemisphere:** stand with your back to the wind; low pressure is on your **right**.

This is Buys-Ballot's law. It follows directly from the balance: the pressure gradient force points towards low pressure and the Coriolis force (to the right in the north) points away from it, so low pressure is on the left of the wind.

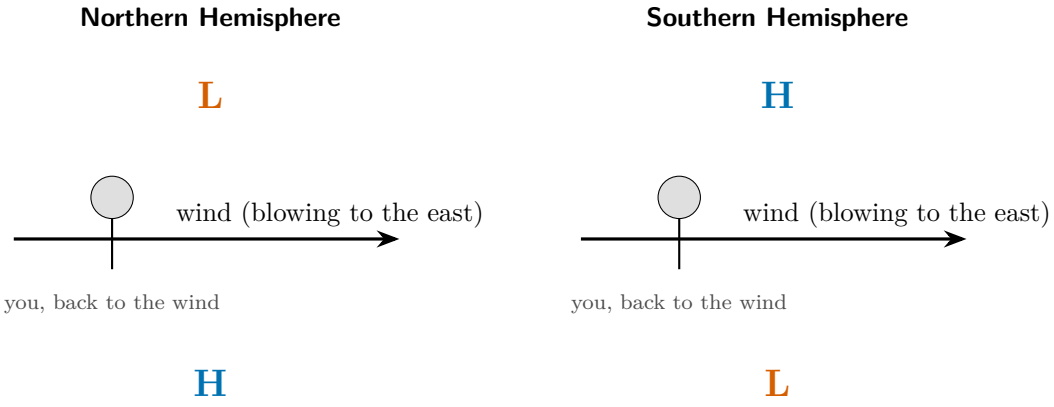


Figure 25.5. Buys-Ballot's law. With your back to the wind, low pressure is on the left in the north and on the right in the south.

25.6 How fast is the geostrophic wind?

Worked example: the wind implied by the lines of Figure 25.1

Suppose the two height lines, 60 m apart, are 500 km apart on the chart, at 45° N. Then

$$f = 2 \times 7.292 \times 10^{-5} \sin 45^\circ = 1.03 \times 10^{-4}\ \text{s}^{-1},\quad V_g = \frac{g\,\Delta z}{f\,\Delta n} = \frac{9.81 \times 60}{1.03 \times 10^{-4} \times 5 \times 10^5} = 11.4\ \text{m s}^{-1}$$

which is 22 knots or 41 km h^{-1}. If the lines were closer, 300 km apart, the wind would be 19 m s^{-1}; at the same spacing but 60° N it would be 9.3 m s^{-1}; at 30° N, 16.1 m s^{-1}.

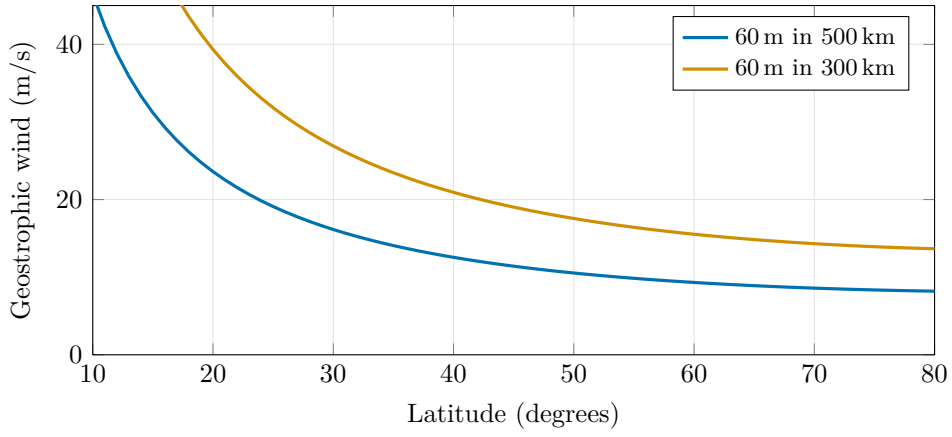


Figure 25.6. The geostrophic wind for a given height difference rises as the lines get closer together and as the latitude decreases (because f falls). Near the equator $f \to 0$ and the formula breaks down.

25.7 When is the wind not geostrophic?

The balance is an excellent approximation for the winds a kilometre or more above the ground at middle and high latitudes, but not always.

- **Near the ground,** friction acts, and the balance changes (Chapter 26).
- **Near the equator,** f is so small that the balance fails.
- **Where the isobars curve strongly,** as round a hurricane or a tight low, the centrifugal force matters; the result is the *gradient wind*.
- **When the pattern changes quickly,** the air accelerates, as in a developing storm.

25.8 Questions that come up

A question that comes up

Is the geostrophic wind along or across the isobars?
Along. Because the two forces are at right angles to the wind, and the pressure gradient force is at right angles to the isobars, the wind must be parallel to the isobars. This is a much better description of upper-air winds than the picture of air flowing "from high to low".

A question that comes up

On Figure 25.1, why is H at the bottom? Surely high pressure is up?
"Up" on the page is north, not height. The H and L mark the pressure on a horizontal map at the 500 hPa level: higher pressure to the south (bottom of the page), lower to the north (top).

25.9 Try it yourself: Practice Set 25

Practice Set 25: geostrophic wind

25.1 The 500 hPa height lines 5460 m and 5520 m are 400 km apart at 50° N. Find the geostrophic wind.

25.2 In the Southern Hemisphere the 500 hPa heights are 5600 m at 40° S and 5500 m at 50° S. Which way does the wind blow?

25.3 At what latitude would the lines of Figure 25.1 (60 m in 500 km) give a geostrophic wind of $20\,\mathrm{m\,s^{-1}}$?

25.4 In the Northern Hemisphere a pilot flying with the wind directly behind notes that the low pressure is on the pilot's left. Is this consistent with Buys-Ballot's law?

Answers to Practice Set 25

25.1 $f = 2 \times 7.292 \times 10^{-5} \sin 50^\circ = 1.117 \times 10^{-4}$; $V_g = \dfrac{9.81 \times 60}{1.117 \times 10^{-4} \times 4 \times 10^5} = 13.2\ \mathrm{m\,s^{-1}}$ (26 kt).

25.2 The heights fall towards the pole, so the low is to the south. In the Southern Hemisphere low pressure is on the right of the wind, so the wind blows from **west to east** (a westerly), as in the Southern Ocean.

25.3 $\sin\varphi = \dfrac{g\Delta z}{2\Omega V_g \Delta n} = \dfrac{588.6}{1458} = 0.404$, so $\varphi = 23.8°$.

25.4 Yes: with your back to the wind, low pressure on the left is exactly Buys-Ballot's law for the Northern Hemisphere.

25.10 The geostrophic wind in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def coriolis_parameter(lat_deg):
    """Coriolis parameter (per second) at a latitude."""
    return 2.0 * OMEGA * math.sin(math.radians(lat_deg))
```

```
def geostrophic_wind(dz_m, dn_m, lat_deg):
    """Geostrophic wind (m/s) from the height change of a pressure surface over a distance."""
    return G / coriolis_parameter(lat_deg) * dz_m / dn_m
```

`geostrophic_wind(60, 5e5, 45)` returns about $11.4\,\mathrm{m\,s^{-1}}$.

Chapter 26

Wind near the ground: friction

This chapter answers Motion Q8 to Q11.

26.1 Question 8: what does a slower wind do to the Coriolis force?

Close to the ground friction acts on the air, always against the wind, slowing it. The first question about it is: *what effect will a reduction in surface wind speed have on the Coriolis force?*

Answer to Motion Q8

The Coriolis force becomes **weaker**, because it depends on the wind speed. A slower wind means a smaller Coriolis force; a faster wind a larger one.

The reason is that the Coriolis force per unit mass is fV, proportional to the wind speed V. If friction cuts the speed to 70 % of the geostrophic value, the Coriolis force falls to 70 % as well. But the pressure gradient force does not depend on the wind at all. So at the surface the Coriolis force can no longer balance the pressure gradient force on its own. Something has to make up the difference, and that is where the three-force balance comes from.

26.2 The balance of three forces

Near the ground three forces act on the air. If the wind is steady they add up to zero.

- the **pressure gradient force**, from high to low pressure;
- the **Coriolis force**, at right angles to the wind (to its right in the Northern Hemisphere);
- **friction**, directly against the wind.

Picture one box of air with these arrows. To see why the balance forces the wind to cross the isobars, imagine that the wind were parallel to the isobars as in Chapter 25. Then the Coriolis force would point directly against the pressure force, and friction (against the wind, along the isobars) would be left with nothing to balance it. The only way out is for the wind to turn a little *towards low pressure*, so that the pressure force has a component along the wind that cancels friction, and the Coriolis force (now pointing partly backwards) cancels the rest.

26.3 Question 9: the forces on a box

The question

Indicate with arrows the three forces acting on the Northern Hemisphere box (pressure gradient in red, Coriolis in blue, friction in green); the black arrow shows the result.

Picture two isobars, 1012 mb at the top and 1016 mb at the bottom. The pressure is lower at the top, so the pressure gradient force points *up*.

Method: draw the arrows in this order.

Step 1. Red: the pressure gradient force. From the box, straight towards the lower pressure (up the page), at right angles to the isobars.

Step 2. Black: the wind. It blows across the isobars, towards low pressure but turned to the right, so draw it up and to the right (north-east). Its angle to the isobars is typically 20 to 45 degrees; this example uses 45°.

Step 3. Blue: the Coriolis force. At right angles to the wind, on its right-hand side. For a north-east wind that is to the south-east (down and to the right).

Step 4. Green: friction. Opposite to the wind: pointing south-west (down and to the left).

Step 5. Check. Join the three coloured arrows tip to tail: they should make a closed triangle. If they do, the forces cancel and the wind is steady.

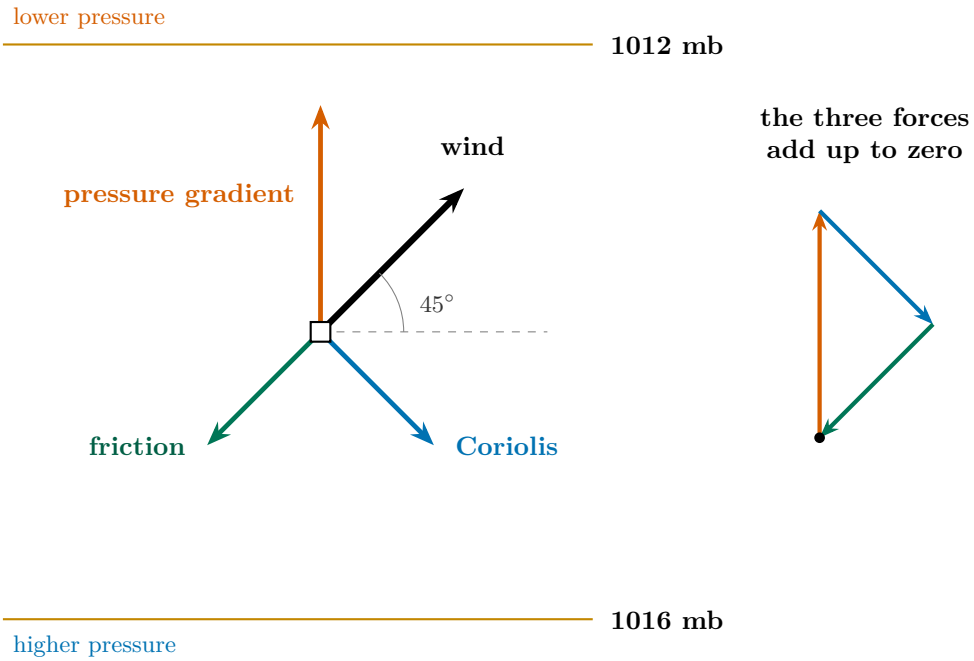


Figure 26.1. The box completed. Red: pressure gradient force; blue: Coriolis force; green: friction; black: the wind. The three forces, joined tip to tail on the right, close into a triangle: they cancel. The black arrow is the *wind*, the direction in which the air actually moves.

Answer to Motion Q9

Pressure gradient force (red): up, towards 1012 mb. Wind (black): up and to the right. Coriolis force (blue): at right angles to the wind, down and to the right. Friction (green): opposite to the wind, down and to the left.

A question that comes up

Is the black arrow, the result, a force?
In a steady wind the three forces add up to zero, so there is *no* resultant force: that is exactly why the wind is steady. What the black arrow shows is the resulting wind: the direction the air moves when the forces balance. It is the right arrow to draw; just do not think of it as a fourth force.

26.4 The numbers: how much does friction turn the wind?

Represent friction as a drag proportional to the wind, with coefficient k. The balance of the three forces then gives, for a pressure gradient force P,

$$\alpha = \arctan\frac{k}{f}, \qquad V = V_g\cos\alpha, \qquad V_g = \frac{P}{|f|}. \tag{26.1}$$

where

α	angle at which the wind crosses the isobars, towards low pressure (degrees)	calculated
k	friction coefficient (s^{-1})	larger over land than sea
f	Coriolis parameter (s^{-1})	
$V,\ V_g$	wind speed with friction, and geostrophic wind speed ($\mathrm{m\,s}^{-1}$)	
P	pressure gradient force per unit mass ($\mathrm{m\,s}^{-2}$)	Eq. (23.1)

The ratio k/f measures how strong friction is compared with the Coriolis force. Figure 26.1 corresponds to $k/f = 1$.

Table 26.1. Friction and the surface wind (the ratio k/f is a guide; real values depend on the surface and the weather).

Surface	k/f	Angle across isobars	Speed V/V_g	Coriolis force (relative)
No friction (high above the ground)	0	0°	100 %	100 %
Open sea	0.3	17°	96 %	96 %
Flat grassland	0.6	31°	86 %	86 %
Forest or suburb	1.0	45°	71 %	71 %
Tall city	1.5	56°	55 %	55 %

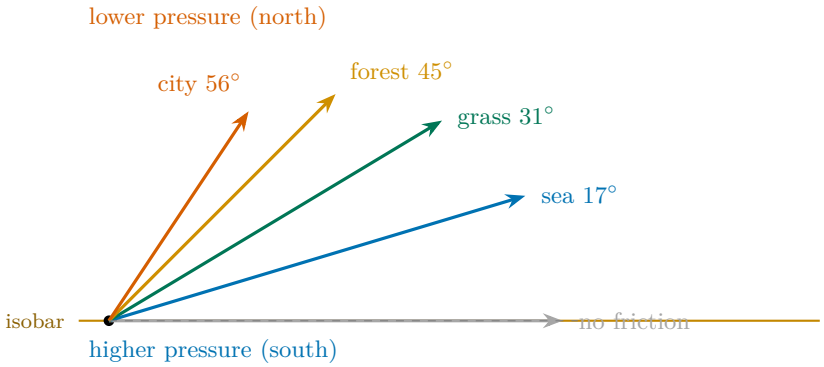


Figure 26.2. The surface wind for different surfaces, for the same pressure gradient force (in the Northern Hemisphere, with low pressure to the north). The rougher the surface, the slower the wind and the greater its angle across the isobars towards low pressure. The length of each arrow is proportional to the wind speed.

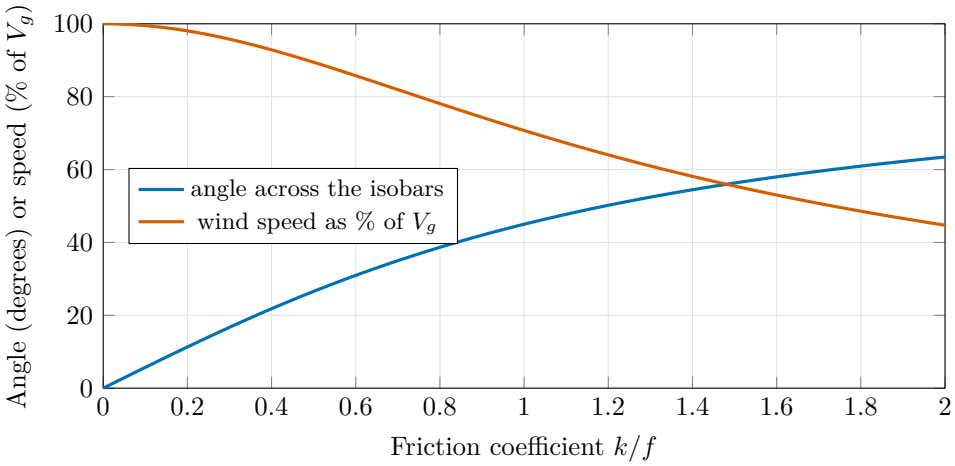


Figure 26.3. The angle across the isobars (blue) and the speed (red) against the friction coefficient. The speed is $V_g \cos\alpha$.

26.5 Question 10: which system is which?

The question

Winds circulate round surface high and low pressure centres in both hemispheres. Name the hemisphere and pressure system for each of four examples: (a) spiralling inwards, anticlockwise; (b) spiralling outwards, clockwise; (c) spiralling inwards, clockwise; (d) spiralling outwards, anticlockwise.

Two observations identify each system.

1. **Does the air spiral inwards or outwards?** Friction makes it cross the isobars towards low pressure, so it spirals *in* towards a low and *out* from a high.
2. **Which way does it turn?** In the Northern Hemisphere the Coriolis force deflects the air to the right, so the circulation round a low is *anticlockwise* and round a high *clockwise*. In the Southern Hemisphere it is the other way round.

Table 26.2. The four systems.

System	Rotation	Air spirals	Pressure gradient force
Northern low	anticlockwise	in	towards the centre
Northern high	clockwise	out	away from the centre
Southern low	clockwise	in	towards the centre
Southern high	anticlockwise	out	away from the centre

Answer to Motion Q10

(a) **Northern Hemisphere low**; (b) **Northern Hemisphere high**; (c) **Southern Hemisphere low**; (d) **Southern Hemisphere high**.

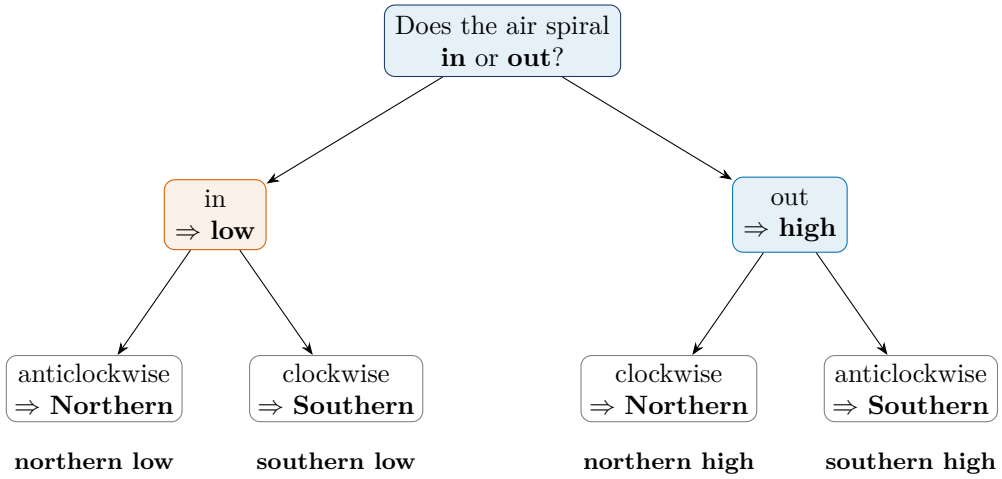


Figure 26.4. A two-question key to the four systems: in or out first, then the sense of rotation.

26.6 Question 11: the forces on one box in each system

The question

Choose one box on each example and draw the pressure gradient force in red, the Coriolis force in blue, and the frictional force in green. The black arrow shows the result.

Method. For the box you choose:

Step 1. Draw the pressure gradient force (red): towards the centre for a low, away from it for a high.

Step 2. Draw the wind (black): follow the arrows already on the figure, spiralling in for a low and out for a high.

Step 3. Draw the Coriolis force (blue) at right angles to the wind: to its right in the Northern Hemisphere, to its left in the Southern Hemisphere.

Step 4. Draw friction (green) directly opposite to the wind.

Step 5. Check that the three arrows could be joined tip to tail into a triangle.

Answer to Motion Q11

On every box the red arrow points from high to low pressure; the black wind arrow is turned from it, to the right in the north and the left in the south, and pulled towards low pressure by friction; the blue arrow is at right angles to the wind on the same side as the turn; the green arrow is exactly opposite to the wind. The pattern is identical at every position round the system, so any box gives the same picture rotated.

26.7 Why this matters for weather

Because the air spirals in towards a low, more air flows in at the bottom than flows out sideways, so the surplus must rise: surface **convergence** gives rising air, cloud and rain. In a high the air spirals out, so air sinks from above to replace it, which warms and dries it and gives clear skies. This is why lows (depressions) bring cloud and rain and highs bring settled weather, and it is the link between this chapter and the weather map you analyse in Chapter 29.

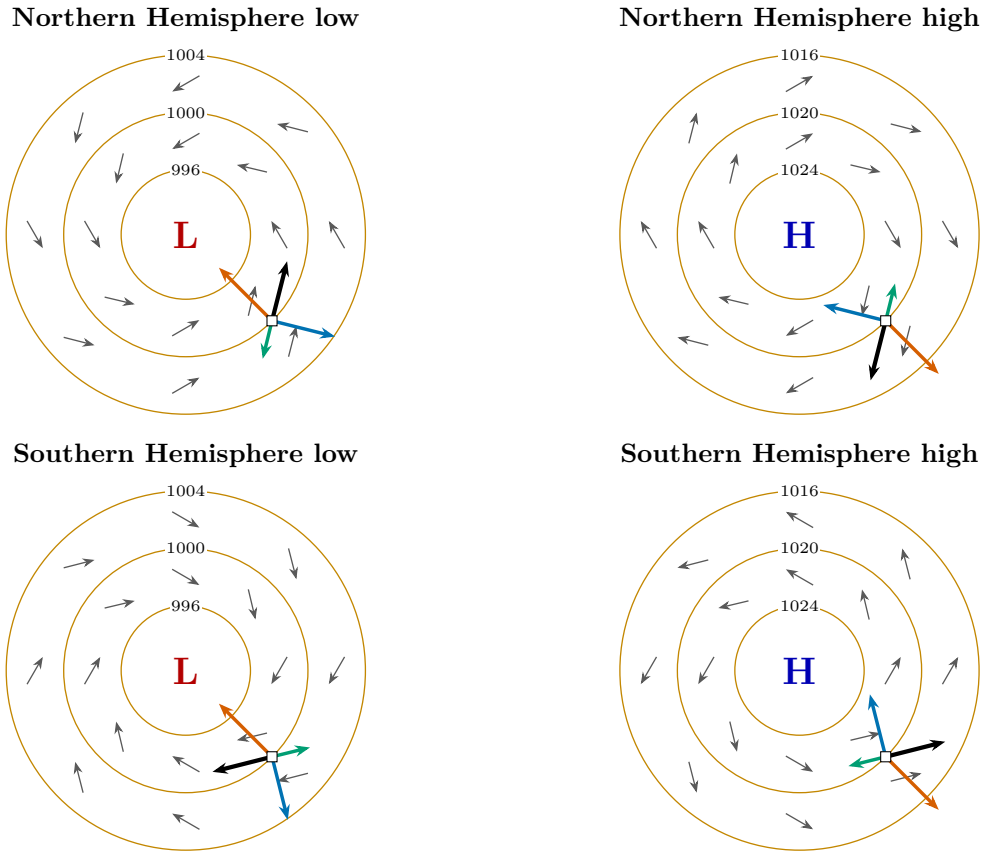


Figure 26.5. The four systems with their forces. In each panel the grey arrows show the surface wind, spiralling in (low) or out (high), and the circles are isobars. On the box at the south-east of each system: red, the pressure gradient force (always along the line to the centre); blue, the Coriolis force (right of the wind in the north, left of the wind in the south); green, friction (against the wind); black, the wind. Any other box would show the same pattern turned round the system.

26.8 Questions that come up

A question that comes up

Why does the wind cross the isobars *towards* low pressure and not away from it?
Because friction removes some of the Coriolis balance. The pressure gradient force is now larger than the Coriolis force, and the surplus pushes the air down the pressure gradient. The result is a wind with a component towards low pressure.

A question that comes up

Why is the angle bigger over land than over the sea?
Land is rougher (trees, buildings, hills), so friction is stronger and k/f is larger. Over the open sea the angle is only 10 to 20 degrees; over land it can be 30 to 45 degrees or more.

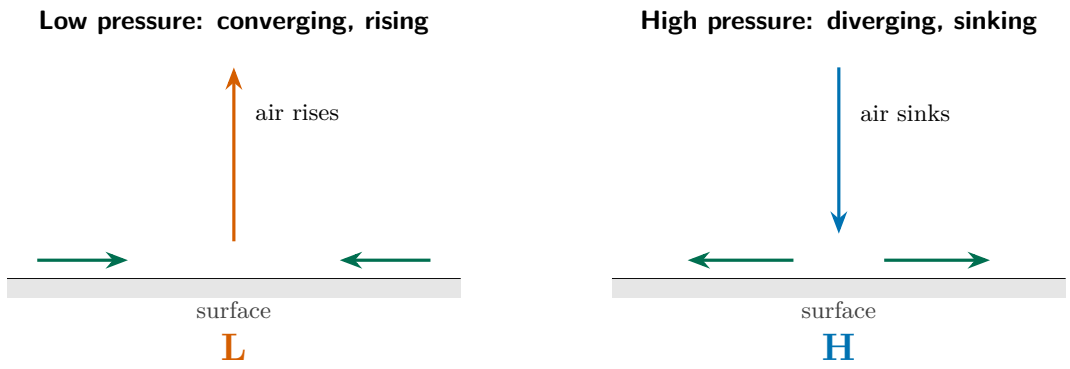


Figure 26.6. Cross-sections. Near the surface the winds converge into a low and diverge from a high; the continuity of air then requires rising motion over the low and sinking over the high.

26.9 Try it yourself: Practice Set 26

Practice Set 26: friction

26.1 Over a flat grassy plain ($k/f = 0.6$) the geostrophic wind is $15\,\mathrm{m\,s^{-1}}$. Find the surface wind speed and its angle to the isobars.

26.2 In a Southern Hemisphere low, in which direction does the air circulate, and does it spiral in or out?

26.3 Describe the winds round a system in which the air spirals outwards in an anticlockwise sense. Which hemisphere and which kind of system is it?

26.4 Over a city the surface wind is $6\,\mathrm{m\,s^{-1}}$ and the geostrophic wind above it is $11\,\mathrm{m\,s^{-1}}$. Roughly what is k/f?

Answers to Practice Set 26

26.1 $\alpha = \arctan 0.6 = 31°$; $V = 15\cos 31° = 12.9\ \mathrm{m\,s^{-1}}$.

26.2 Clockwise, spiralling **in**.

26.3 Anticlockwise and outward is a **Southern Hemisphere high** (Table above).

26.4 $V/V_g = 6/11 = 0.55 = \cos\alpha$, so $\alpha \approx 57°$ and $k/f = \tan 57° \approx 1.5$: a tall-city value.

Chapter 27

Reading a station model

This chapter answers Map Q12.

27.1 Why a station model?

About 10 000 weather stations around the world measure as many as 18 quantities at each observation: temperature, dew point, pressure, wind, cloud, weather, and more. To show all that on one map the World Meteorological Organization (WMO) fixed a compact way of drawing an observation, the **station model**. Every number and symbol has a fixed position around a small circle that marks the station. Once you know the positions, a map becomes a table you can read at a glance.

27.2 The anatomy of a station model

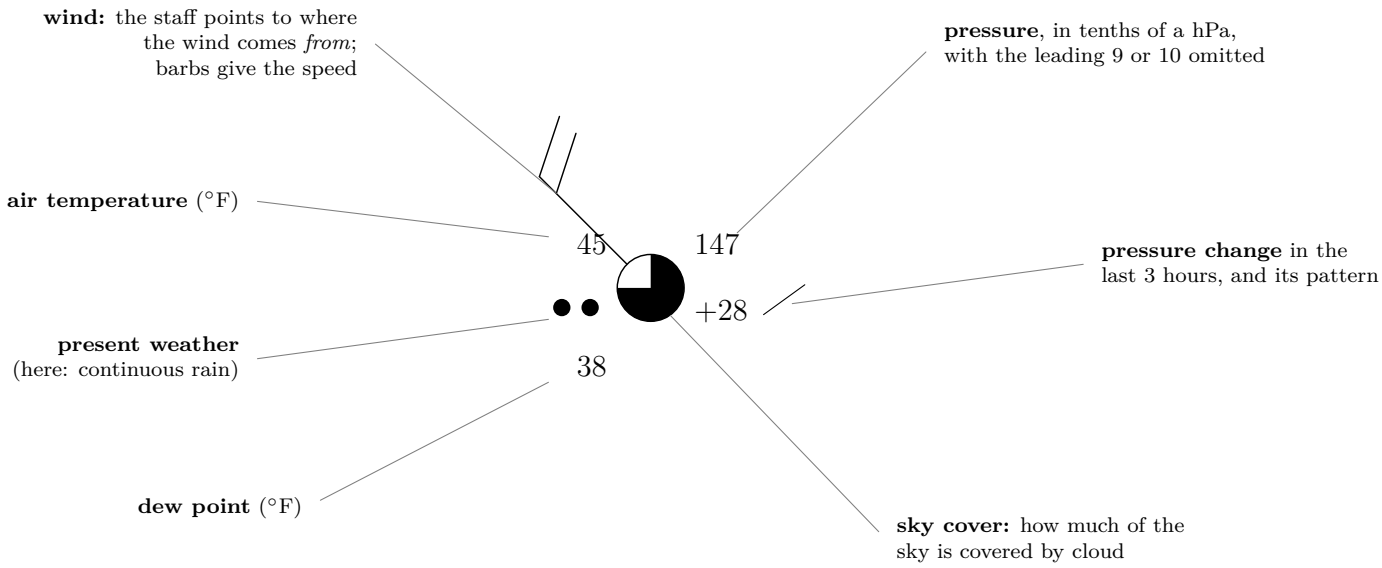


Figure 27.1. A station model with every field labelled. This example reads: temperature 45 °F, dew point 38 °F, pressure 1014.7 hPa, rising by 2.8 hPa in the last three hours, three-quarters of the sky covered, continuous rain, and a wind of 20 knots from the north-west.

27.3 Decoding the pressure

Only three digits are plotted, in tenths of a hectopascal, and the leading digits are left off. Sea-level pressure almost always lies between about 950 and 1050 hPa, so the missing digits can be restored without doubt.

Table 27.1. Where each quantity sits around the circle.

Position	What it shows
Top left	air temperature, in °F here
Bottom left	dew-point temperature
Top right	pressure: three digits (tenths of a hectopascal)
Right, below the pressure	pressure change over the last 3 hours (with a symbol for its pattern)
Left of the circle	present weather symbol
The circle itself	sky cover (how much is filled)
A line from the circle	wind: the staff points **from** where the wind blows, barbs give the speed

The pressure rule

1. Put a decimal point before the last digit: 117 → 11.7.
2. If the three-digit number is **less than 500**, put **10** in front: 11.7 → 1011.7 hPa.
3. If it is **500 or more**, put **9** in front: 998 → 99.8 → 999.8 hPa.

Table 27.2. Examples of the pressure rule (the first four are the models of Question 12).

Code	Decimal point	< 500 or ≥ 500?	Prefix	Pressure
024	02.4	< 500	10	**1002.4 hPa**
998	99.8	≥ 500	9	**999.8 hPa**
117	11.7	< 500	10	**1011.7 hPa**
997	99.7	≥ 500	9	**999.7 hPa**
147	14.7	< 500	10	1014.7 hPa

27.4 Temperature and dew point

Both are in degrees Fahrenheit here. To convert to Celsius, subtract 32 and multiply by 5/9:

$$T(^\circ\mathrm{C}) = \tfrac{5}{9}\,[T(^\circ\mathrm{F}) - 32]. \tag{27.1}$$

where

$T(^\circ\mathrm{C})$, $T(^\circ\mathrm{F})$ the same temperature in Celsius and in Fahrenheit °F from the station model

So 37 °F is 2.8 °C, 63 °F is 17.2 °C, and 21 °F is −6.1 °C.

The **dew point is the water-vapour measure of Chapter 22**, and the gap between the two, $T - T_d$, tells you how close the air is to saturation. A gap of 3 °F means nearly saturated air (fog, rain or thick cloud likely); a gap of 20 °F means dry air. You can turn the pair into a relative humidity with the formula $\mathrm{RH} = 100\, e_s(T_d)/e_s(T)$.

27.5 Sky cover

The circle is filled in proportion to the cloud cover. Here it is in tenths. The shading grows clockwise from the top, as in Figure 27.2.

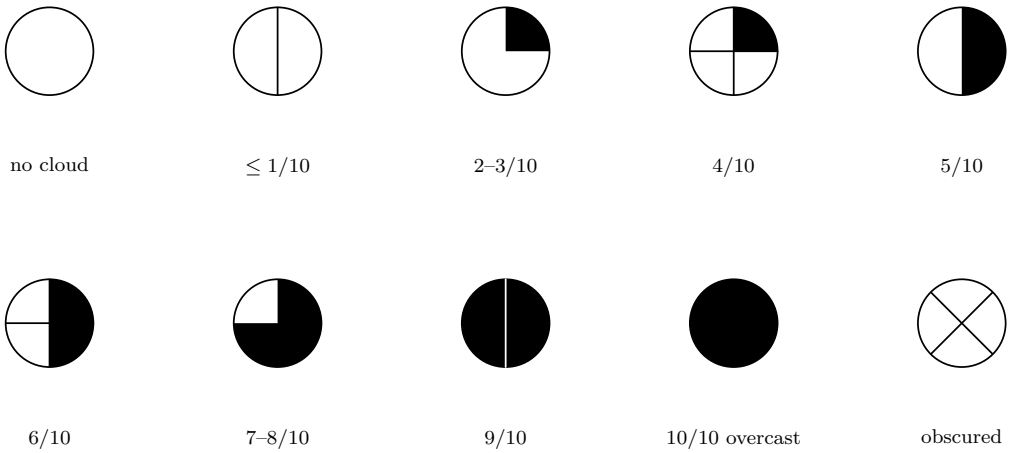


Figure 27.2. The sky-cover symbols: from clear (empty) through half (right half filled) to overcast (filled), and an obscured sky (a cross). The "9/10" symbol is filled with a vertical gap: overcast with small openings.

27.6 Wind: the barbs

The wind is drawn as a straight line, the **staff**, that starts at the circle and points **towards the direction the wind comes from**, as a weather vane does. At the free end are short lines, the **barbs**, that tell you the speed in knots.

- a **half barb**: 5 knots;
- a **full barb**: 10 knots;
- a **pennant** (a filled triangle): 50 knots;
- no barb but a staff: 1 to 2 knots; a circle round the station and no staff: calm.

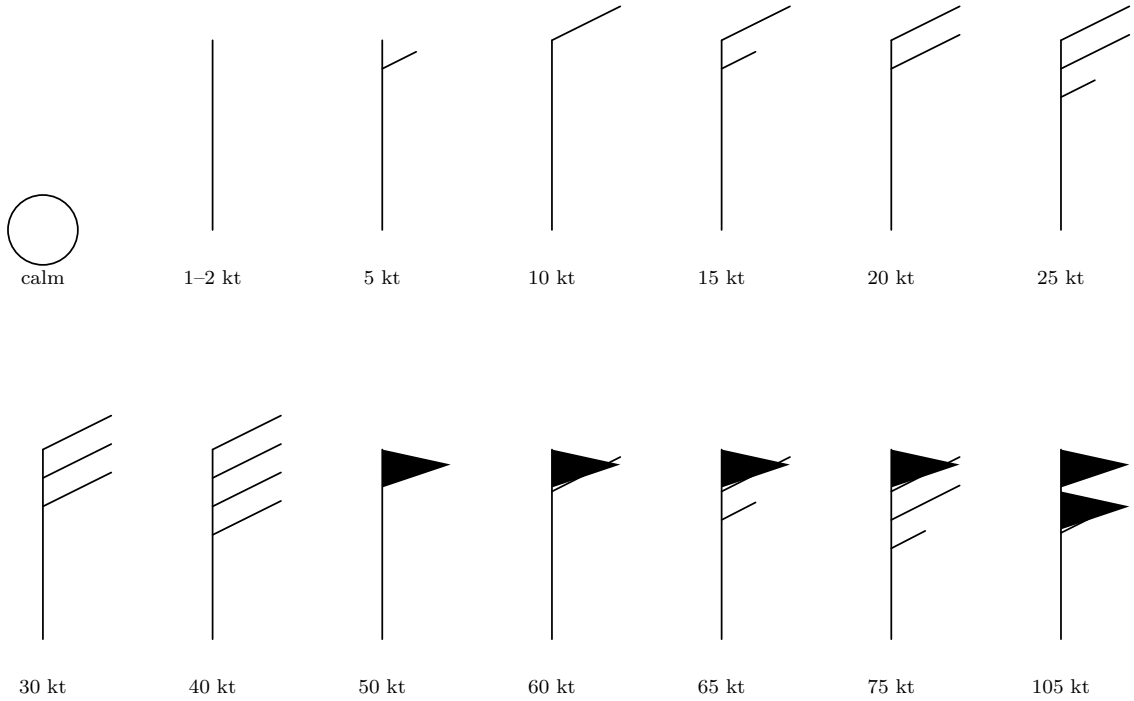


Figure 27.3. Wind barbs for the Northern Hemisphere, drawn with the staff pointing up (a north wind) and the barbs on the right. Add the barbs: 10 + 10 + 5 = 25 knots.

To read a barb.

Step 1. Find the staff and follow it from the circle to its free end. The wind comes *from* that direction: a staff pointing to the lower left is a wind from the south-west.

Step 2. Count the barbs at the free end: pennants (50), full barbs (10), half barbs (5). Add them.

Step 3. Convert if you wish: $1\,\text{knot} = 0.514\ \text{m}\,\text{s}^{-1}$, so $25\,\text{knots} = 12.9\ \text{m}\,\text{s}^{-1}$.

A question that comes up

The staff points where the wind comes from, but the air is moving the other way. Which way do I draw a streamline?
Away from the barbs, i.e. in the direction *opposite* to the staff. A staff pointing to the south-west means the air is heading north-east. Keep that in mind in Chapter 28.

27.7 Present weather and pressure tendency

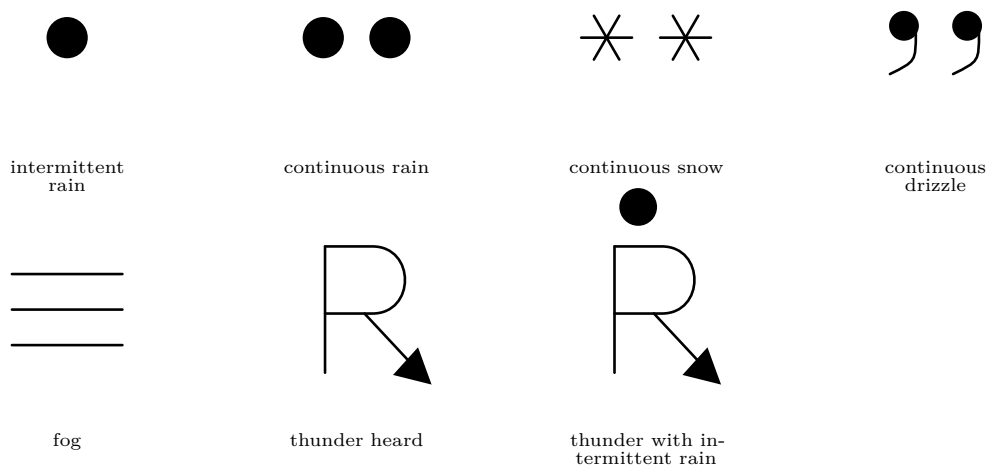


Figure 27.4. The present-weather symbols used here.

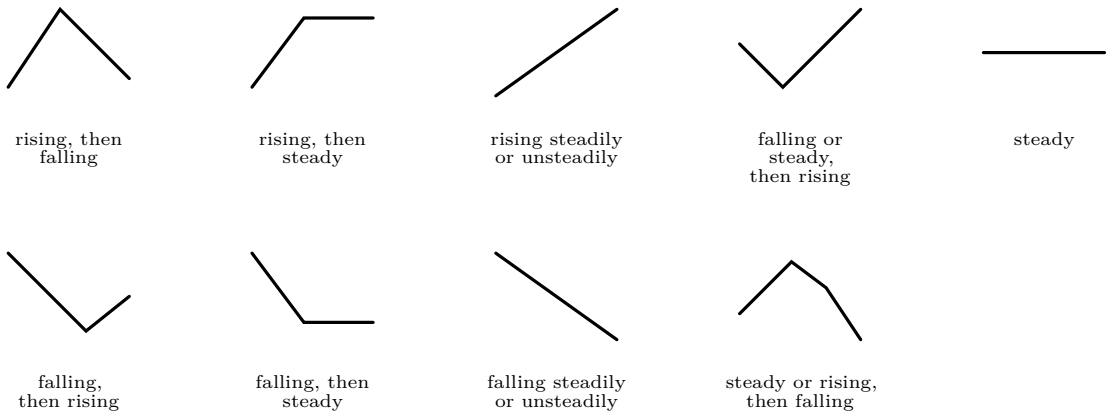


Figure 27.5. The barometric tendency symbols: how the pressure has changed in the last three hours. A line rising to the right means rising pressure, one falling to the right means falling pressure.

The number next to the symbol, for example +17 or −11, is the pressure change in tenths of a hectopascal: +17 is a rise of 1.7 hPa in three hours, −11 a fall of 1.1 hPa. Rapid falls often warn of a storm approaching; rising pressure usually brings improving weather.

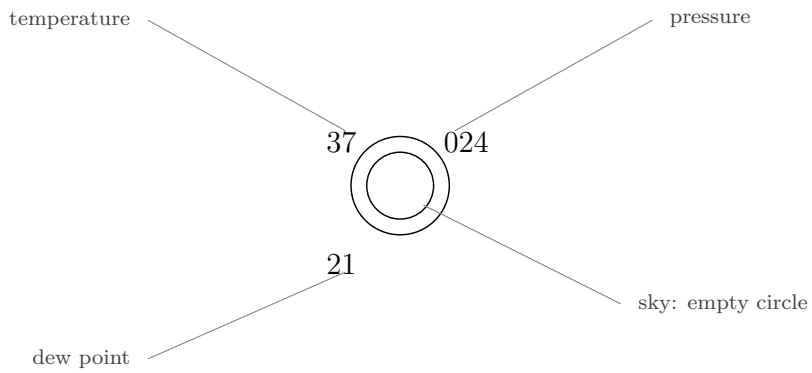
27.8 Question 12: the four station models

The question

Decode the information from each of the following station models.

Method. For each model work down this list: pressure; temperature; dew point; sky cover; and, where shown, current weather, wind speed, wind direction, pressure change and tendency.

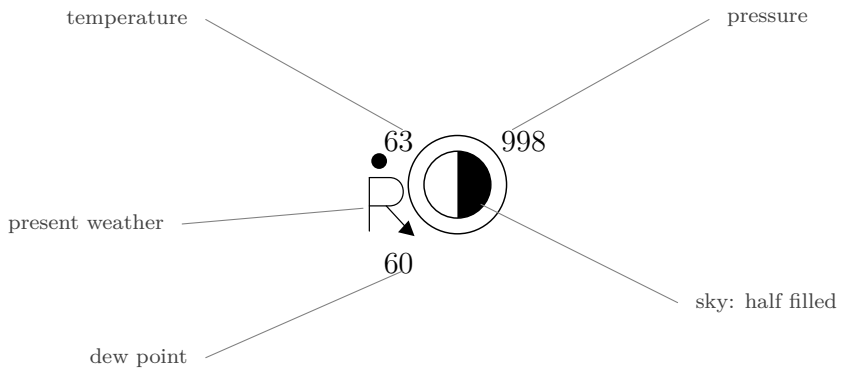
Model 1



Model 1

Pressure: 024 → 02.4 → **1002.4 hPa**
Temperature: **37** °F (2.8 °C) **Dew point:** **21** °F (−6.1 °C)
Sky cover: empty circle: **0 %** (clear). No staff, only a ring round the circle: **calm**. RH ≈ 52 %.

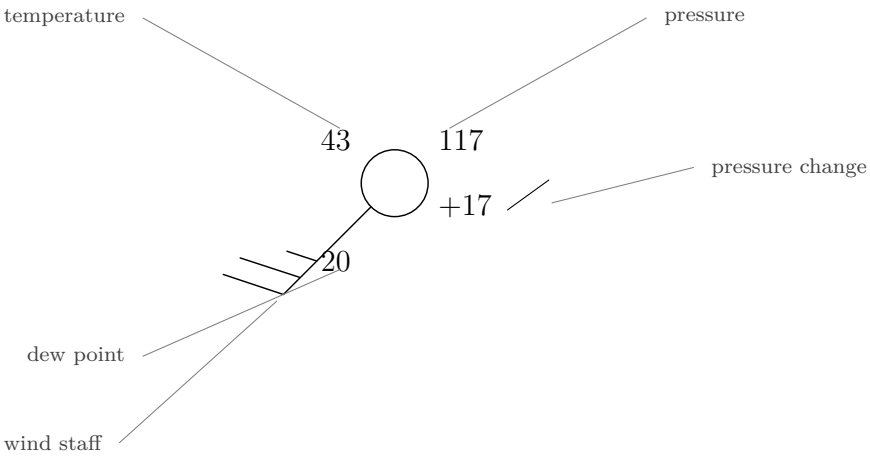
Model 2



Model 2

Pressure: 998 → 99.8 → **999.8 hPa**
Temperature: **63** °F (17.2 °C) **Dew point:** **60** °F (15.6 °C)
Current weather: thunder with intermittent rain. Sky half filled (5/10); no staff: calm. RH ≈ 90 %.

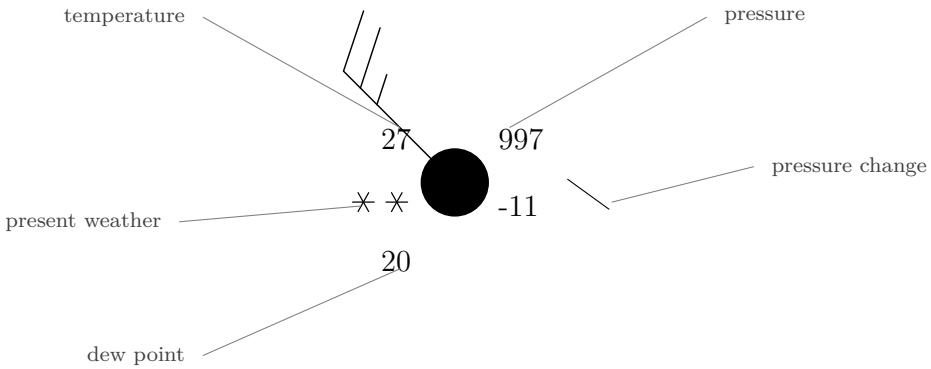
Model 3



Model 3

Pressure: 117 → 11.7 → **1011.7 hPa**
Temperature: 43 °F (6.1 °C) **Dew point: 20** °F (−6.7 °C) **Sky cover:** empty circle, **0 %**
Wind: staff to the lower left: **from the south-west**; two full barbs and a half barb: **25 knots**
Pressure change: +17: +1.7 **mb**, **rising**

Model 4



Model 4

Pressure: 997 → 99.7 → **999.7 mb**
Temperature: 27 °F (−2.8 °C) **Dew point: 20** °F (−6.7 °C)
Sky cover: filled circle, **100 %** **Current weather: continuous snow**
Wind: staff to the upper left: **from the north-west**, **25 knots**
Pressure change: −11: −1.1 **mb**, **falling**

27.9 What the models tell you

Putting the numbers together gives each station a story.

- **Model 1:** cold, dry (the dew point is 16 °F below the temperature), clear and calm: a cold, high-pressure night.

Table 27.3. All four models decoded.

	Model 1	Model 2	Model 3	Model 4
Barometric pressure (mb)	1002.4	999.8	1011.7	999.7
Temperature (°F)	37	63	43	27
Dew point (°F)	**21**	60	20	20
Sky coverage (%)	0	50	0	100
Current weather	—	thunder, interm. rain	—	continuous snow
Wind speed (knots)	calm	calm	25	25
Wind direction (from)	—	—	south-west	north-west
Pressure change in 3 h (mb)	—	—	+1.7	−1.1
Tendency	—	—	rising	falling

- **Model 2:** warm and very humid (the dew point only 3 °F below the temperature), with thunder and rain: unstable, thundery air.
- **Model 3:** clear and windy with rising pressure: air moving away from a low towards a high, after a front has passed.
- **Model 4:** overcast, snowing, 27 °F, wind 25 knots and falling pressure: a depression is arriving or overhead.

27.10 Try it yourself: Practice Set 27

Decode each of the three models in Figure 27.6 (pressure, temperature and dew point in °F and °C, relative humidity, sky cover, weather, wind speed and direction, pressure change and tendency), then check against the answers below.

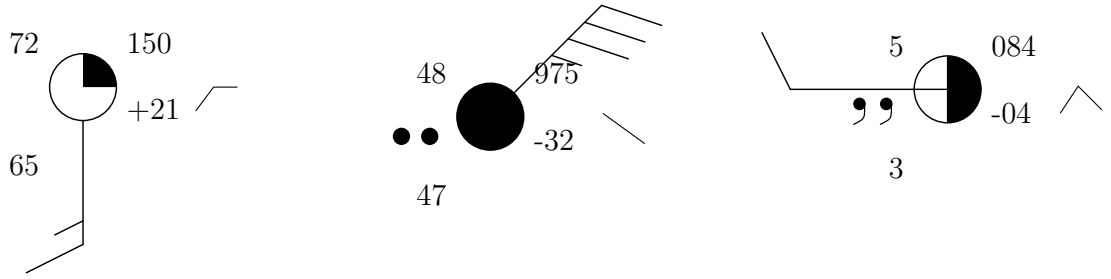


Figure 27.6. Three station models to decode.

Answers to Practice Set 27

27.1 **150** → 15.0 → **1015.0 hPa**. $T = 72\,°\text{F}$ (22.2 °C), $T_d = 65\,°\text{F}$ (18.3 °C), RH ≈ 79 %. Sky: 2–3/10 (about 25 %). Wind: staff pointing straight down: **from the south**, one full barb and a half barb: **15 knots**. Tendency: +21 → up 2.1 hPa, rising then steady. No weather symbol: fair, warm and humid.

27.2 **975** (≥ 500) → 97.5 → **997.5 hPa**. $T = 48\,°\text{F}$ (8.9 °C), $T_d = 47\,°\text{F}$ (8.3 °C); RH ≈ 96 %. Overcast (100 %). Continuous rain. Wind from the **north-east**, three full barbs and a

half barb: **35 knots**. Pressure −32: down 3.2 hPa, falling steadily.

27.3 **084** (< 500) $\to$ 08.4 $\to$ **1008.4 hPa**. $T = 5\,°\text{F}$ ($-15\,°\text{C}$), $T_d = 3\,°\text{F}$; RH $\approx 91\,\%$. Sky: 6/10. Continuous drizzle. Wind from the **west**, one full barb: **10 knots**. Pressure −4: down 0.4 hPa, rising then falling.

27.11 Station models in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def fahrenheit_to_celsius(t_f):
    """Temperature in deg C from deg F."""
    return (t_f - 32.0) * 5.0 / 9.0
```

```
def decode_pressure(code):
    """Sea-level pressure (hPa) from a three-digit station-model code."""
    return (1000.0 if code < 500 else 900.0) + code / 10.0
```

```
def knots_to_ms(knots):
    """Wind speed in m/s from knots."""
    return 0.514 * knots
```

`decode_pressure(247)` returns 1024.7 and `decode_pressure(987)` returns 998.7.

Chapter 28

Streamlines

This chapter answers Map Q13.

28.1 What a streamline is

A wind barb tells you the wind at one station. A **streamline** joins the stations: it is a line that is everywhere **parallel to the wind**. If you could photograph a river with a long exposure, the streaks of floating leaves would be streamlines. On a map they show at a glance where the air is coming from, where it is going, and where it swirls, converges or spreads out.

- At every point of a streamline, the line is tangent to the wind there.
- Streamlines **never cross**, because the wind has only one direction at one place.
- They begin at an **upwind** station and carry an **arrow head** downwind, usually where the flow changes direction.
- They show direction only. Their spacing does not show speed unless you choose to draw them that way.

28.2 The method

Step 1. **Pick a starting point** at an upwind station: one where the wind is blowing towards the area you want to follow.

Step 2. **Read the barb correctly.** The staff points to where the wind comes *from*. The air moves the other way, *away* from the staff. A barb with its staff pointing to the lower left is a south-west wind and the streamline there runs towards the upper right.

Step 3. **Draw a short line in that direction** and move downwind until you are near the next station.

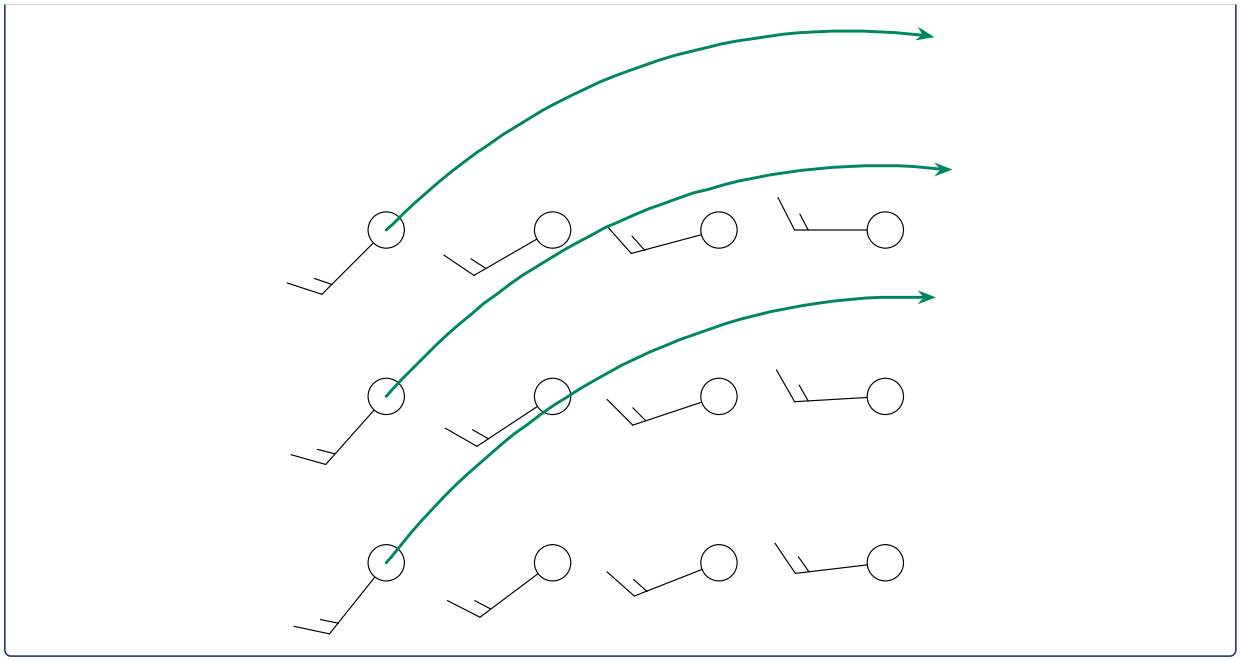
Step 4. **Bend the line smoothly** so that it follows the direction of the barb at the next station, then the next.

Step 5. **Continue** until the stations run out or the flow leaves the map, then draw an **arrow head** at the downwind end.

Step 6. **Repeat from another starting point** until you have the five or six streamlines the question asks for. Keep them smooth and make sure no two cross.

Worked example: a small flow

Twelve stations report winds from the south-west on the left of the map and from the west on the right (the staffs point lower-left on the left and to the left on the right). Starting at a left-hand station the air moves towards the north-east. As you move east the barbs rotate clockwise, so the streamline bends to the right and ends up heading east. Each line starts at a station and stays parallel to the barbs it passes.



A question that comes up

How do I know whether to draw the arrow at the start or the end?
At the *downwind* end: the end towards which the air is moving, that is, away from the stations' barbs. A good test: put your pencil on the station, look at the barb, and ask "which way is this air going next?"

28.3 The stations

Figure 28.1 shows 34 stations in the central United States, plotted with only the sky cover and the wind barb. Each station numbered so that you can refer to it. Look at the barbs before you draw anything: where does the wind blow from in each part of the map?

Table 28.1. The wind by region (the staff points to where the wind comes from).

Region	Stations	Wind from
Dakotas, Nebraska, Kansas, Oklahoma, north Texas	1, 4, 12, 15, 21, 22, 25, 27, 30	north-west (WNW–NW), about 10 knots
Iowa, Missouri (west of the low)	5, 9, 16	north-west to west-north-west, 10–25 knots
Southern Missouri, Arkansas, Texas (east), Mississippi, Alabama	23, 28, 29, 31, 32, 33, 34	south-south-west to west-south-west, about 10 knots
Illinois, Kentucky, Tennessee, Indiana (south)	13, 17, 18, 19, 20, 24, 26	south-south-west to south, 10–25 knots
Wisconsin, Michigan, Ohio, Indiana (north), northern Illinois	2, 3, 6, 7, 8, 10, 11, 14	north-east to south-east, 2–20 knots

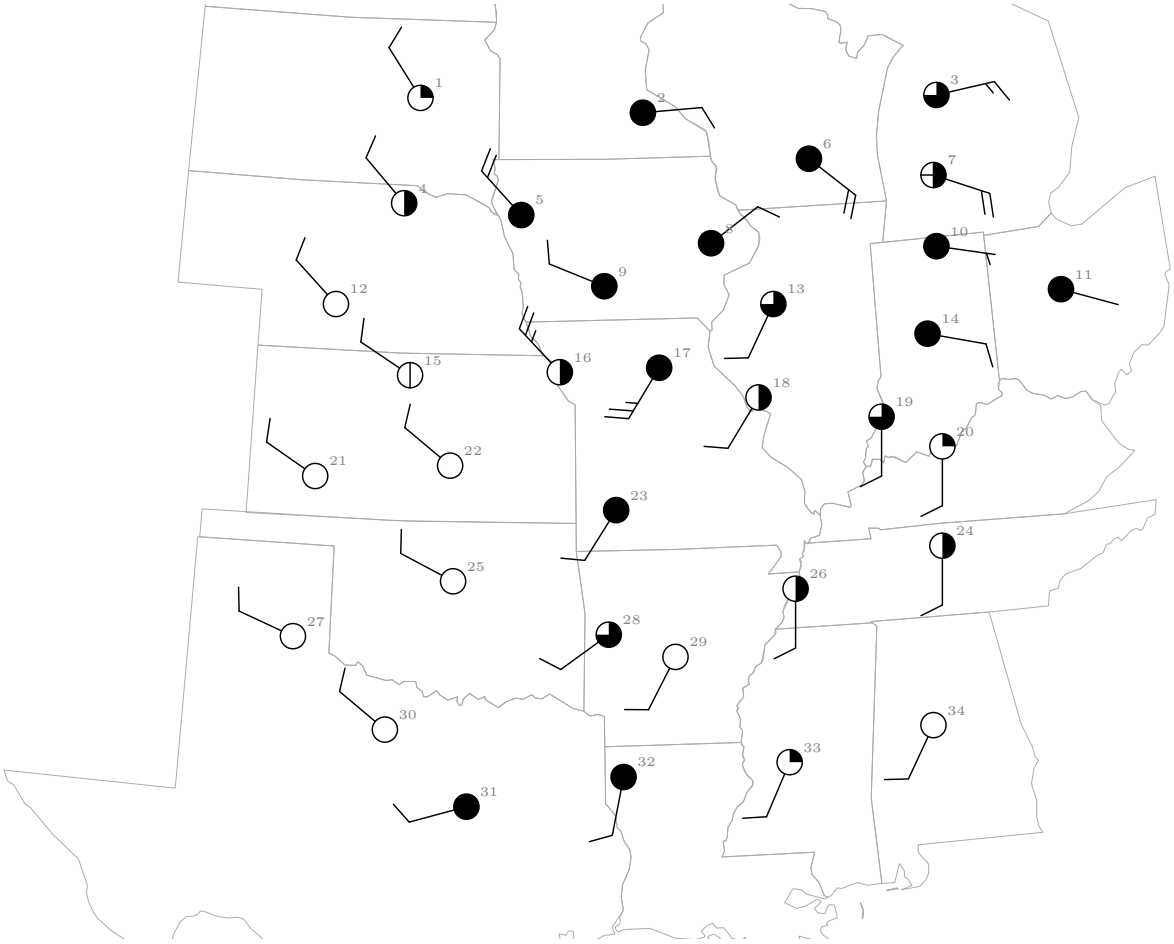


Figure 28.1. The stations (circle for the sky cover, staff and barbs for the wind; station numbers in grey). The state outlines are real boundaries drawn in a Lambert conformal conic projection.

28.4 Question 13: the streamlines

The question

Streamlines begin at an upwind location and are drawn as long lines with an arrow where the wind shifts abruptly. Complete the analysis by drawing 5 to 6 streamlines to illustrate the general wind flow.

Two streamlines are drawn as examples. One runs south-east from near the Nebraska–Kansas border across Kansas into Oklahoma, in the north-westerly wind. The other starts in north Texas and curves north and then north-east through Arkansas and Missouri into Illinois, in the southerly wind.

Answer to Map Q13

Five or six smooth lines, each starting upwind and ending with an arrow, such as those in Figure 28.2: north-westerly flow in the west curving south-east then east; southerly flow in the south curving north and then north-west; easterly flow in the north curving south-west. They all swirl **anticlockwise** and inwards towards one centre in Iowa and northern Illinois.

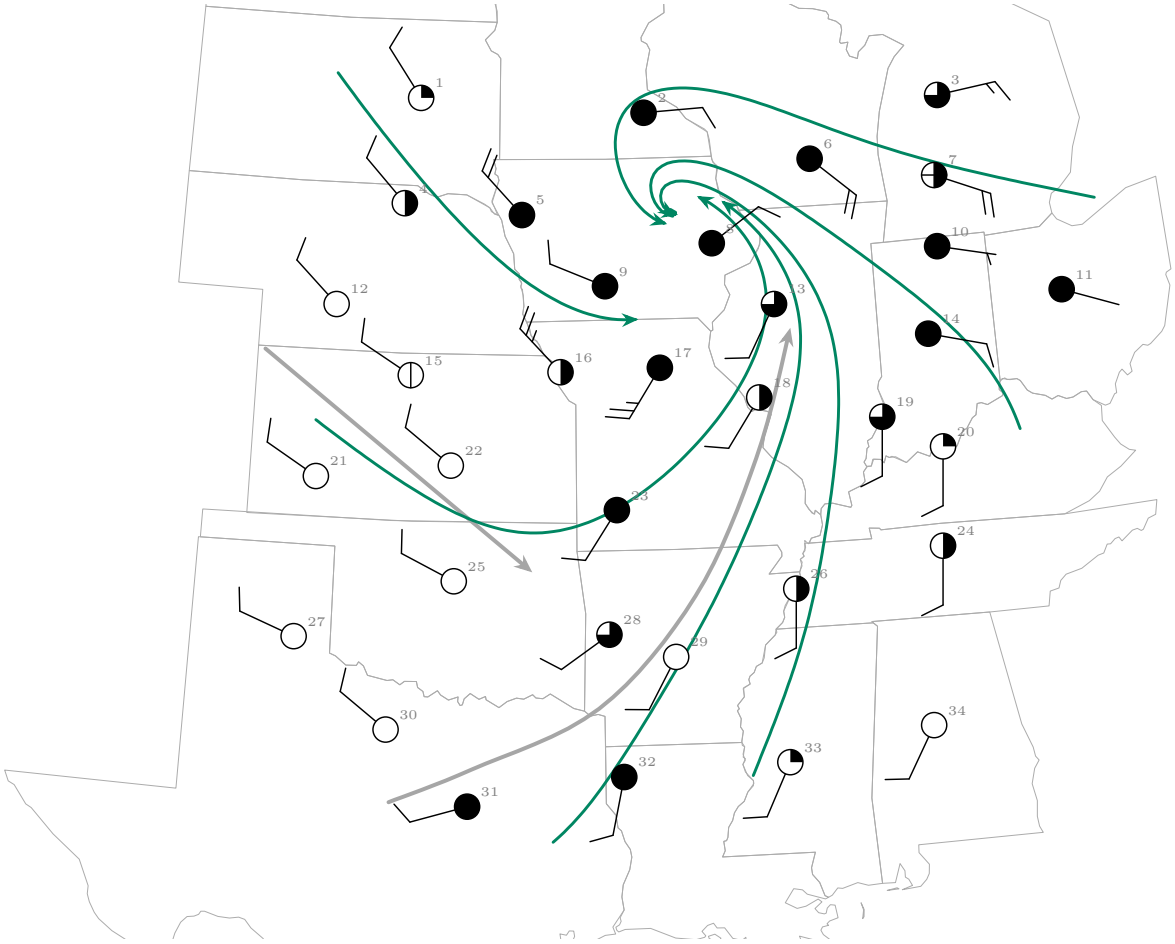


Figure 28.2. The map completed. The two grey streamlines are the examples. The six green streamlines were drawn through the barbs of all 34 stations, starting upwind in the Dakotas, Kansas, Texas, Mississippi, Indiana and Michigan. They curve round a centre in Iowa and northern Illinois.

28.5 What the streamlines show

The pattern is the signature of a Northern Hemisphere low-pressure system (Chapter 26).

- The wind is **north-westerly** to the west of the centre, **southerly** to the south and east of it, and **easterly** to the north: an anticlockwise circulation.
- The streamlines curve **inwards**, because friction makes the air cross the isobars towards low pressure. They converge near the centre, so air must rise there: this is where the cloud and rain of Question 14 are.
- The streamlines from the south carry warm, humid air northwards; those from the north-west carry cold, dry air southwards. Where the two meet are the fronts.
- The centre of the swirl is where the pressure is lowest. The station reports confirm it: the lowest pressures on the map, 1002.4 and 1002.6 hPa, are at stations 8 and 9, next to the centre of the swirl.

28.6 Common mistakes

- **Drawing the arrow the wrong way round,** along the staff rather than away from it. Always ask where the air is going.

- **Joining stations with straight segments** instead of a smooth curve that follows every barb.
- **Letting streamlines cross.** If two lines are about to cross you have misread a barb.
- **Starting downwind.** A streamline needs a clear upwind start.
- **Forgetting the arrow heads.** Put an arrow on every line.

28.7 Try it yourself: Practice Set 28

Practice Set 28: reading wind barbs

28.1 A station has its staff pointing to the right (east) with one full barb and a half barb. From which direction does the wind blow, how strong is it, and in which direction does a streamline through the station run?

28.2 The staff points to the top of the page (north) with two full barbs. State the wind and the direction of the streamline.

28.3 A station near the centre of a low shows a wind from the north-east, another to its east shows a wind from the south-east and another to its south a wind from the south-west. Is the circulation clockwise or anticlockwise?

28.4 In Figure 28.2 what is the wind at station 31 (south-west Texas)? What is the direction of the streamline there?

Answers to Practice Set 28

28.1 From the **east**, 15 knots; the streamline runs **towards the west**.

28.2 From the **north**, 20 knots; the streamline runs **towards the south**.

28.3 Going from the north-east side, to the east, to the south the winds are from NE, SE, SW. A wind from the NE blows to the SW, from the SE to the NW, from the SW to the NE: the air circles anticlockwise, as round a Northern Hemisphere low.

28.4 Station 31 shows a wind from the west-south-west (255°) of about 10 knots. The streamline there runs towards the east-north-east (it is the start of the north-easterly curve in the figure).

Chapter 29

Fronts and the weather map

This chapter answers Map Q14.

29.1 Air masses and fronts

An **air mass** is a very large body of air, hundreds of kilometres across, whose temperature and humidity are much the same throughout. It takes on the character of the surface it has rested on: air that has sat over a cold, dry continent is cold and dry; air from the tropical ocean is warm and humid. A **front** is the boundary between two unlike air masses. You can recognise one by any of the following:

- a sharp change of **temperature**;
- a sharp change of **moisture** (the dew point);
- a sharp change of **wind direction**.

Fronts are classified by how they move.

Warm front A warm air mass advances over a cooler one. Drawn as a red line with semicircles pointing the way the front moves.

Cold front A cold air mass advances under a warmer one. Drawn as a blue line with triangles pointing the way the front moves.

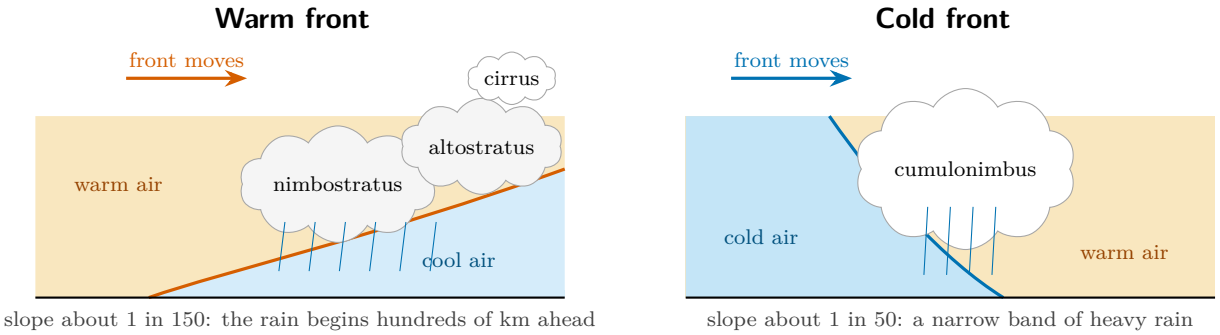


Figure 29.1. Cross-sections through a warm front (left) and a cold front (right), with the vertical scale greatly exaggerated. In both the front is a sloping surface along which the warm air rises over the cold air.

Wind veering. Notice that in both cases the wind *veers*, that is, turns clockwise, as the front passes. Notice how the wind direction shifts at the frontal boundary.

29.2 A low with a warm front and a cold front

Figure 29.2 shows a low-pressure centre (L) with a warm front curving east and south from it and a cold front trailing to the south-west. Four stations sample the air round them, which gives a small case in which to learn how to read a front before tackling the whole map.

Table 29.1. What happens as each kind of front passes a station in the Northern Hemisphere.

	Warm front	**Cold front**
Before	pressure falls; cloud thickens from cirrus to nimbostratus; wind backs to south-east	pressure falls; warm, humid, south-westerly wind
Passage	steady rain or drizzle; wind veers to south-west; temperature rises	heavy showers, perhaps thunder; wind veers sharply to north-west
After	warm, humid air, cloud breaks; pressure steady	colder, drier air; pressure rises; clearing sky

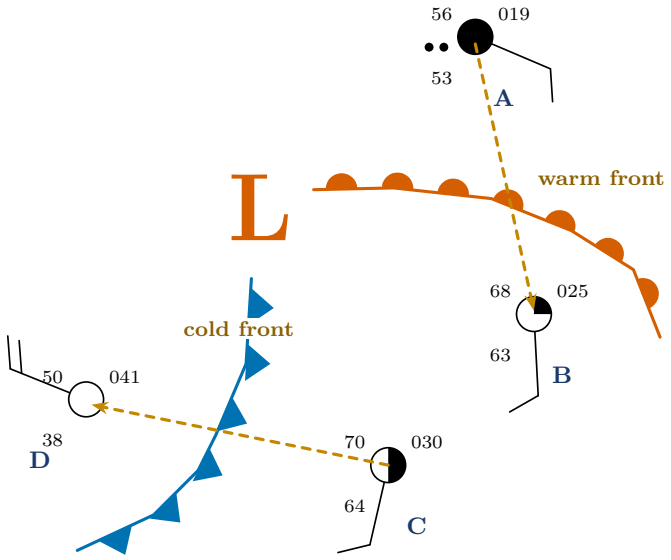


Figure 29.2. Station A is north of the warm front, B and C are in the warm sector between the fronts, D is behind the cold front. The dashed orange arrows mark the two crossings studied below: C to D across the cold front, and A to B across the warm front.

Step 1. The warm sector (B and C). The air between the fronts is warm (68 and 70 °F) and humid (dew points 63 and 64 °F, relative humidity above 80 %), and the wind is from the south.

Step 2. Behind the cold front (D). The air is cooler (50 °F), much drier (dew point 38 °F, RH 63 %), the sky is clear and the wind has swung to the west-north-west.

Step 3. Crossing the cold front, C to D. The temperature falls 20 °F, the dew point falls 26 °F, the wind turns from south-south-west (193°) to west-north-west (292°): clockwise by about 100°. The pressure rises and the cloud clears.

Step 4. Crossing the warm front, A to B. The temperature rises 12 °F, the dew point 10 °F, and the wind turns from east-south-east (113°) to south (177°): clockwise by about 64°. The sky changes from overcast and raining to almost clear.

Table 29.2. The four stations of Figure 29.2.

Stn	Temp. (°F)	Dew pt (°F)	RH (%)	p (hPa)	Wind from	Sky
A	56	53	90	1001.9	ESE 10 kt	overcast, rain
B	68	63	84	1002.5	S 10 kt	2–3/10
C	70	64	81	1003.0	SSW 10 kt	5/10
D	50	38	63	1004.1	WNW 20 kt	clear

What the four stations teach

A front is marked by *simultaneous* changes of temperature, moisture and wind. At **both** fronts the wind **veers** (turns clockwise). The cold front brings the bigger changes in temperature and dew point; the warm front brings the continuous rain.

29.3 Question 14: the full analysis

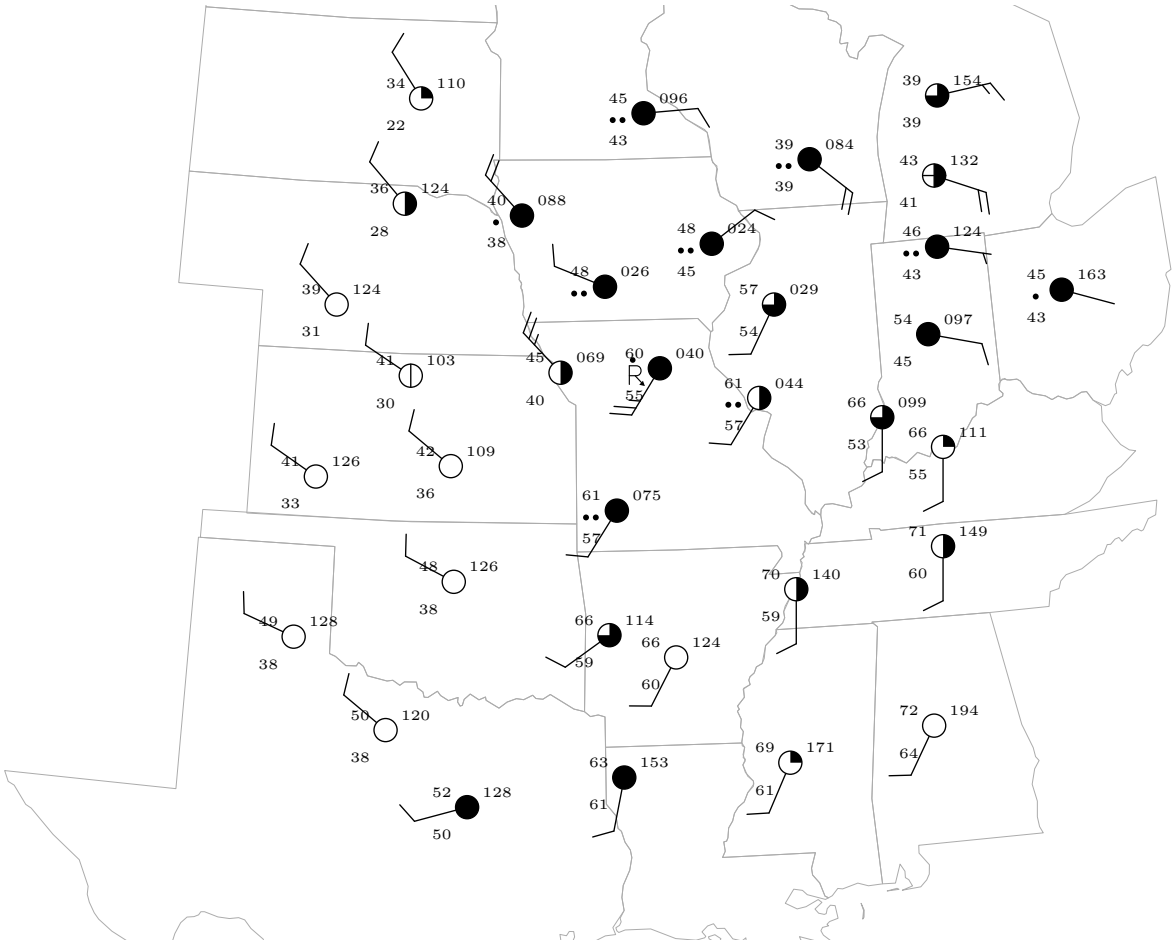


Figure 29.3. The 34 stations with all their values: temperature, dew point, pressure, weather, sky cover and wind. They are listed in Appendix C.2.

The question

Using the map of Figure 29.3: draw isobars at 4 mb intervals (e.g. 1004, 1008, 1012 mb); label the low-pressure centre with an "L"; draw the warm and cold fronts; outline the area where cloud cover exceeds 75 %; shade the areas receiving precipitation.

Figure 29.3 shows the same 34 stations as Figure 28.1, but with all the numbers: temperature, dew point, pressure and weather, as well as the sky cover and the wind. The recommended order is the order of the bullets; each step helps the next.

29.3.1 Step 0: read the pressures

Decode every pressure with the rule of Chapter 27 (a three-digit number below 500 gets 10 in front, otherwise 9), and write the full value next to each station.



Figure 29.4. Step 0: the decoded sea-level pressures (hPa) at the 34 stations. The lowest values, 1002.4 and 1002.6, are in Iowa and Illinois (stations 8 and 9); the highest, 1019.4, is in Alabama (station 34).

29.3.2 Step 1: isobars every 4 mb

The isobars are 1004, 1008, 1012, 1016 and 1020 mb. (The lowest pressure is 1002.4, so there is no 1000 mb isobar. The highest, 1019.4 in Alabama, falls just short of 1020; the 1020 isobar lies a little beyond it, where the trend continues, and may be left out.)

Method: interpolating.

1. Look at two neighbouring stations and ask whether a multiple of 4 lies between their pressures.
2. If it does, find where it falls on the straight line between them *in proportion*.
3. Mark that point. Do it for every pair of neighbours.
4. Join the points of equal value with a smooth curve. Do the same for the next value.

Worked example: where does 1008 mb cross between stations 18 and 19?

Station 18 reads 1004.4 hPa and its neighbour station 19 reads 1009.9 hPa. The pressure changes by $1009.9 - 1004.4 = 5.5$ hPa between them, and 1008 is $1008 - 1004.4 = 3.6$ hPa above the first station:

$$\text{fraction of the distance} = \frac{3.6}{5.5} = 0.65.$$

The 1008 mb isobar crosses the line joining the stations 65 % of the way from station 18 towards station 19.

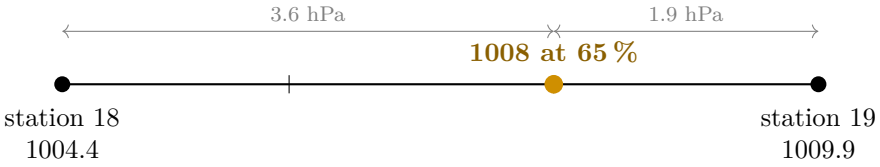


Table 29.3. More interpolations to practise on.

Stations	Pressures (hPa)	Isobar	Calculation	Fraction
9 and 5	1002.6, 1008.8	1004	$(1004 - 1002.6)/6.2$	23 %
9 and 5	1002.6, 1008.8	1008	$(1008 - 1002.6)/6.2$	87 %
16 and 5	1006.9, 1008.8	1008	$(1008 - 1006.9)/1.9$	58 %
20 and 24	1011.1, 1014.9	1012	$(1012 - 1011.1)/3.8$	24 %

Rules for drawing isobars

- Isobars are smooth curves. Each connects points of exactly the same pressure.
- Isobars **never cross** and never branch.
- They never end inside the map: they either close into a loop (round a low or a high) or leave the map edge.
- The pressure rises or falls in steps of 4 mb from one isobar to the next; if the next isobar is much further away the gradient is gentle there.
- Label each isobar with its value.

29.3.3 Step 2: the L

The low-pressure centre is where the isobars close round the lowest pressure. The two lowest pressures on the map are 1002.4 hPa (station 8, northern Illinois) and 1002.6 hPa (station 9, Iowa). The 1004 mb isobar encloses them, together with stations 13 and 17 (1002.9 and 1004.0). Write the **L** between stations 8 and 9, near the centre of the closed isobar, at about 1002 hPa.

Answer: the L

The L goes in northern Illinois and eastern Iowa, between stations 8 and 9, in the middle of the innermost isobar. It is also the centre of the swirl of streamlines you drew in Question 13.

29.3.4 Step 3: the fronts

A front lies where the temperature, the dew point and the wind direction change rapidly. Compare pairs of neighbouring stations: if two neighbours differ strongly in all three, a front runs between them.

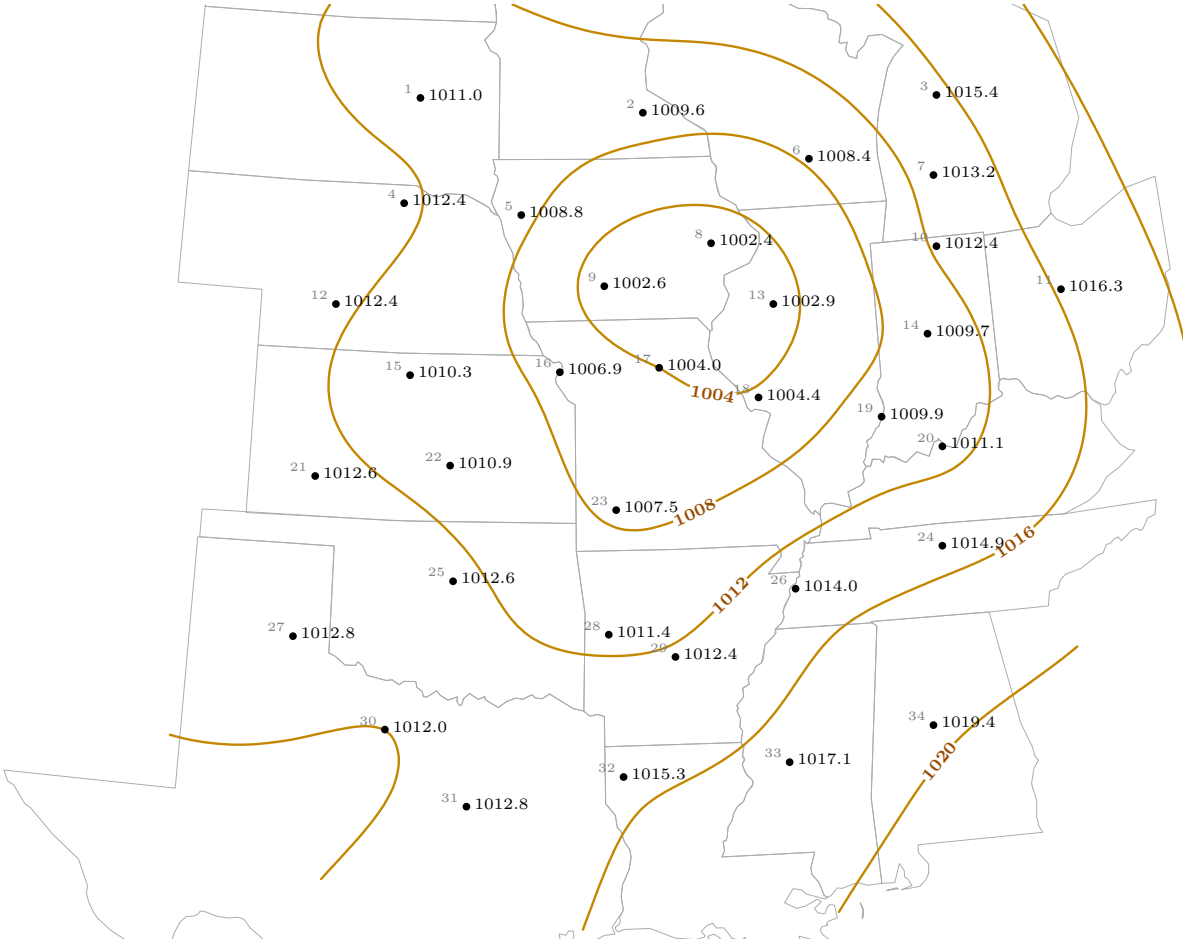


Figure 29.5. Step 1: isobars at 4 mb intervals through the station pressures. The 1004 mb isobar closes round the lowest pressures; the 1008 and 1012 mb isobars follow it outwards; 1016 and 1020 mb lie to the east and south-east. (The curves here were drawn as a smooth surface through all 34 pressures.)

The cold front. West of the low, the stations behind the front are cool (42 to 52 °F), drier (dew points 36 to 50 °F) and have winds from the west to north-west; those ahead are warm (60 to 66 °F), humid (dew points 55 to 61 °F) and have winds from the south-south-west to south-west. The pairs 16–17, 22–23, 25–28 and 31–32 straddle it, with temperature jumps of 11 to 19 °F. The front runs from the low towards the south-south-west, down through Missouri, Arkansas and east Texas.

The warm front. East of the low, the stations north of the front are cool (46 to 54 °F) with winds from the north-east or east; those south of it are warm (57 to 66 °F) with winds from the south. The pairs 8–13, 14–19 and 10–20 straddle it, with temperature jumps of 9 to 20 °F and the wind turning from east or north-east to south. The front runs east from the low through Indiana and Ohio.

Answer: the fronts

A **cold front** (blue, with triangles pointing east, the way it moves) from the L towards the south-south-west to Texas; a **warm front** (red, with semicircles pointing north-east) from the L eastwards through Indiana and Ohio. The warm sector, the wedge of warm air between them, lies to the south-east.

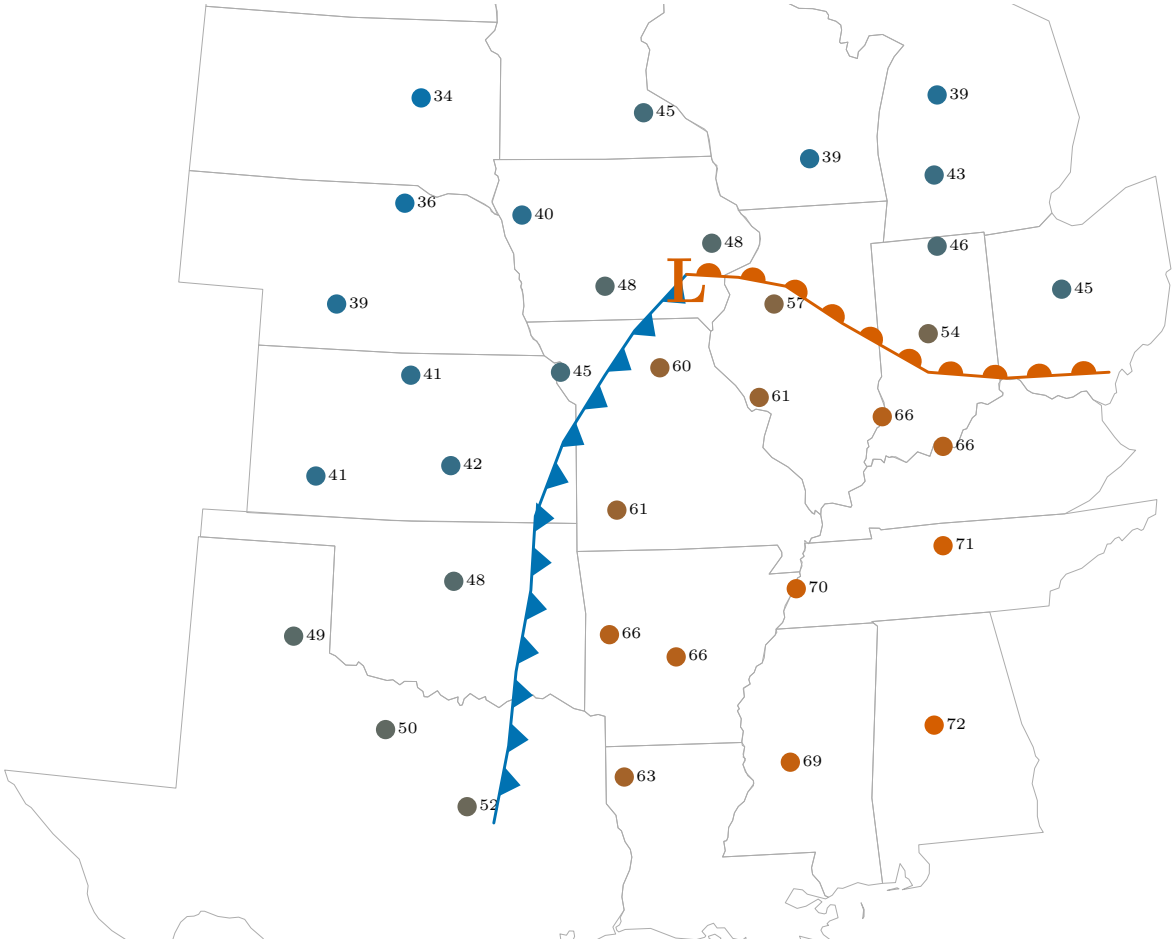


Figure 29.6. Step 3: air temperature (°F) at each station, coloured from blue (cold, 32 °F) to red (warm, 72 °F). The warm air fills the south-east and a tongue reaches towards the low: the warm sector. Cold air covers the north-west. The cold front (blue, triangles) separates them in the west; the warm front (red, semicircles) in the east.

29.3.5 Step 4: cloud cover above 75 %

Cloud cover above 75 % means the stations drawn with a three-quarters, nine-tenths or completely filled circle. Looking at the sky circles: stations 2, 5, 6, 8, 9, 10, 11, 14, 17, 23, 31 and 32 are completely overcast; stations 3, 13, 19 and 28 have 7 to 8 tenths. Draw a line round the group, including the area between them. The result is a broad area wrapped round the low and extending along the cold front to the south-west.

29.3.6 Step 5: the precipitation

Shade round every station that reports precipitation: the weather symbol is a dot or two dots (rain), or the thunderstorm symbol with a dot. These are stations 2, 5, 6, 8, 9, 10, 11, 17, 18 and 23: a band to the north and east of the low, along the warm front, and a narrow band along the cold front in Missouri.

Table 29.4. Evidence for the fronts: pairs of stations on either side of each front (cold side → warm side).

Front	Stations	Temperature (°F)	Dew point (°F)	Wind from
cold	16 → 17	45 → 60	40 → 55	NW → SSW
cold	22 → 23	42 → 61	36 → 57	NW → SSW
cold	25 → 28	48 → 66	38 → 59	WNW → SW
cold	31 → 32	52 → 63	50 → 61	WSW → S
warm	8 → 13	48 → 57	45 → 54	NE → SSW
warm	14 → 19	54 → 66	45 → 53	E → S
warm	10 → 20	46 → 66	43 → 55	E → S

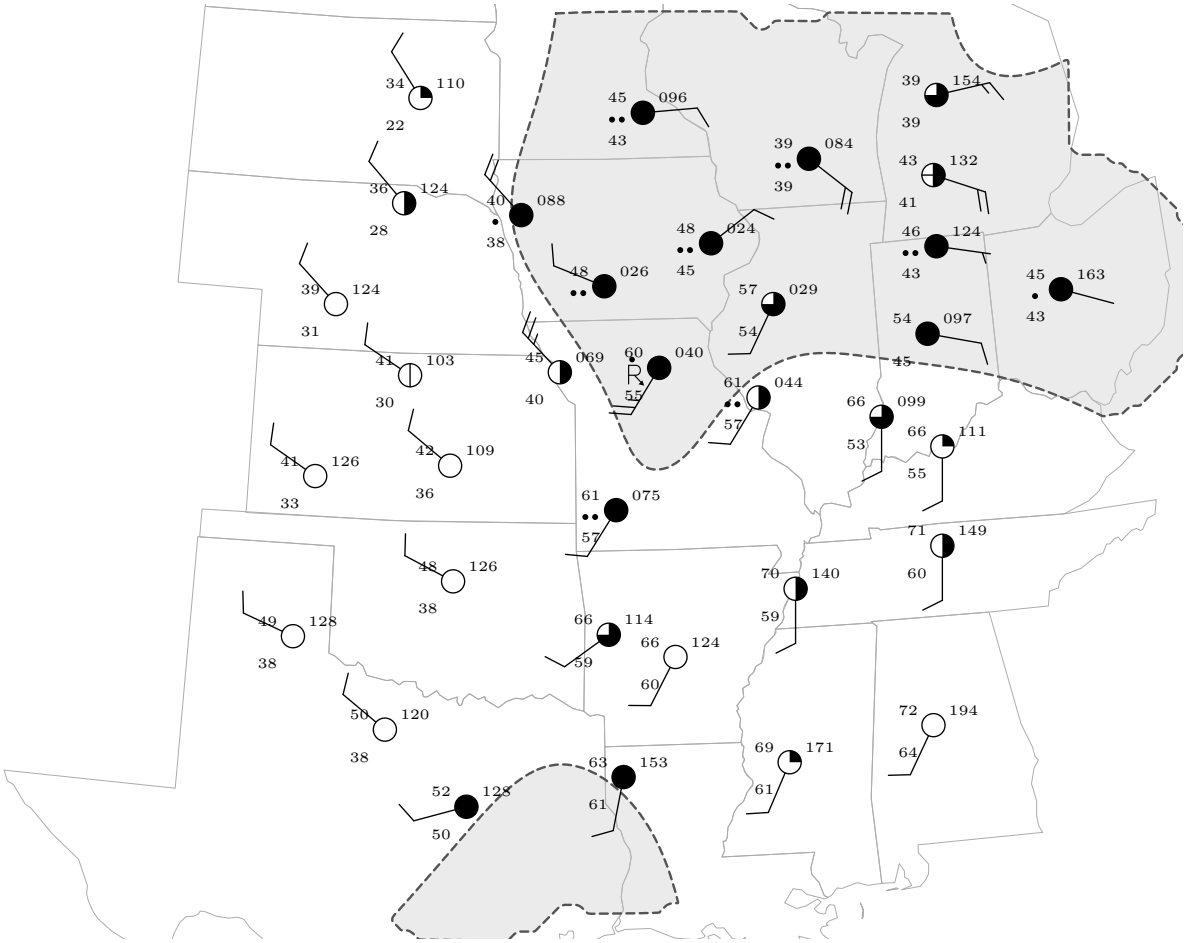


Figure 29.7. Step 4: the area where the cloud cover exceeds 75 % (dashed outline and grey tint). It wraps round the low on its north and east sides and tails away to the south-west along the cold front.

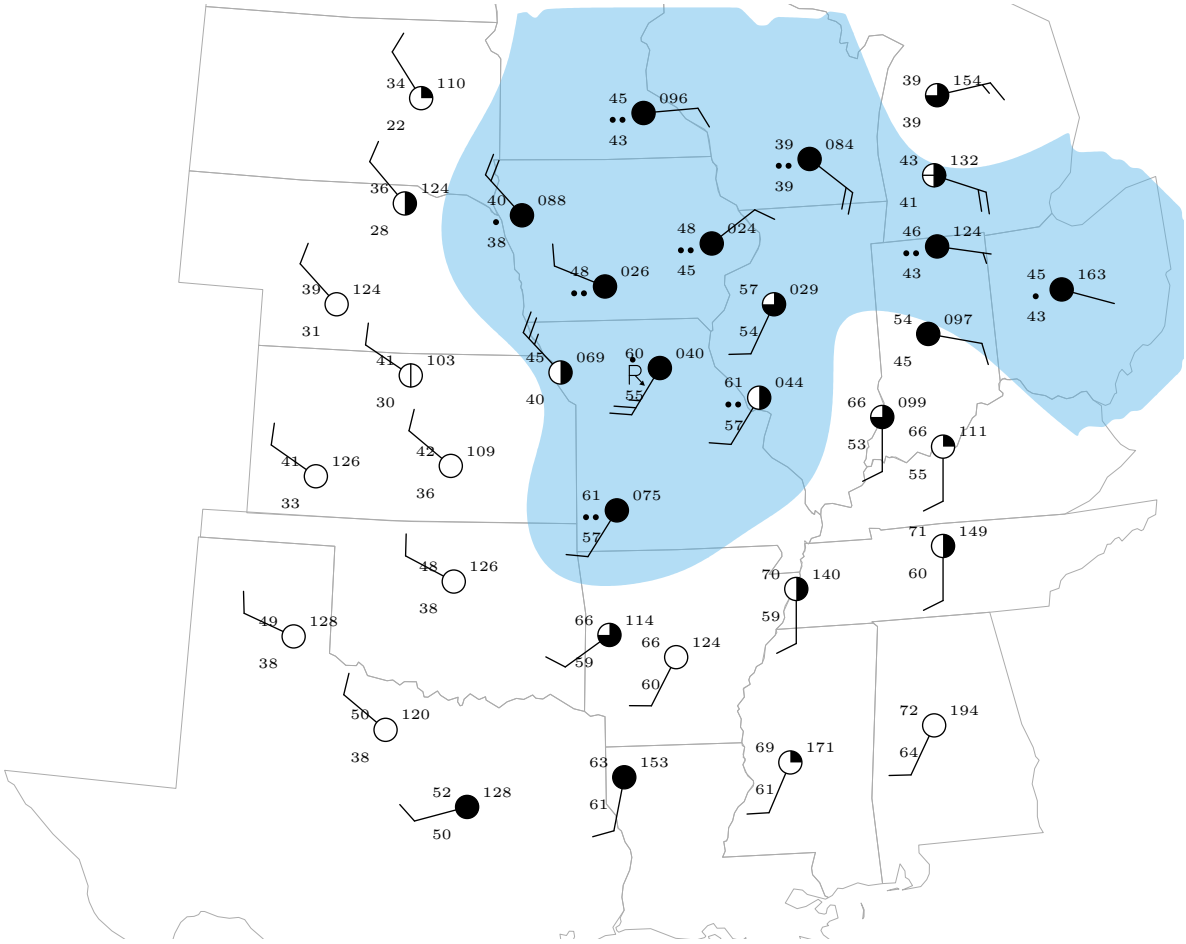


Figure 29.8. Step 5: the areas receiving precipitation (blue shading) at the ten stations that report rain or thunderstorms.

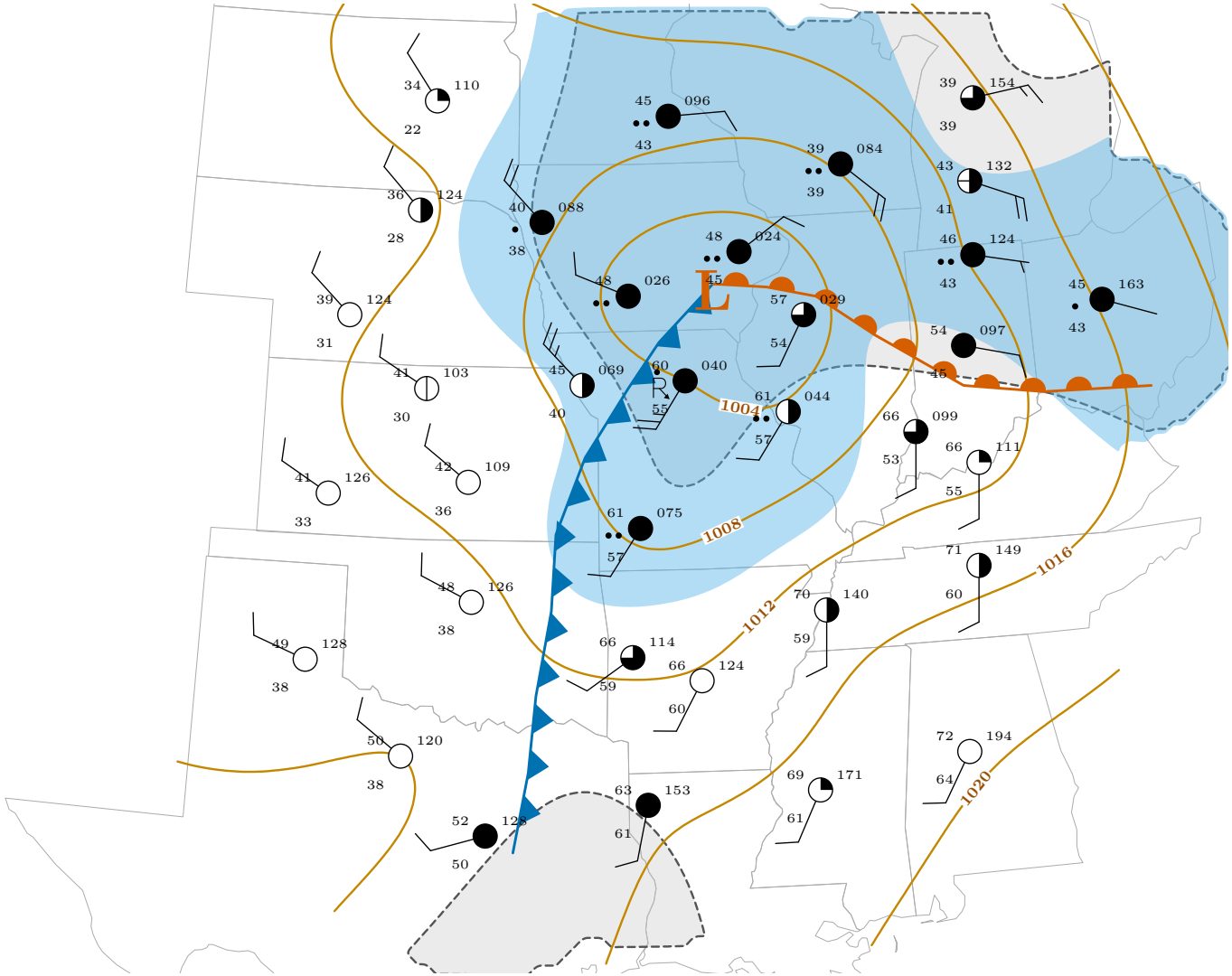


Figure 29.9. Question 14 complete: isobars every 4 mb (orange), the low (L), the cold front (blue triangles) and the warm front (red semicircles), the area with more than 75 % cloud (dashed outline) and the precipitation (blue shading), over the 34 stations. The state outlines are real boundaries. The fronts are an interpretation, drawn from the temperature, dew-point and wind contrasts.

29.4 Reading the finished picture

The map is a text-book mid-latitude depression.

- The low is over Iowa and northern Illinois, at about 1002 hPa.
- The **warm sector** is the tongue of warm, humid air in the south and east, between the two fronts: temperatures in the 60s and 70s °F and dew points of 55 to 64 °F.
- The cold air in the north-west is dry (dew points of 22 to 38 °F) and mostly clear.
- The cloud and rain are on the fronts and near the low, where the air is being lifted.
- The winds circle the low anticlockwise (Chapter 28) and cross the isobars towards low pressure (Chapter 26).

Worked example: is the wind consistent with the isobars?

Between stations 9 (1002.6 hPa) and 5 (1008.8 hPa), about 216 km apart, the pressure changes by 6.2 hPa. The geostrophic wind for that gradient at 41.5° N ($f = 9.7 \times 10^{-5}\,\mathrm{s}^{-1}$) is

$$V_g = \frac{1}{\rho f}\frac{\Delta p}{\Delta n} = \frac{620}{1.2 \times 9.7 \times 10^{-5} \times 2.16 \times 10^{5}} = 25\ \mathrm{m\,s^{-1}} \approx 48\ \text{knots}.$$

The observed surface winds are 10 to 25 knots. That is about what friction does: it slows the surface wind to a half or two thirds of the geostrophic value and turns it across the isobars (Chapter 26).

29.5 Questions that come up

A question that comes up

Where exactly should I put the front?
There is no single right answer: a front analysis is an interpretation. Put the line along the zone where the temperature, dew point and wind change fastest, bend it to follow the stations, and begin it at the low. Check that the stations on the cold side are colder and drier than those on the warm side.

A question that comes up

Why do the isobars bend sharply where a front crosses them?
Because the wind changes direction at a front, and the wind follows the isobars, the isobars must change direction there too. A front lies along a trough of low pressure, with the isobars forming a V pointing away from the low.

A question that comes up

Which stations count as "cloud above 75 %"?
Those whose circle is filled to at least three quarters: 7–8 tenths, 9 tenths (filled with a gap) or overcast. Stations with half or less are outside the outline.

29.6 Try it yourself: Practice Set 29

Practice Set 29: fronts and isobars

29.1 Stations X (1011.1 hPa) and Y (1014.9 hPa) are neighbours. Where does the 1012 mb isobar cross the line between them?

29.2 A station reports a temperature of 61 °F, a dew point of 57 °F and a south-south-west wind of 10 knots. A neighbour to its west reports 42 °F, 36 °F and a north-west wind. What kind of front lies between them?

29.3 Station P is 30 °F with dew point 28 °F, east wind, overcast and snow. Station Q to the south is 41 °F with dew point 38 °F and south-east wind. Is a warm or cold front likely to lie between them? Why?

29.4 Why is there no 1000 mb isobar on Figure 29.3?

Answers to Practice Set 29

29.1 $(1012 - 1011.1)/(1014.9 - 1011.1) = 0.9/3.8 = 0.24$: 24 % of the way from X to Y.

29.2 A **cold front**: the temperature falls 19 °F, the dew point 21 °F, and the wind turns clockwise from south-south-west to north-west as you cross it from east to west.

29.3 A **warm front**: the warmer, moister air (41 °F, 38 °F) lies to the south with a south-east wind, riding over the colder air with east wind to the north, where it is snowing. The wind turns clockwise from east to south-east.

29.4 No station reports a pressure as low as 1000 hPa (the lowest is 1002.4), and an isobar should only be drawn where the data support it.

Chapter 30

Putting Part III together

30.1 One chain of ideas

The three sections of this part are one argument, and they close the argument begun in Parts I and II:

1. Air holds water vapour up to a limit set by its temperature. Relative humidity compares the vapour with that limit; the mixing ratio and the dew point measure the vapour itself, and cooling air to its dew point makes dew, fog or cloud (Moisture Q1–Q6, Chapters 21 and 22).
2. Warm air columns are taller than cold ones (the scale-height law of Part I, Chapter 3), so pressure surfaces slope down from the warm tropics towards the cold poles, more steeply with height (Motion Q1–Q6, Chapter 24). The uneven warming that causes this is the sun-angle effect of Part II (Chapter 18).
3. The slope produces a pressure gradient force; on a rotating Earth the Coriolis force turns the moving air until the two balance and the wind blows along the isobars, the geostrophic wind (Motion Q7, Chapter 25).
4. Near the ground friction slows the wind and turns it across the isobars towards low pressure, so air spirals into lows, rises, and forms cloud and rain (Motion Q8–Q11, Chapter 26).
5. A weather map shows all of this at once: station models record the observations, streamlines show the flow, and isobars, fronts, cloud and rain show the weather systems it organises (Map Q12–Q14, Chapters 27 to 29).

30.2 All the answers in one table

30.3 Mistakes to avoid

1. **Ranking by relative humidity** instead of the actual mixing ratio (Moisture Q3).
2. **Reading the dew point from the wrong row** of Table 21.1, or at the wrong temperature: always check that the air could hold *at least* the sample's vapour at the temperature you give (Moisture Q6: 26, not 25).
3. **Putting H and L the wrong way round** on the pressure surfaces: the lower pressure surface (colder column) is the L (Motion Q4).
4. **Drawing the Coriolis force to the wrong side:** right in the north, left in the south, always at right angles to the wind.
5. **Drawing the pressure gradient force away from low pressure.** It points *towards* low pressure.
6. **Forgetting that the staff of a wind barb points to where the wind comes from** (Map Q12 and Q13).
7. **Mis-reading the pressure code:** 998 is 999.8, not 1099.8 or 998.0 (Map Q12).
8. **Isobars that cross,** or that stop in the middle of the map (Map Q14).

Table 30.1. Every answer of Part III, with the chapter where it is worked out.

Question	Answer	Ch.
Moisture Q1	RH = 49.3, 88.8, 26.0, 10.4, 19.9 %	21
Moisture Q2	Humidity falls sharply: the vapour content is unchanged but the capacity of the warmer air is much larger	21
Moisture Q3	Mixing ratios 9.126, 8.974, 7.684, 11.078, 10.54 g kg^{-1}; ranks 3, 4, 5, 1, 2	21
Moisture Q4	RH does not say how much vapour is actually in the air	21
Moisture Q5	Dew point $\approx$ 14 °C (13.8)	22
Moisture Q6	w_s = 35.134 g kg^{-1}; w = 21.78 g kg^{-1}; dew point $\approx$ 26 °C	22
Motion Q1	Lines 5400→5700 m (500 mb) and 8700→9500 m (300 mb)	24
Motion Q2	Saskatoon	24
Motion Q3	Saskatoon is colder at each height; the cold column is thinner	24
Motion Q4	z has the higher pressure: H at z (Monterrey side), L at x; arrow from z to x	24
Motion Q5	At 9000 m (800 m against 140 m of slope)	24
Motion Q6	Wind speeds are greater at 9000 m	24
Motion Q7	North: H at 5520 m, L at 5460 m; south: H at 5640 m, L at 5580 m; force arrow to L, Coriolis arrow to H, wind between them	25
Motion Q8	The Coriolis force is weakened, because it depends on wind speed	26
Motion Q9	Red up (to low pressure), black up-right, blue down-right (right of wind), green down-left (against wind)	26
Motion Q10	NH low, NH high, SH low, SH high (examples a to d)	26
Motion Q11	On each box: red along the line to or from the centre; blue at right angles to the wind (right in the north, left in the south); green against the wind	26
Map Q12	Model 1: 1002.4, 37, **21**, 0 %. Model 2: 999.8, 63, 60, thunder with intermittent rain. Model 3: 1011.7, 43, 20, 0 %, 25 kt from SW, +1.7, rising. Model 4: 999.7, 27, 20, 100 %, continuous snow, 25 kt from NW, −1.1, falling	27
Map Q13	Five or six smooth streamlines swirling anticlockwise round a centre in Iowa and Illinois	28
Map Q14	Isobars at 4 mb; L in northern Illinois/Iowa; cold front to the south-south-west, warm front to the east; cloud above 75 % wrapped round the low; rain north and east of the low and along the cold front	29

30.4 Self-test

Try these first. The answers follow.

Self-test: one line each

S1. Air at 20 °C has a mixing ratio of 7.5 g kg^{-1}. What is its relative humidity?

S2. That air is cooled to 10 °C. What is its relative humidity now, and what is its dew point?

S3. Which holds more water vapour: air at 35 °C with RH 30 %, or air at 25 °C with RH 60 %?

S4. Find the dew point of air at 30 °C and RH 50 %, and the height at which cloud would form if it were lifted.

S5. True or false: the dew point can be higher than the air temperature. Explain.

S6. Why do pressure surfaces slope down towards cold air?

S7. The 300 mb surface is at 8700 m over place S and 9500 m over place M. In which direction does the pressure gradient force act at 9000 m?

S8. Two 500 hPa height lines 90 m apart are 600 km apart at 35° N. Find the geostrophic

wind.

S9. In the Southern Hemisphere a wind blows towards the west. Where is the low pressure, to its north or south?

S10. What does friction do to (a) the speed and (b) the direction of the surface wind?

S11. Over land the surface wind is $8\,\mathrm{m\,s^{-1}}$ and crosses the isobars at $40°$. Estimate the geostrophic wind.

S12. Air spirals inward and clockwise. Which hemisphere, which pressure system?

S13. Decode the pressure codes 087 and 962.

S14. Convert $50\,°\mathrm{F}$ and $14\,°\mathrm{F}$ to °C.

S15. A wind barb has two full barbs and one pennant. What is the speed?

S16. Behind a front the temperature falls $18\,°\mathrm{F}$ and the wind turns clockwise from south-west to north-west. Which kind of front?

S17. A staff points to the north-east. In which direction does the air move?

S18. The 1012 mb isobar lies between stations at 1009.6 and 1013.2 hPa. Where?

S19. Why does a strong wind go with closely spaced isobars?

S20. What can relative humidity not tell you?

Answers to the self-test

S1. $7.5/14.956 = 50\,\%$.

S2. $w_s(10) = 7.762$: RH $= 7.5/7.762 = 97\,\%$. Dew point: Table 21.1 gives 6.771 at $8\,°\mathrm{C}$ and 7.762 at $10\,°\mathrm{C}$, so $T_d = 8 + 2(7.5 - 6.771)/(7.762 - 6.771) = 9.5\,°\mathrm{C}$.

S3. $w_s(35) \approx 37.3$ gives $0.30 \times 37.3 = 11.2\,\mathrm{g\,kg^{-1}}$; $w_s(25) \approx 20.5$ gives $0.60 \times 20.5 = 12.3\,\mathrm{g\,kg^{-1}}$. The $25\,°\mathrm{C}$ air holds more.

S4. $w = 0.5 \times 27.694 = 13.85\,\mathrm{g\,kg^{-1}}$: between $18\,°\mathrm{C}$ (13.162) and $20\,°\mathrm{C}$ (14.956), $T_d = 18 + 2(13.85 - 13.162)/1.794 = 18.8\,°\mathrm{C}$. Cloud base $\approx 125 \times (30 - 18.8) = 1400\,\mathrm{m}$.

S5. False (in open air): it would mean the air is oversaturated and the surplus vapour would condense at once. At most $T_d = T$ at saturation.

S6. Cold air is denser, so a column of it is thinner: the same pressure drop uses up less height, so the pressure surfaces are lower over cold air.

S7. From the high (the 300 mb surface is higher over M, so z over M has the higher pressure) to the low: from M towards S.

S8. $f = 2 \times 7.292 \times 10^{-5} \sin 35° = 8.37 \times 10^{-5}$; $V_g = \dfrac{9.81 \times 90}{8.37 \times 10^{-5} \times 6 \times 10^5} = 17.6\,\mathrm{m\,s^{-1}}$ (34 kt).

S9. To the **north**: in the Southern Hemisphere the low is on the right of the wind (Buys-Ballot), and the right of a westward wind is north.

S10. (a) It reduces the speed. (b) It turns the wind across the isobars towards low pressure.

S11. $V = V_g \cos\alpha$, so $V_g = 8/\cos 40° = 10.4\,\mathrm{m\,s^{-1}}$.

S12. Inward and clockwise: Southern Hemisphere low.

S13. 087 → 1008.7 hPa; 962 → 996.2 hPa.

S14. $50\,°\mathrm{F} = 10\,°\mathrm{C}$; $14\,°\mathrm{F} = -10\,°\mathrm{C}$.

S15. $2 \times 10 + 50 = 70$ knots.

S16. A cold front.

S17. The air moves towards the south-west.

S18. $(1012 - 1009.6)/(1013.2 - 1009.6) = 0.67$: two thirds of the way from the lower to the higher station.

S19. Close isobars mean a large pressure gradient force, which needs a large wind to balance it with the Coriolis force (and friction).

S20. How much vapour is actually in the air (or how the air compares with air at another temperature): it only compares the vapour with the capacity at the present temperature.

30.5 A checklist

- ☐ I can look up, and interpolate, in Table 21.1.
- ☐ I can find an RH, a mixing ratio and a dew point from any two of T, w, RH.
- ☐ I can explain why pressure surfaces slope and why wind increases with height.
- ☐ I can draw the three forces correctly in both hemispheres.
- ☐ I can name the four systems of Figure 26.5 from their rotation.
- ☐ I can decode a station model: pressure, temperature, dew point, sky, wind, weather, tendency.
- ☐ I can draw streamlines (arrows downwind) and isobars (interpolating, never crossing).
- ☐ I can say how to recognise a front, and which way the wind turns across it.

Appendix C

Part III: formulas, stations, derivations and symbols

C.1 Formulas

Table C.1. The relationships used in this part.

Quantity	Formula	Where, and what for
Saturation vapour pressure	$e_s(T) = 6.112 \exp\left[\dfrac{17.67\,T}{T+243.5}\right]$ hPa (T in °C)	the table, dew point
Saturation mixing ratio	$w_s = \dfrac{622\,e_s}{p - e_s}$ g kg^{-1}, with $p = 995.5$ hPa	the table
Mixing ratio from RH	$w = w_s \times \mathrm{RH}/100$	Eq. (21.4); Moisture Q3, Q6
Relative humidity	$\mathrm{RH} = 100\,w/w_s = 100\,e/e_s$	Eq. (21.3); Moisture Q1
Vapour pressure from w	$e = \dfrac{w\,p}{622 + w}$	Eq. (21.2); dew point (formula)
Dew point	$T_d = \dfrac{243.5\,\ln(e/6.112)}{17.67 - \ln(e/6.112)}$	Moisture Q5, Q6
Rule of thumb	$\mathrm{RH} \approx 100 - 5\,(T - T_d)$	Eq. (22.1); checks
Cloud base	$z \approx 125\,(T - T_d)$ m	Eq. (22.3); cloud height
Condensed water	$w - w_s(T)$	Eq. (22.2); below dew point
Hypsometric equation	$\Delta z = \dfrac{R_d\,\overline{T}}{g} \ln\dfrac{p_1}{p_2}$	Eq. (24.1); Motion Q1–Q6
Pressure gradient force	$-\dfrac{1}{\rho}\dfrac{\Delta p}{\Delta n} = -g\dfrac{\Delta z}{\Delta n}$ (on a pressure surface)	Eq. (23.1); Chapters 23–26
Coriolis parameter	$f = 2\Omega \sin\varphi$, $\Omega = 7.292 \times 10^{-5}$ s^{-1}	Motion Q7
Coriolis force	fV per unit mass	Motion Q8
Geostrophic wind	$V_g = \dfrac{1}{\rho f}\dfrac{\Delta p}{\Delta n} = \dfrac{g}{f}\dfrac{\Delta z}{\Delta n}$	Eq. (25.1); Motion Q6, Q7
Wind with friction	$\alpha = \arctan(k/f)$, $V = V_g \cos\alpha$	Eq. (26.1); Motion Q9–Q11
Temperature	$T(°\mathrm{C}) = \frac{5}{9}[T(°\mathrm{F}) - 32]$	Eq. (27.1); Map Q12
Pressure code	3 digits: decimal before last; prefix 10 if < 500, 9 if ≥ 500	Map Q12
Knots	1 kt $= 0.514$ m s$^{-1} = 1.85$ km h^{-1}	Map Q12
Interpolation	$\text{fraction} = \dfrac{v_{\text{wanted}} - v_1}{v_2 - v_1}$	Map Q14

Constants. $g = 9.81$ m s^{-2}; $R_d = 287$ J kg^{-1} K^{-1}; $\rho \approx 1.2$ kg m^{-3} near the ground; $\Omega =$

$7.292 \times 10^{-5}\,\mathrm{s}^{-1}$; Earth's radius 6371 km.

C.2 The 34 stations of Figures 8 and 10

Temperatures and dew points are in °F; the pressure is the three-digit code and the decoded value in hPa; cloud cover is in percent (the filled fraction of the circle); the wind is the direction it comes **from** and its speed in knots (counted from the barbs, so approximate). Positions are good to about 10 km. A dash means that no value is available (station 9 has no dew point).

#	Position	Temp.	Dew	Code	hPa	Sky	Wind	Weather
1	44.6°N, 98.4°W	34	22	110	1011.0	25	NNW 10	
2	44.3°N, 92.9°W	45	43	096	1009.6	100	E 10	rain (continuous)
3	44.3°N, 85.6°W	39	39	154	1015.4	75	ENE 15	
4	42.7°N, 98.7°W	36	28	124	1012.4	50	NW 10	
5	42.5°N, 95.9°W	40	38	088	1008.8	100	NW 20	rain (intermittent)
6	43.4°N, 88.8°W	39	39	084	1008.4	100	SE 20	rain (continuous)
7	42.9°N, 85.8°W	43	41	132	1013.2	60	ESE 20	
8	42.0°N, 91.3°W	48	45	024	1002.4	100	NE 10	rain (continuous)
9	41.2°N, 93.9°W	48	–	026	1002.6	100	WNW 10	rain (continuous)
10	41.6°N, 86.0°W	46	43	124	1012.4	100	E 5	rain (continuous)
11	40.6°N, 83.1°W	45	43	163	1016.3	100	ESE 2	rain (intermittent)
12	40.8°N, 100.3°W	39	31	124	1012.4	0	NW 10	
13	40.8°N, 89.9°W	57	54	029	1002.9	75	SSW 10	
14	40.1°N, 86.4°W	54	45	097	1009.7	100	E 10	
15	39.6°N, 98.5°W	41	30	103	1010.3	10	NW 10	
16	39.7°N, 95.0°W	45	40	069	1006.9	50	NW 25	
17	39.8°N, 92.7°W	60	55	040	1004.0	100	SSW 25	thunder, rain
18	39.1°N, 90.4°W	61	57	044	1004.4	50	SSW 10	rain (continuous)
19	38.7°N, 87.6°W	66	53	099	1009.9	75	S 10	
20	38.0°N, 86.3°W	66	55	111	1011.1	25	S 10	
21	37.7°N, 100.5°W	41	33	126	1012.6	0	NW 10	
22	38.0°N, 97.5°W	42	36	109	1010.9	0	NW 10	
23	37.2°N, 93.7°W	61	57	075	1007.5	100	SSW 10	rain (continuous)
24	36.3°N, 86.5°W	71	60	149	1014.9	50	S 10	
25	36.0°N, 97.3°W	48	38	126	1012.6	0	WNW 10	
26	35.7°N, 89.8°W	70	59	140	1014.0	50	S 10	
27	34.9°N, 100.8°W	49	38	128	1012.8	0	WNW 10	
28	35.0°N, 93.9°W	66	59	114	1011.4	75	SW 10	
29	34.6°N, 92.5°W	66	60	124	1012.4	0	SSW 10	
30	33.3°N, 98.7°W	50	38	120	1012.0	0	NW 10	
31	31.9°N, 97.0°W	52	50	128	1012.8	100	WSW 10	
32	32.5°N, 93.7°W	63	61	153	1015.3	100	S 10	
33	32.6°N, 90.2°W	69	61	171	1017.1	25	SSW 10	
34	33.1°N, 87.0°W	72	64	194	1019.4	0	SSW 10	

The map figures in this part are drawn from this table.

C.3 Derivations for the curious

Nothing in this appendix is needed to answer the questions, but each result is used in the chapters. They are short.

C.3.1 Mixing ratio and vapour pressure

Treat the dry air and the water vapour as ideal gases at the same temperature T and volume. Their densities are $\rho_d = (p - e)/(R_d T)$ and $\rho_v = e/(R_v T)$, where p is the total pressure, e the vapour pressure and R_d, R_v the gas constants of dry air and of water vapour. The mixing ratio is

$$w = \frac{\rho_v}{\rho_d} = \frac{R_d}{R_v}\,\frac{e}{p-e} = \varepsilon\,\frac{e}{p-e}, \qquad \varepsilon = \frac{R_d}{R_v} = \frac{18.02}{28.96} = 0.622.$$

Multiplied by 1000 this is the $\mathrm{g\,kg^{-1}}$ of Table 21.1: $w = 622\,e/(p-e)$.

C.3.2 Saturation vapour pressure

The Clausius–Clapeyron equation relates the change of the saturation vapour pressure of water to temperature:

$$\frac{de_s}{dT} = \frac{L_v\,e_s}{R_v\,T^2},$$

where $L_v = 2.5\times10^6\,\mathrm{J\,kg^{-1}}$ is the latent heat of vaporisation. At 15 °C this gives $\frac{1}{e_s}\frac{de_s}{dT} = 6.9\,\%\,\mathrm{K^{-1}}$, which is why the curve of Figure 21.1 is steep and bends upward. Integrating with a constant L_v gives an exponential in $-1/T$; fitting the result over the range of the atmosphere gives the convenient empirical form

$$e_s(T) = 6.112\exp\!\left[\frac{17.67\,T}{T+243.5}\right]\ \mathrm{hPa} \qquad (T \text{ in } ^\circ\mathrm{C}),$$

which reproduces the values of Table 21.1 to better than 0.2 % between −20 and 40 °C.

C.3.3 The dew point

The dew point is the temperature at which the saturation vapour pressure equals the actual vapour pressure e. Setting $e_s(T_d) = e$ in the empirical form and solving for T_d gives, with $x = \ln(e/6.112)$,

$$T_d = \frac{243.5\,x}{17.67 - x}.$$

For the sample of Question 6, $w = 21.78\,\mathrm{g\,kg^{-1}}$, $p = 995.5\,\mathrm{hPa}$: $e = 21.78 \times 995.5/(622 + 21.78) = 33.68\,\mathrm{hPa}$, $x = 1.7066$, $T_d = 26.03\,^\circ\mathrm{C}$.

C.3.4 Cloud base from the dew point

A rising unsaturated parcel cools at the dry adiabatic rate $\Gamma_d = g/c_p = 9.8\,\mathrm{K\,km^{-1}}$. Its dew point falls more slowly, because the vapour pressure falls with the pressure: about $1.8\,\mathrm{K\,km^{-1}}$. The gap $T - T_d$ therefore closes at $9.8 - 1.8 = 8\,\mathrm{K\,km^{-1}}$, so the parcel saturates after rising $(T - T_d)/8$ km $\approx 125\,(T - T_d)$ m.

C.3.5 The hypsometric equation

The vertical balance (hydrostatic balance) says that pressure falls with height by the weight of the air: $dp = -\rho g\, dz$. Using the ideal gas law $\rho = p/(R_d T)$,

$$dz = -\frac{R_d T}{g}\frac{dp}{p}.$$

Integrating from p_1 (lower) to p_2 (upper) with the mean temperature $\overline{T}$ of the layer,

$$z_2 - z_1 = \frac{R_d \overline{T}}{g}\ln\frac{p_1}{p_2}.$$

Thickness is proportional to the mean temperature: that is why pressure surfaces rise over warm air.

C.3.6 The pressure gradient force on a pressure surface

At a fixed height the horizontal pressure gradient force per unit mass is $-\frac{1}{\rho}\frac{\partial p}{\partial n}$. On a surface of constant pressure, the height varies with position, and the hydrostatic relation $\partial p/\partial z = -\rho g$ converts the horizontal pressure gradient at fixed height into a height gradient along the surface:

$$-\frac{1}{\rho}\left(\frac{\partial p}{\partial n}\right)_z = -g\left(\frac{\partial z}{\partial n}\right)_p.$$

The density has disappeared: the force depends only on the slope of the pressure surface. This is why questions about the slope of the surfaces are questions about the force.

C.3.7 The geostrophic wind

For a wind V blowing along the isobars, the pressure gradient force $g\,\Delta z/\Delta n$ and the Coriolis force fV act at right angles to the wind and to each other's line, in opposite directions. Setting them equal,

$$fV_g = g\frac{\Delta z}{\Delta n} \quad\Longrightarrow\quad V_g = \frac{g}{f}\frac{\Delta z}{\Delta n}.$$

In terms of pressure: $V_g = \Delta p/(\rho f\,\Delta n)$.

C.3.8 The steady wind with friction

Take x east, y north, a pressure gradient force (P_x, P_y), Coriolis acceleration $(fv, -fu)$ and a linear drag $(-ku, -kv)$. In a steady state the three add to zero:

$$P_x + fv - ku = 0, \qquad P_y - fu - kv = 0.$$

Solving the pair for u and v,

$$u = \frac{kP_x + fP_y}{k^2 + f^2}, \qquad v = \frac{kP_y - fP_x}{k^2 + f^2}.$$

For the low on the north side and isobars running east–west ($P_x = 0$, $P_y = P > 0$) this gives $u = fP/(k^2 + f^2)$ and $v = kP/(k^2 + f^2)$: a wind with a component towards the low. The angle to the isobars satisfies $\tan\alpha = v/u = k/f$ and the speed is $\sqrt{u^2 + v^2} = P/\sqrt{k^2 + f^2} = V_g \cos\alpha$, with $V_g = P/f$. For $k = 0$ this is the geostrophic wind.

C.3.9 Pressure at a height between two reported levels

In Question 4 the pressure at 9000 m was estimated from the reported levels. Between two levels with pressures p_a and p_b at heights z_a and z_b, the pressure was interpolated linearly in $\ln p$:

$$\ln p = \ln p_a + (\ln p_b - \ln p_a)\frac{z - z_a}{z_b - z_a},$$

which is the exact result for an isothermal layer. Above the highest level an isothermal extension $p = p_{300}\exp[-g(z - z_{300})/(R_d T)]$ was used.

C.4 Weather symbols: quick reference

The symbols used on the maps. Cover the right-hand descriptions and test yourself.

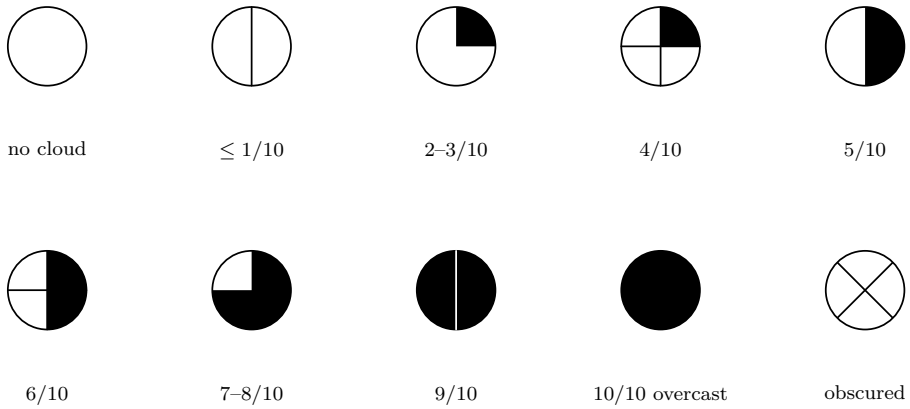


Figure C.1. Sky cover: no cloud, ≤ 1/10, 2–3/10, 4/10, 5/10, 6/10, 7–8/10, 9/10, 10/10 (overcast), obscured.

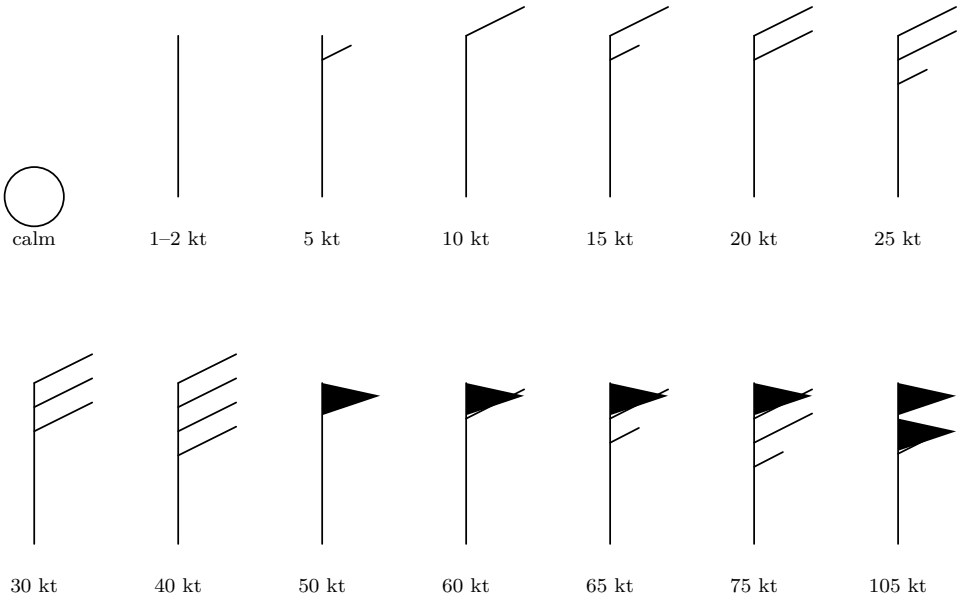


Figure C.2. Wind barbs (staff pointing north, from the north): calm, 1–2, 5, 10, 15, 20, 25, 30, 40, 50, 60, 65, 75 and 105 knots.

C.5 About the figures, the data and further reading

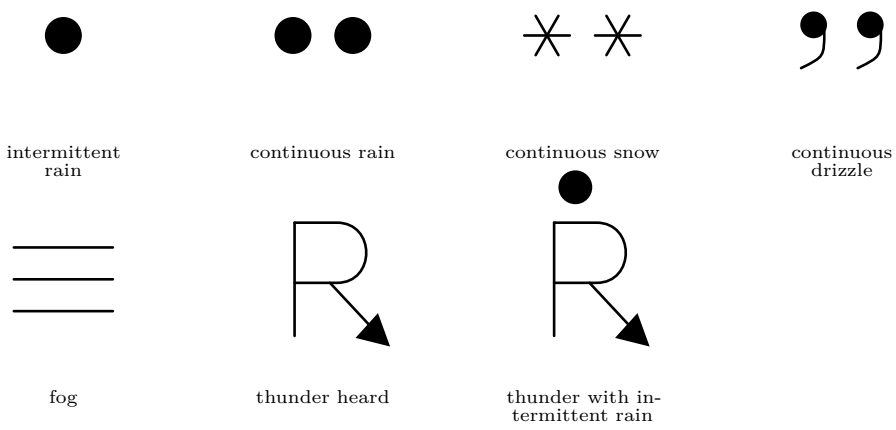


Figure C.3. Present weather: intermittent rain, continuous rain, continuous snow, continuous drizzle, fog, thunder heard, thunder with intermittent rain.

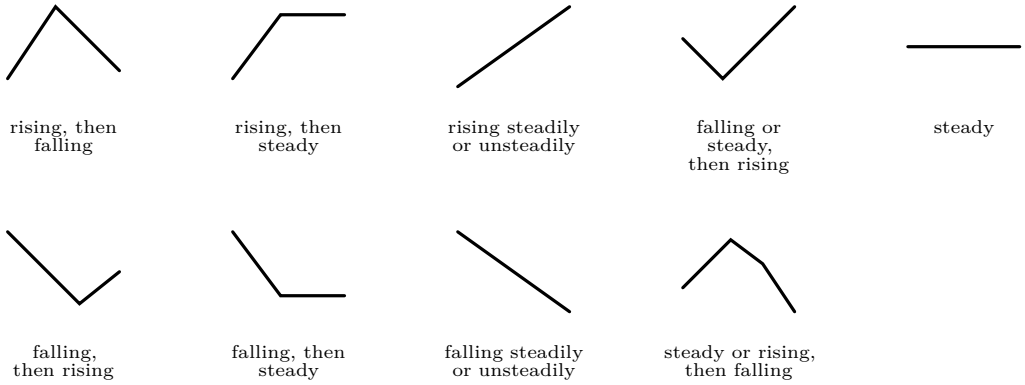


Figure C.4. Barometric tendency over the past three hours.

C.5.1 How the figures were made

Every figure in this part was drawn for it and is vector graphics: the curves and glasses, the force diagrams and the weather maps are all drawn in the document itself with TikZ and pgfplots, so they stay sharp at any size.

- The **saturation curves** are computed from the formulas of Appendix C.1; the dots are the values of Table 21.1.
- The **geostrophic adjustment** (Figures in Chapter 25) is the numerical solution of the equations of motion of a parcel released at rest between two height lines, with a gentle damping that relaxes the velocity towards the geostrophic wind.
- The **four circulation systems** (Chapter 26) are computed from the steady force balance of Appendix C.3, with friction equal to 0.6 times the Coriolis force.
- The **map**: state outlines are boundaries from Natural Earth (public domain), drawn in the Lambert conformal conic projection, fitted so that the stations fall on the right states. The 34 stations are listed in Appendix C.2.
- The **isobars** are contours of a smooth thin-plate-spline surface through the 34 station pressures; the **cloud and rain areas** are contours of distance-weighted averages of the station cover and rain reports; the **streamlines** follow a smoothed wind field made from the 34 barbs. The **fronts** were placed by hand, from the temperature, dew-point and wind contrasts, as an analyst would.

C.5.2 Limits to bear in mind

- Station values were read from a plotted teaching map: wind speeds are estimated by counting barbs, and a few cloud-cover values could differ by one tenth.
- The geostrophic-wind estimates between Saskatoon and Monterrey use only two places and rounded heights. They show the right behaviour (increase of wind with height) but not exact winds.
- The fronts and the isobars are *analyses*: reasonable analysts can draw them slightly differently. What matters is that every rule is respected (isobars never cross, fronts follow the contrasts of temperature, moisture and wind).

C.5.3 Further reading

- Ahrens and Henson (2022): the standard introduction; the station model, humidity and the forces on the wind are all treated at this level.
- Stull (2017): a free textbook that includes the hypsometric equation, the geostrophic wind and station models with worked examples.
- Wallace and Hobbs (2006): thermodynamics of moist air, the balance of forces, and fronts and cyclones.
- Holton and Hakim (2013): the geostrophic wind, thermal wind and friction at a more advanced level.
- Bolton (1980): the saturation vapour pressure formula used here.
- World Meteorological Organization (2019): the definition of the station model and its codes.

Appendix D

Constants, units and conversions

D.1 Constants

Each part works with the rounded values below. The precise values are given for reference.

Table D.1. Constants used in the book.

Quantity	Value used	Precise value	Source	Part
Acceleration due to gravity g	$9.81\,\mathrm{m\,s^{-2}}$	$9.80665\,\mathrm{m\,s^{-2}}$ (standard)	NOAA et al. (1976)	I, III
Gas constant for dry air R_d	$287\,\mathrm{J\,kg^{-1}\,K^{-1}}$	$287.05\,\mathrm{J\,kg^{-1}\,K^{-1}}$	Wallace and Hobbs (2006)	I, III
Specific heat of dry air c_p	$1004\,\mathrm{J\,kg^{-1}\,K^{-1}}$	$1004\,\mathrm{J\,kg^{-1}\,K^{-1}}$	Wallace and Hobbs (2006)	I
Standard sea-level pressure	1000 hPa (Part I)	1013.25 hPa	NOAA et al. (1976)	I
Standard sea-level temperature	15 °C	288.15 K	NOAA et al. (1976)	I
Stefan–Boltzmann constant σ	$5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$	5.670374×10^{-8}	Tiesinga et al. (2021)	II
Wien's constant C	$2898\,\mu\mathrm{m\,K}$	$2897.77\,\mu\mathrm{m\,K}$	Tiesinga et al. (2021)	II
Planck's constant h	–	$6.62607015 \times 10^{-34}\,\mathrm{J\,s}$ (exact)	Tiesinga et al. (2021)	II
Speed of light c	–	$2.99792458 \times 10^{8}\,\mathrm{m\,s^{-1}}$ (exact)	Tiesinga et al. (2021)	II
Boltzmann's constant k_B	–	$1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ (exact)	Tiesinga et al. (2021)	II
Solar constant S	$1368\,\mathrm{W\,m^{-2}}$	$1361\,\mathrm{W\,m^{-2}}$	Kopp and Lean (2011)	II
Earth's axial tilt	23.5°	23.44°	Stull (2017)	II
Earth's rotation rate Ω	$7.292 \times 10^{-5}\,\mathrm{s^{-1}}$	$7.292 \times 10^{-5}\,\mathrm{s^{-1}}$	Holton and Hakim (2013)	III
Kelvin–Celsius offset	273.15 (273.16 in Part II)	273.15 (exact)	Tiesinga et al. (2021)	all

D.2 Units and conversions

Quantity	**Conversion**
Pressure	1 hPa = 1 mb = 100 Pa; 1 mmHg (torr) = 1.333 hPa
Temperature	$T_{\mathrm{K}} = T_{\circ\mathrm{C}} + 273.15$; $T_{\circ\mathrm{C}} = \frac{5}{9}\,(T_{\circ\mathrm{F}} - 32)$
Temperature difference	1 K = 1 °C = 1.8 °F
Wind speed	$1\ \mathrm{kt} = 0.514\ \mathrm{m\,s^{-1}} = 1.852\ \mathrm{km\,h^{-1}}$
Length	1 km = 1000 m; $1\ \mu\mathrm{m} = 10^{-6}$ m
Power per area	$\mathrm{W\,m^{-2}}$: watts per square metre

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